

Trash Catching Vacuum Robot

ME 195B Section 01 - Senior Project Design II

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May 12th, 2026

Executive Summary

The Trash Catching Vacuum Robot is designed to improve the functionality of conventional household cleaning robots by combining autonomous vacuuming with an assistive trash catching feature. While current robotic vacuums are effective at collecting dust and small debris, those robots do not provide a convenient way to dispose of larger waste items. This project proposes a multifunctional robotic platform that can autonomously clean floors while also serving as a mobile trash receptacle capable of intercepting thrown waste. The goal of the system is to reduce the effort required for routine cleaning tasks and provide a more interactive and efficient household cleaning solution. The primary objective of this project is to design and prototype a compact robot capable of operating in two primary modes: vacuuming and trash-catching. The robot uses a differential drive system that allows for tight maneuverability and in place rotation, enabling navigation within typical indoor environments. In vacuum mode, the robot autonomously travels across the floor while collecting debris through a suction mechanism and storing it in a removable dust container. In trash catching mode, the system uses an onboard camera and sensors to detect and track incoming objects, estimate their trajectory, and move to an interception point where the object can be captured in a collection bin. Engineering analysis was performed to determine the mechanical and performance requirements necessary for successful operation. Projectile motion calculations were used to model a scenario in which an object is thrown at approximately 8 mph from a height of 1 meter at a 60° angle. Based on these calculations, the robot must reach a peak interception speed of approximately 3.9 mph in order to successfully catch the object. These results were used to determine the required motor speed and torque for the drive system. Additionally, structural analysis of the collection bin was conducted using finite element analysis to determine the minimum wall thickness needed to withstand the impact forces generated by thrown objects. The proposed robot has a target size of approximately 12.75 inches in height and 14 inches in diameter, with a maximum payload capacity of 6 pounds and an overall system weight of approximately 18 pounds. The electrical system is powered by a 6S LiPo battery, with power distributed through a buck converter to support the motors, sensors, and onboard computing components. Safety considerations, including current limits and proper power management, were incorporated to ensure reliable operation. Experimental testing and validation were conducted to evaluate the performance of the robot's key subsystems, including the dustbin airflow design, object detection model, and navigation system. Airflow simulations and physical testing of the dustbin demonstrated improved suction efficiency and reduced vortex formation with the optimized geometry. The AI-based trash detection system was trained and tested using an external image-label dataset, showing moderate object recognition and bounding box accuracy under controlled conditions. Additionally, integration testing of the camera and motion system confirmed that the robot can detect, track, and respond to target objects by issuing real-time movement commands. LiDAR-based mapping and navigation tests further verified the robot's ability to interpret its surroundings and operate autonomously. Overall, this project demonstrates the integration of mechanical design, electronics, and sensing technology to create a multifunctional household robot. By combining vacuuming and trash catching capabilities into a single system, the design expands the functionality of traditional cleaning robots and highlights the potential for future developments in assistive home automation.

Acknowledgement

The members of this team would like to express their gratitude to the Mechanical Engineering Department for funding the project so the group may have more leeway with the spending on the project and reduce the financial burden amongst the team. The team appreciates the support they have gotten from the people who supported the project through encouragement and financial support to complete this project. The team expresses their appreciation towards Dr. Raghu Agarwal for sharing his knowledge and watching over the project to not only refine it, but also achieve support for its goals. The team would also like to thank their classmates for the criticism and review that have been conducted during presentations, so that improvements can happen early on rather than later. Finally, a special thanks to the people that work in SJSU's Makerspace for 3D Printing the parts necessary to make this project function, helped reduce time and created a framework the team can work with for the project. This project has not only been supported financially, but with words as well.

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Chapter 1: Introduction

Section 1.1: Background and Motivation

Maintaining a clean room is essential for health, safety, and the preservation of surfaces, yet routine cleaning remains time-consuming and labor-intensive. Throughout history, there have been inventions and tools that would help alleviate those tasks. For example, clothes and water to mops and special cleaning agents. Although there may be alternative cleaning methods to make homes or places environmentally safe. It demands the user's involvement in order to maintain that clean environment. Some consumers have trouble with such tasks: senior citizens, the physically unable, and people who don't have the time to spend on cleaning. Currently, in today's market, there are floor cleaning robots called Roombas that alleviate the need to mop or clean floors manually, but don't succeed in the task of collecting larger debris, such as crumpled paper, food scraps, or any irregular trash that is not small enough for vacuum clearance.

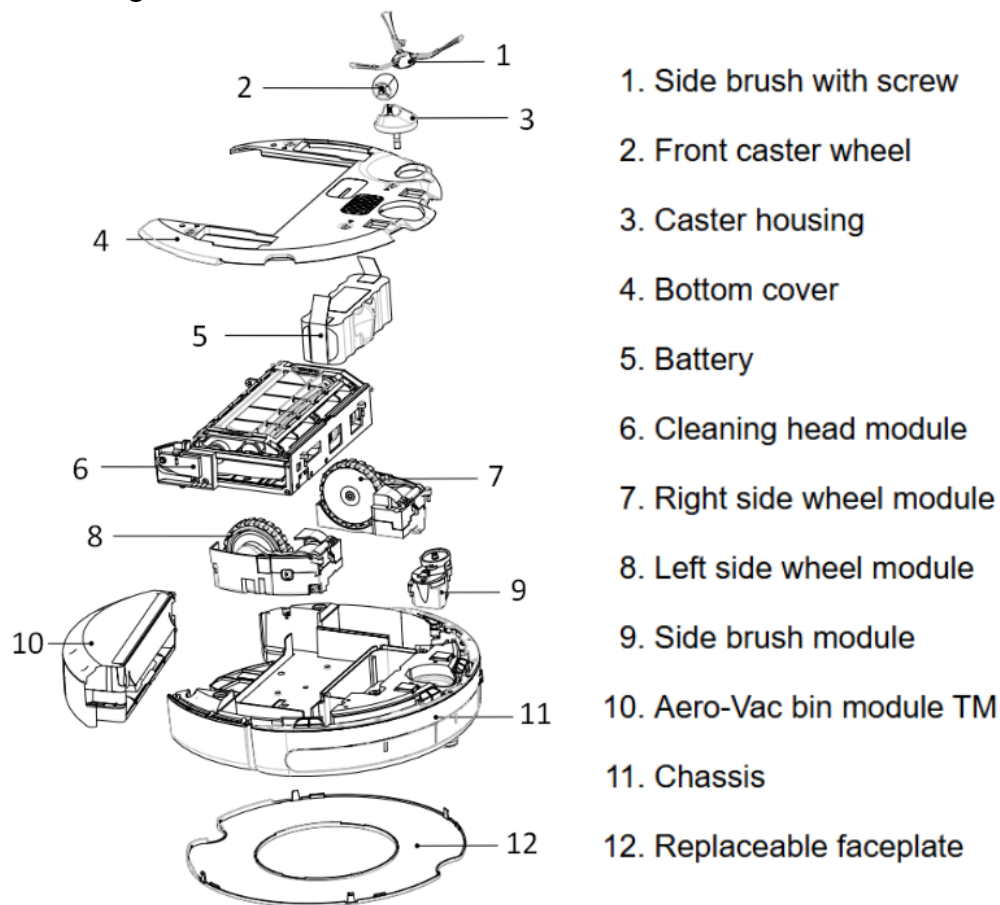


Figure 1: Exploded View of a Roomba 600 Series with Labeled Components

Figure 1 illustrates an exploded model of a Roomba 600 series. The design assists in floor cleaning, but does not help with disposing of larger debris, as it does not have a compartment with enough volumetric space or the function to do so. This results in additional time and energy being required from the user to dispose of these items by heading to the nearest bin for item removal. Even when the Roomba's sole purpose is to clean floors, it is not consistent, as the Roomba would need to clean the same area multiple times due to its innate coding of randomized pathing to consider it acceptable. The Roomba 600 series is a tripping hazard, as the only sensor the Roomba 600 has is a bump sensor, which is a bump-only detection through direct contact, which then triggers obstacle avoidance by changing pathing to a random direction. This became a motivation for the design aspect

and functionality of the robot to assist in the field of consistent cleaning and help carry larger debris for the consumers. An add-on to solve the user's problem of needing to walk to the nearest bin would be the robot's capabilities to catch any thrown object for convenience and less labor-intensive work. This is important as it will save the time and energy for the user when disposing of unwanted objects, as the robot will remain within the vicinity of the user to assist with trash removal. This relieves the requirement of labor-intensive work for consumers who don't have the physical means and schedule to maintain their space.

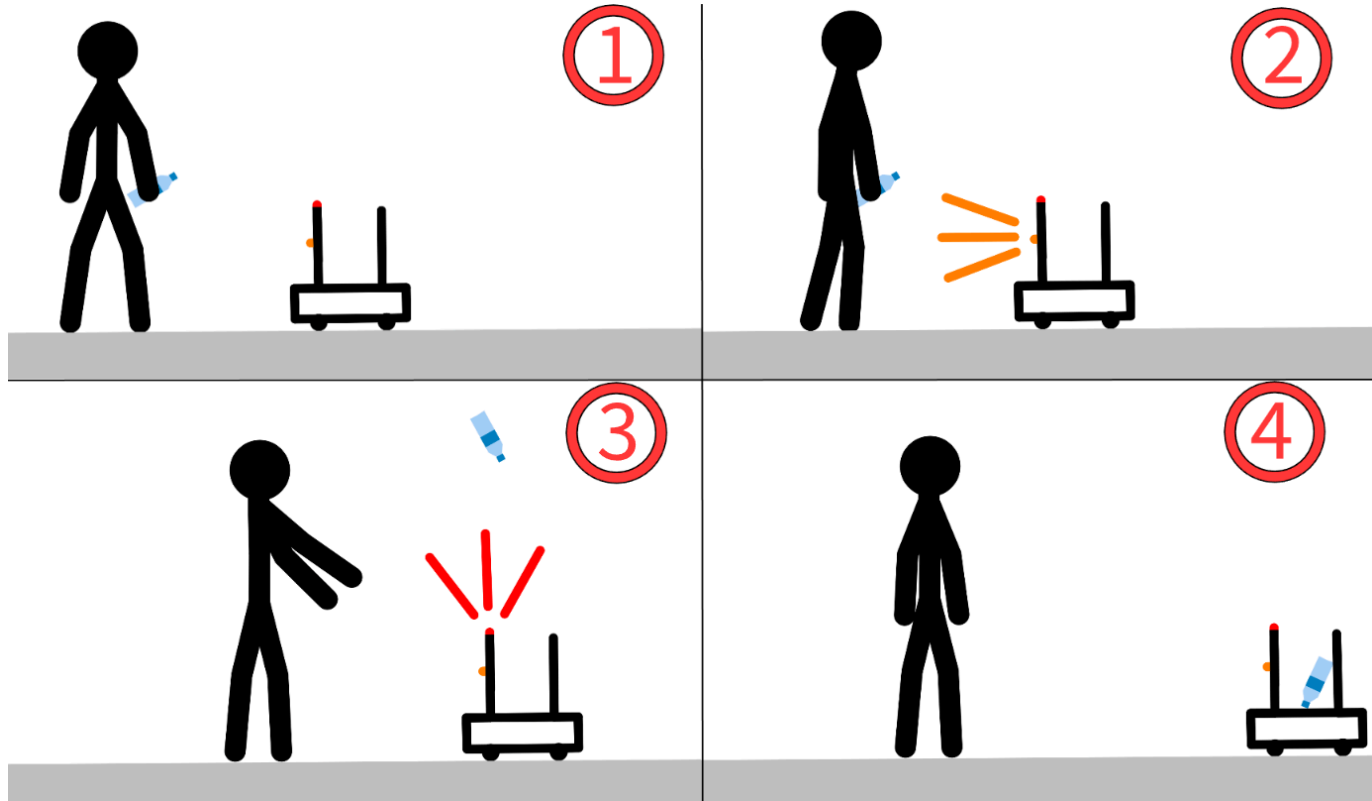


Figure 2: Story Board for Project's Purpose

Figure 2 presents a storyboard numbered for the sequence of events. It is to show a visual representation of what the robot will do when dealing with a thrown object from the user. The robot will also feature safety aspects such as obstacle avoidance, which comes with the challenge of staying near the user at a safe distance to prevent itself from being a tripping hazard and navigating the cleaning space when vacuuming the floors. Optimized pathing is another challenge the robot will face for efficient cleanup.

Section 1.2: Project Objective and Design Specifications

The team's main objective is to design, build, and program a household assistant that can independently vacuum and serve as an aim-assist trash bin. To achieve this, the project requires the need of a drive system capable of quick mobility, along with vacuum components such as a fan to generate suction and a litter compartment. There are two state-of-the-art design concepts that relate to the team's project, one is the autonomous vacuum cleaner popularized by the Roomba, and the other is the trash catcher prototype created by HTX Studio, a design studio based in Hangzhou, China. These are two different types of cleaning, and this project does it differently by combining both systems into one. Because multiple modes are integrated, a concept for mode switching must be added. For this, the project will implement a remote control to switch between modes, and since both systems have different drive systems, one must be picked. The drive system of the robot requires

two-wheel inline with each other, reminiscent of the Roomba. Since the project is intended as a general-use assistant, it will combine these two concepts into one prototype. In addition to the two design functions, some other objectives that the team wants to implement are that it should also be easy and practical to operate. For this, a remote to switch between the varying functions and modes of the robot is needed. These functions will allow the user to command actions like starting automated vacuuming or idling nearby to be ready to catch tossed objects. Not to mention, unlike traditional autonomous cleaning robots, this system combines continuous vacuuming with targeted debris capture from humans using the ROS2 architecture and artificial intelligence. Allowing communication from multiple data inputs for publisher nodes to subscriber nodes to complete the tasks. This product intends to cause a market shift to cater towards users who want a designated area cleaned at all times without the need to approach the nearest garbage bin with a push of a button. The end goal of the project is to create a product that reduces physical effort and time required to maintain a clean residential and workspace environment, particularly this would be targeting users with limited mobility or a lack of time. With the added extra feature, it should be easier to maintain the user's environment that they wish to clean. The team believes that the global impact of such a product would change automated home cleaning robots in ways that enhance cleaning and leisure time for consumers. Environmentally, the product does help clean, but at the same time, manufacturing a product would increase one's carbon footprint. This is a trade the product gives as it combines already marketed cleaning robot ideas into one to provide generally more help in cleaning, and is targeted towards consumer targets of wanting garbage disposed of right then and there, within proximity at all times. The following list is the general engineering construct and system that is required to make this come into fruition.

1.2.1 General Constraints and Practical Objectives

- Size: 12.75" Tall, 14" Diameter
- Voltage Source: 24 V
- Budget: ~\$1300
- Weight:
 - Payload Capacity: 6 lbs
 - Max weight: 18 lbs

1.2.2 Mobility Objectives

- Maneuverability
 - The robot needs to be able to rotate 360° while stationary, just like the Roomba, and this is to allow the robot to navigate the room easily during its vacuum mode. To achieve this, the driven wheels need to be in line with each other across the center of the robot's base.
- Speed
 - In vacuuming mode, the speed of the robot should be about one foot per second.
 - For the trash-catching mode, it is designed to catch an object that is thrown at 8 mph, a height of 1 m, and at an angle of 60° from the ground. The max speed that the robot reaches to intercept the thrown object is 3.9 mph. The oversized motor selected allows the robot to reach a speed of 9.6 mph.
- Obstacle handling
 - The robot should be able to navigate household environments while avoiding obstacles.

1.2.3 Trash-Catching (Aim-assist) Objectives

- Capture Success
 - The robot is expected to safely catch a 1 lb object without breaking.
 - The success rate of catching is 75%

- Detection Range
 - Due to the constraints of the selected Raspberry Pi AI camera, the view range to see a thrown object is 76° .
- Reaction Behavior
 - The robot will begin accelerating within 0.25s of an object crossing the center of its view range.

1.2.4 Vacuuming Performance Objectives

- Suction Generation
 - Generate sufficient suction to collect dirt and dust particles
- Dust Bin Volume
 - The dust bin should have a minimum volume of 300 mL
- Operation Time
 - Operate continuously for at least 60 minutes

1.2.5 User Interaction and Control Objectives

- Control method
 - A remote control to switch between vacuuming mode and trash-catching mode

The robot is designed to weigh approximately 13.5 - 18 lb, depending on whether there is a load or not inside the bin. It features a compact footprint suitable for indoor navigation only, as the sensors are easily affected by sunlight. Moreover, it uses 4-inch radius wheels to support both smooth and fast maneuvering required for trash-catching. In the first mode, vacuum mode, the robot should be able to operate at around 3 mph to 4 mph. Though this speed can be modified through software to ensure a more precise and safe cleaning. In the second mode, trash-catching, it must accelerate at a reasonable yet efficient acceleration of 4 m/s^2 to cover lateral distances depending on how far the trash was thrown above the robot. Additionally, the power system uses a 6S LiPo battery with a nominal voltage of 22.2 V and a capacity of 10,200 mAh, which is capable of delivering peak currents at 49 A to the dual high-torque motors, supplying the other components that require minimal current, such as the vacuum motor, and onboard electronics such as sensors, cameras, and so forth. The electronic components mentioned above will have the 22.2 voltage stepped down to 5 V and 12V through a buck converter that can withstand the input current and voltage to prevent any fire hazard and device failure. Fast-acting fuses will be implemented for the high-current components, such as the battery and motor drivers, which are connected to the power distribution hub. This is to protect the whole system from failure by shutting down the system power, preventing short-circuiting. Expected runtime in mixed operation is roughly 1 hour to 2 hours, based on what mode is chosen and for how long. Functionally, the robot should be able to vacuum typical small debris (rice grains, crumbs, etc) effectively and achieve a trash-catching success rate of at least 80% during testing, assuming the sensors and cameras have small tolerances in accuracy. Note that standardized throws from a defined distance and angle will be applied. Other safety specifications include appropriate insulated wiring, where the system will use 8 AWG, 12 AWG, and 18 AWG stranded tinned copper wire according to codes and standards based on the current used for each subsystem, and appropriate terminal blocks that can handle the maximum current and voltage. Note that ferrules will be applied to secure a stable connection through the terminal blocks. This product will come with an accessible kill switch to power off the robot and allow the user to charge the LiPo battery safely.

Section 1.3: Customer Needs Summary

Table 1: Customer Needs Summary

Customer Needs	Rank order of Customer Needs (#)	Subjective Weight WT
Category 1 - Design		
Durability	4	3.25
Lightweight	4	5.75
Life Span	5	3.25
Stability	5	4.00
Trash Volume	3	3.75
Product Size	4	4.00
Category 2 - Cost		
Reasonable Pricing	3	2.75
Category 3 - Functionality		
Accuracy to Catch Trash	5	1.50
Vacuum Effectiveness	1	1.25
User Friendly	4	2.00
Category 4 - Technology/Sustainability		
Noise level	8	6.25
Artificial Intelligence (AI)	2	5.75

Section 1.4: Project Significance and Uniqueness

The Trash-Catching Vacuum Robot is significant because it combines two different functions into one integrated system. Most household robots are designed for a single purpose, such as vacuuming floors or navigating autonomously. This project goes beyond that by combining autonomous vacuuming with a real-time trash-catching feature. This requires the integration of mechanical design, electrical systems, and sensor-based control into one coordinated platform. What makes this project unique is the dual-function capability. The robot must not only move independently and clean floors, but also detect thrown objects, calculate their position, and

accelerate quickly enough to intercept them. This adds a level of responsiveness that traditional robotic vacuums do not have. The system relies on sensor fusion using LiDAR, cameras, and ultrasonic sensors, along with motor calculations and structural analysis to ensure stable and safe operation.

By combining motion control, object detection, and assistive functionality into a single device, this project demonstrates an innovative approach to household robotics. It represents a practical application of engineering principles to create a multifunctional robotic system.

Section 1.5: Society and Technology in Relation to the Project

The Trash-Catching Vacuum Robot reflects the growing role of technology in everyday life. Society is increasingly adopting smart-home devices and automated systems to improve convenience and efficiency. Robotic vacuums, voice assistants, and home automation platforms are becoming common in modern households. This project builds on that trend by expanding the idea of what a household robot can do. From a societal perspective, there is a rising demand for technologies that support aging populations and individuals with limited mobility. Many people prefer to live independently, but daily tasks can become physically demanding over time. Assistive robotics has the potential to reduce strain and improve accessibility within the home. This project aligns with that direction by focusing on reducing physical effort and improving convenience. Technologically, the project demonstrates how modern sensors, microcontrollers, and automation systems can work together to create intelligent behavior. It reflects the broader movement toward systems that can sense, process information, and respond in real time. Utilizing 3D-printed PLA for the structural assembly, the project considers the environmental footprint of manufacturing. By combining these societal needs with responsible engineering choices, this project illustrates how modern automation can improve daily life while remaining ethically and environmentally conscious.

Section 1.6: Teamwork

- ▲ Project Manager - Schedules meetings, tracks deadlines, and oversees progress.
- Designer/CAD - 3D design and drawings
- Analysis/Simulation - Finite Element Analysis, SOLIDWORKS FEA, Program Simulation
- ✕ Manufacturer/Assembly - 3D prints, assembly with mounting features
- ◆ Programmer - program development, utilizing sensors
- ◆ Documentation - report writing, photo documentation

Table 2: Member Responsibilities

Patrick Wong	▲ ■ ● ◆
Branden Tran	■ ✕ ◆
Gustavo Ornelas	◆ ◆ ●
Alejandro Cedeno	✕ ◆
Kristopher Do	◆ ◆

- Detailed tasks will be assigned with deadlines
- Progress will be reviewed during weekly meetings to ensure accountability
- Weekly or subteam meetings will serve as an opportunity to adjust tasks or workload in a fair manner

Section 1.7: Gantt Chart

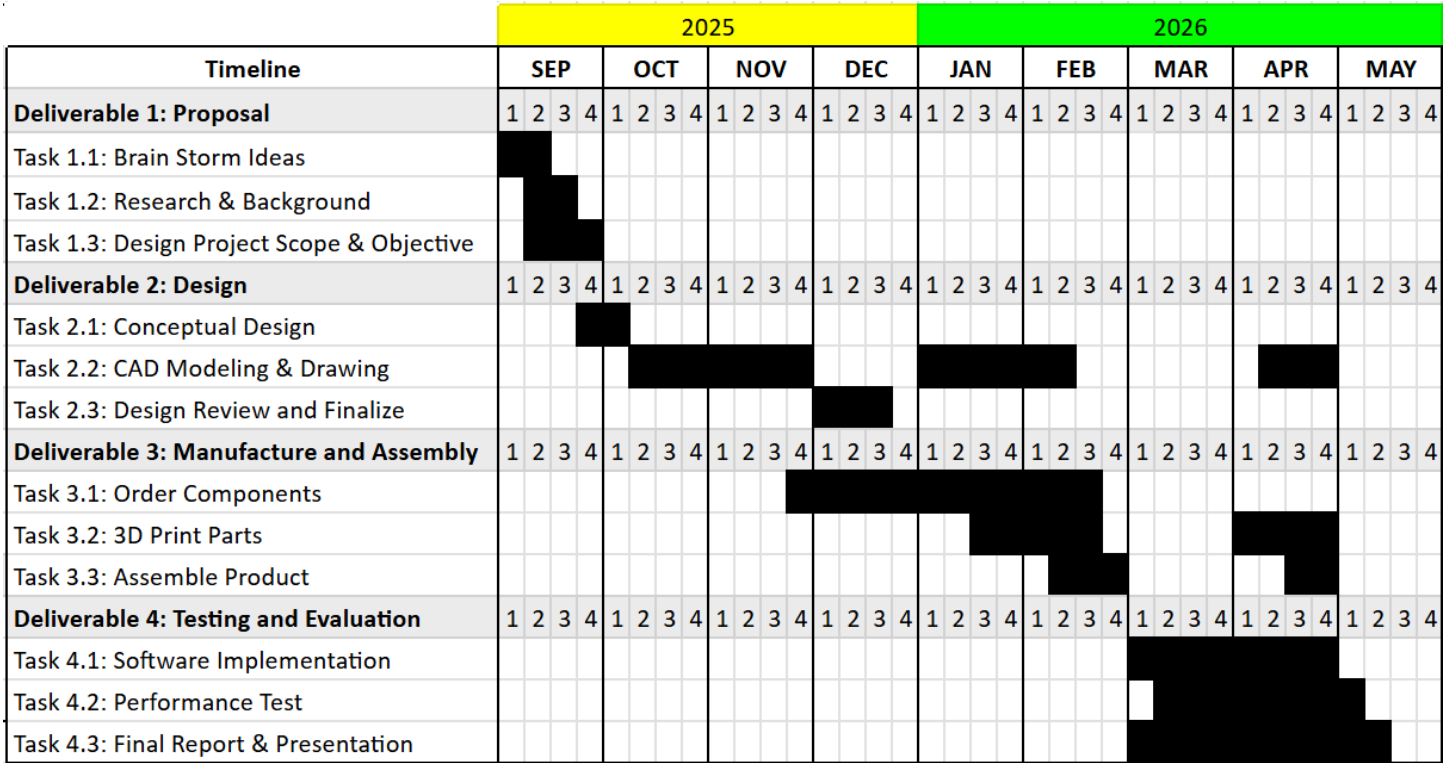


Figure 3: Gantt Chart

Chapter 2: Literature Review, Current Status, and Theoretical Background

Section 2.1: Theoretical Background & Engineering Principles

Speed Requirements and Motor Selection

A major part of the design process for this project is to determine how fast the robot needs to go to catch a human-tossed object. For that, the parameters of the throw or projectile must be defined before any calculations can proceed. The parameters that need to be defined are the speed magnitude of the throw, initial height or displacement, and angle with respect to the floor. After this is done, the basic kinematic equations shown in (1) and (2) are used to graph the trajectory of the projectile and find the final velocity at the intersection point. Assuming that the robot is initially at rest, it will start accelerating as soon as the object crosses the center of the robot, and the required acceleration to intercept the projectile can be found using (2).

$$x_f = x_i + v_i t + \frac{1}{2} a t^2 \quad (1)$$

$$v_f = v_i + a t \quad (2)$$

Now knowing the velocity and acceleration components of the robot, the calculations regarding the motor parameters can proceed after understanding the free-body diagram of a driven wheel. Shown in Figure 4, the components relating to motion and motor torque are highlighted in red. It is seen that the frictional force on a driven wheel is what drives the wheel in the direction of velocity. This frictional force, which can be calculated using (3), is the only force creating torque about the center of the wheel. Using (4), the torque that the motor needs to apply to produce the required acceleration can be found.

$$F = m a \quad (3)$$

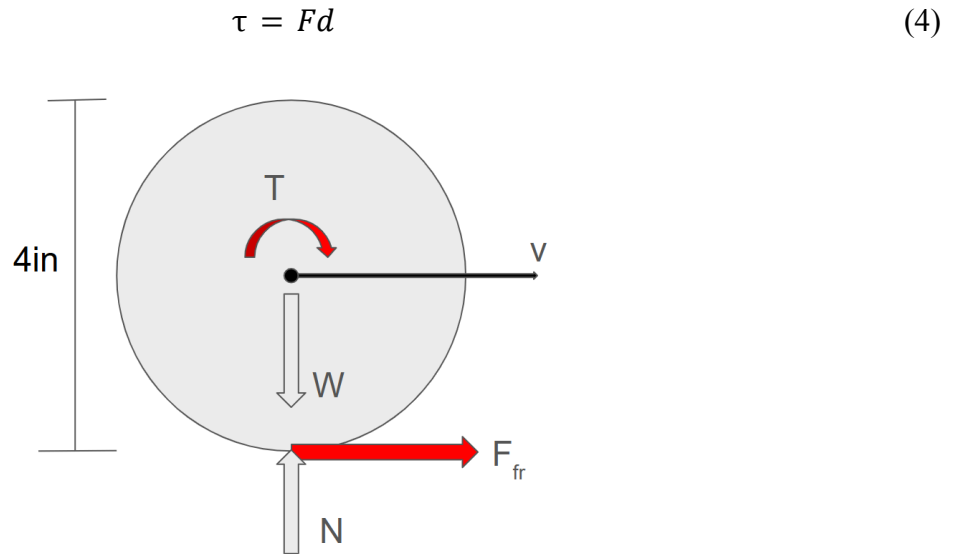


Figure 4: FBD of a Driven Wheel

Electrical motors are typically advertised with angular velocity. Therefore, (5) will be used to convert linear velocity to angular velocity.

$$\omega = v/r \quad (5)$$

Electrical Analysis for System

For the robot to operate reliably, the electrical system must be stable and properly wired, which requires identifying key electrical parameters in advance, including the operating voltage, rated current, and stall current of each DC motor, as well as the power requirements of other components such as cameras, sensors, the microprocessor, and motor drivers. These values are obtained from each component's datasheet and are necessary to estimate the total current drawn from the battery and to properly size power and protection components such as buck converters, motor drivers, kill switches, and fuses in order to prevent overheating, electrical failure, and potential fire hazards. The calculated electrical requirements are also used to select an appropriate LiPo battery based on its voltage, current-supply capability, expected runtime, and overall performance. Using equation (6), the total system devices' current is determined by summing the current demands of all system devices, while equation (8) is used to calculate the required buck converter power by multiplying the total current by the operating voltage. This required power is then compared to the rated power of the selected buck converter to ensure safe operation; if the rated power exceeds the required power, the converter can operate reliably, whereas an undersized converter may overheat, malfunction, or fail and potentially damage other system components.

A critical aspect of the electrical design is determining the input current drawn by the buck converter, which directly corresponds to the current drawn from the battery. Using Equation (8), the input current is calculated by solving for current based on the previously determined power requirement and the required maximum operating voltage. This value is then adjusted upward to account for efficiency losses, since the buck converter does not operate at 100% efficiency and the input voltage may fluctuate. Once the buck converter input current is established, the total battery current is calculated using equation (7). Because the system uses two wheel motors, their stall currents are doubled, while the stall current of the single vacuum motor is included once. Continuous current is not used in this analysis because it does not represent the worst-case load; instead, stall

current defines the maximum current demand, which can occur for several seconds if the motors become obstructed or prevented from rotating.

$$I_{\text{devices}} = I_1 + I_2 + I_3 + I_4 + I_5 \quad (6)$$

$$I_{\text{Battery}} = I_{\text{Buck Converter}} + I_{\text{Wheel Motors}} + I_{\text{Vacuum Motor}} \quad (7)$$

$$P = IV \quad (8)$$

With the total current drawn from the battery now known, verification is performed to ensure that the selected battery can supply the minimum required stall current while also providing the maximum voltage demanded by the entire electrical system, not just individual devices. If the battery fails to meet any of these criteria, it will be unable to deliver sufficient power, resulting in reduced motor torque and speed as well as unreliable operation of sensors, cameras, and other electronics. To verify battery capability, the continuous current rating is calculated using the battery specifications and equation (9), and this value is compared to the minimum total current drawn from the battery. If the battery's continuous current rating exceeds the required current, it meets the design criteria; otherwise, a higher-capacity battery must be selected. In addition, battery load is evaluated to assess overall performance, as excessive loading can lead to voltage sag, overheating, and reduced efficiency. Using equation (10), the battery load is expressed as a percentage of the battery's maximum capability. A lower load percentage indicates minimal voltage sag, cooler operation, and improved ability to maintain full motor torque and speed over longer operating periods.

$$A_{\text{Continuously}} = Ah \times C \quad (9)$$

$$\text{Battery Load} = \frac{I_{\text{Draw continuous}}}{A_{\text{battery continuous}}} \times 100 \quad (10)$$

Lastly, once the battery has been verified to safely support the robot's electrical system, the selected power distribution hub and terminal block must be capable of handling the total maximum current, total continuous current, and operating voltage supplied by the battery. Using equations (11) and (12), the current ratings provided in the power distribution hub, terminal block, and kill switch datasheet are compared against the calculated total continuous current and stall current to ensure the hub can safely distribute power without becoming a safety hazard due to excessive current or voltage. Ensuring these ratings are sufficiently higher allows power to be distributed reliably across all subsystems. After verifying the power distribution hub, terminal block, and killswitch, fuse selection is performed using equation (13) to protect against current overloads. This equation was used to determine appropriate slow-blow fuses for the battery, as well as fuses for the motor drivers connected to the wheel and vacuum motors. Protecting the motor drivers is critical for reliable robot operation, as these components must continuously supply the required current to the DC motors. It is important to note that fast-acting fuses must be rated to handle higher transient currents, and the motor drivers must support continuous operation while also having a maximum current rating exceeding 24.5 A to ensure safe and reliable performance.

Note: PDH → Power Distribution Hub, TB → Terminal Block, KS → Killswitch

$$I_{\text{PDH, TB, KS}} > 2 \times I_{\text{Continuous Wheel Motors}} + I_{\text{Continuous Vacuum Motor}} + I_{\text{Devices}} \quad (11)$$

$$I_{\text{PDH, TB, KS}} > 2 \times I_{\text{Stall Wheel Motors}} + I_{\text{Stall Vacuum Motor}} + I_{\text{Stall Devices}} \quad (12)$$

$$F_{\text{use Battery, motor driver}} = 1.25 \times I_{\text{Continuous}} \quad (13)$$

Stress Analysis for Material Thickness

With the robot having its job to catch items thrown by its users, it is important that it is able to handle the impacts of the thrown items. This is to ensure that the robot can safely dispose of trash without breaking, and can be continuously used without having to worry about its lifespan. This will be applied generally to the overall design of the robot and more specifically to the custom bin that's specially made for the robot. The robot's material will be made out of 3D printed PLA, which its spec's will be based on Figure 5, shown below. This will be used to apply the theory of the allowable stress to ensure the safety of the design. To meet safety, a factor of safety of 4 will be used for the overall project, as it is known to be the number used to represent continuous usage of a product or continuous loading. Applying equation (14) will determine the allowable stress that can be worked with for the material.

<u>Mechanical properties (*)</u>	<u>Injection molding</u>		<u>3D printing</u>	
	Typical value	Test method	Typical value	Test method
Tensile modulus	-	-	2346.5 MPa	ISO 527 (1 mm/min)
Tensile stress at yield	-	-	49.5 MPa	ISO 527 (50 mm/min)
Tensile stress at break	-	-	45.6 MPa	ISO 527 (50 mm/min)
Elongation at yield	-	-	3.3 %	ISO 527 (50 mm/min)
Elongation at break	-	-	5.2 %	ISO 527 (50 mm/min)
Flexural strength	-	-	103.0 MPa	ISO 178
Flexural modulus	-	-	3150.0 MPa	ISO 178
Izod impact strength, notched (at 23°C)	-	-	5.1 kJ/m ²	ISO 180
Charpy impact strength (at 23°C)	-	-	-	-
Hardness	-	-	-	-

Figure 5: Mechanical Properties for 3D Printed PLA

$$\sigma_{allowable} = \frac{\sigma_u}{SF} \quad (14)$$

Some parameters must be taken into account, especially the direction the impact of the object will be coming from in contact with the bin's wall, the velocity of the object that is heading towards impact with the wall, and the mass of the object. Contact time to the wall, which means how long the object has stayed in contact with the wall, is determined based on the elasticity of the object, as the more elastic, the lesser the contact time will be to the wall. Equations (15), (16), and (17) are formalized to determine the impact force with respect to the wall. The following figure (6) helps visualize and come up with an equation that will help determine the force acting on the wall perpendicularly.

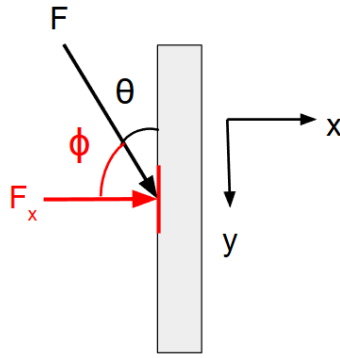


Figure 6: Impact Force Analysis

$$v_n = v \cos(\phi) \quad (15)$$

$$\Delta p = mv \quad (16)$$

$$\Delta p = Ft_c \quad (17)$$

Once the forces have been determined, Hoop's law can be used to find out the force acting on the circumference of the walls of a shell. This is dependent on the number of parameters a problem has; in this case, one too many parameters are missing. This means Hoop's Stress for thin walls is applicable, as it is not required to know both the outer diameter and inner diameter of a cylindrical shell. The parameters required to solve this problem are knowing the average diameter size, which in this case can be any diameter size, and the contact area of the object to the wall. These parameters and calculated findings are applied to the following equations (18) and (19), which will help determine the thickness required to handle the impacts of the object.

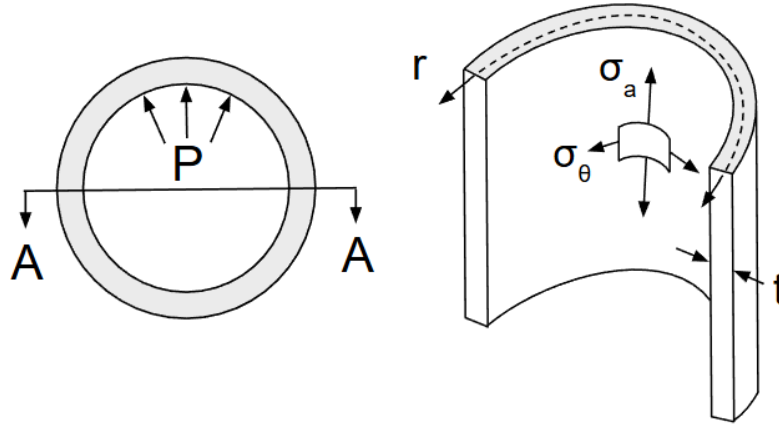


Figure 7: Hoop's Thin Wall Theory

$$P = \frac{F}{A} \quad (18)$$

$$\sigma_\theta = \frac{P * r}{t} \quad (19)$$

Section 2.2: Existing Solution

This project will require literature and state-of-the-art reviews on two difficult problems that this project needs to address: trash catching and autonomous vacuum cleaning. The literature review of the trash catcher will be based on *HTX Studio's* video on their prototype. Autonomous vacuum cleaners are more popular, so the state-of-the-art brand, the Roomba, will be the basis of this literature review.

HTX's Catching Concept

<https://www.youtube.com/watch?v=H0XYANRosVo>

The only documented media of a mobile trash catcher robot seems to be on HTX Studio's auto-aiming trash can [1]. From this four-minute video, the main components that are relevant to this project are the prototype's drive system and camera vision that is able to locate a projectile. This is not an open-source project, meaning that the parts and programming used to create the robot were not documented. This means that the methods and parts of the prototype have to be reverse-engineered or identified visually through the video.

HTX's prototype drive motors

The motor parameters for the robot are an important part of the robot's ability to catch trash, as if it doesn't have the speed or the torque to carry the robot to the same or faster speed of the projectile, it would not be able to catch it. While unsure of the specific motors used in the video, their three motor configurations shown in Figure 8 provided speeds up to 10 mph in their video. The configuration that they used did not seem applicable to this project, as the autonomous vacuuming design requires two motors in line with each other across the center of the base, and this is to produce a 360-degree rotation that is required for the vacuum to efficiently navigate a room. And because of their lightweight design, the motors seemed to be specifically designed to only handle low torque conditions. Therefore, this project aims not to use the exact motors used in this prototype, but a high-torque motor seems to be more applicable due to the added heavy components for the autonomous vacuum system.

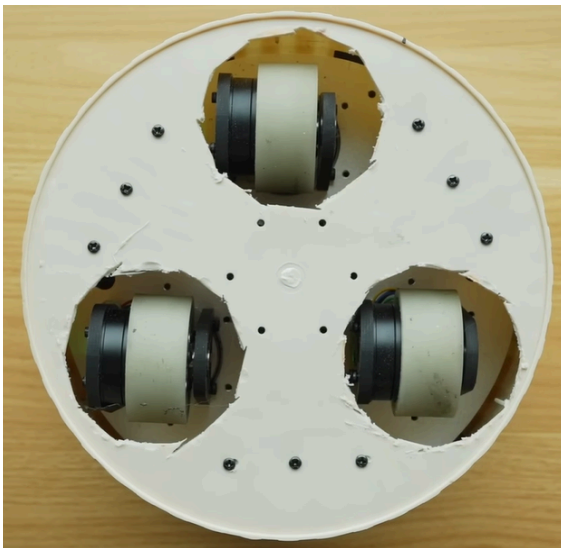


Figure 8: HTX Prototype's Drive Motors



Figure 9: Wireless Camera Module

HTX's vision reaction system

A crucial component for HTX's prototype to have the ability to catch is the vision system. Their prototype uses a wireless camera module attached to the top of its bin to visualize objects that are flying above it, seen in Figure 9. The camera then feeds into a microcontroller that can process the movement of an object tossed over it. The results they achieved with this camera and reaction system are not clear, as there was no discussion of its accuracy. The only statement on this system was that it required lots of coding and fine-tuning. While unsure of the specific camera module used in the prototype, the Trash Catching Vacuuming Robot will use an AI Raspberry Pi camera module for its compatibility with the Raspberry Pi.

In the video, there was also little discussion on the logic behind the catching programming. At first, it was believed that the control system was based on a position error control, where the robot would try to accelerate

faster towards the object the more it deviated from the center of the robot. But if this system were used, it would mean that the robot would always try to accelerate towards the object. This is not the ideal case because in the case of a projectile thrown over the robot, it would first accelerate towards the person throwing the object, then accelerate in the opposite direction, which would be counterproductive. An important detail in the video was that the robot only started accelerating after the projectile was just above the robot.

The codes and standards that apply to this project are not well documented, as the prototype was made under a DIY studio rather than a professional company.

iRobot Robotic Vacuums

Robotic vacuums, primarily that being the Roomba, represent the current state of the art when it comes to consumer floor cleaning technology. Models such as the iRobot Roomba 600 Series typically use 2 independent wheels and 1 caster wheel, use differential drive motion to turn in any direction, have low-profile bodies to be able to vacuum the floor, and multiple sensors, which include bump sensors, ultrasonic sensors, infrared, and so on, to navigate their surroundings [2]. Figure 10 displays the basic anatomy of the Roomba. More advanced cleaning robots use cameras and LiDAR to fully simulate and map their surroundings. The performance of these robotic vacuums has proved themselves to be capable of entering the market. Although these advanced robots are effective and will continue to be for the foreseeable future when it comes to cleaning small debris, they are not designed to handle irregular or large pieces of trash; disposing of them requires the cost of manual labor from the target audience.

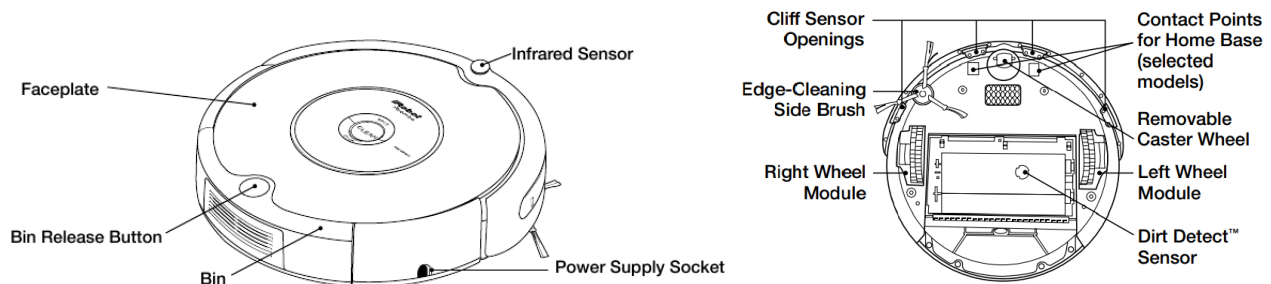


Figure 10: iRobot Anatomy Diagrams

The design will be able to account for those irregular and larger disposables that the underprivileged portion of the target audience, the disabled and elderly, may struggle with.

The codes and standards that apply to the Roomba are summarized below

- Electrical safety - UL, IEC, UN
- Wireless - FCC
- Environment - RoHS, REACH
- Mechanical safety- ISO, UL

In the engineering context, these standards drive the housing thickness, limit motor power, dictate battery enclosure design, and influence sensor placement.

Eufy Robotic Vacuums

A strong competitor of a similar class to this project and iRobot includes Eufy's own robotic vacuums. One of their most popular products is the Eufy Robotic Vacuum E25. This robotic vacuum is radically more advanced than the aforementioned examples, distinguishing the product from other competitors with a HydroJet System for mopping capabilities, a CornerRover arm to sweep corners, and the usage of the Eufy app to control its functions. Please refer to Figures 11-13 down below for a more visual configuration of components from the official user manual. These advanced systems allow the robot to adapt to various situational indoor environments. Although this product goes beyond the scope of the project, the Eufy Robotic Vacuum E25 serves as the ceiling for the project to aim and the basis of the concept of applying multiple functions and a human input system into a robotic vacuum.

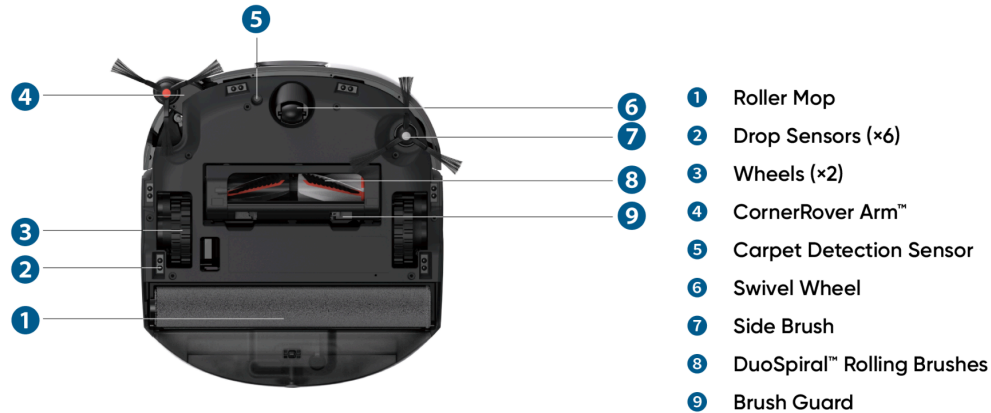


Figure 11: Eufy Bottom Section Itemized Labeling

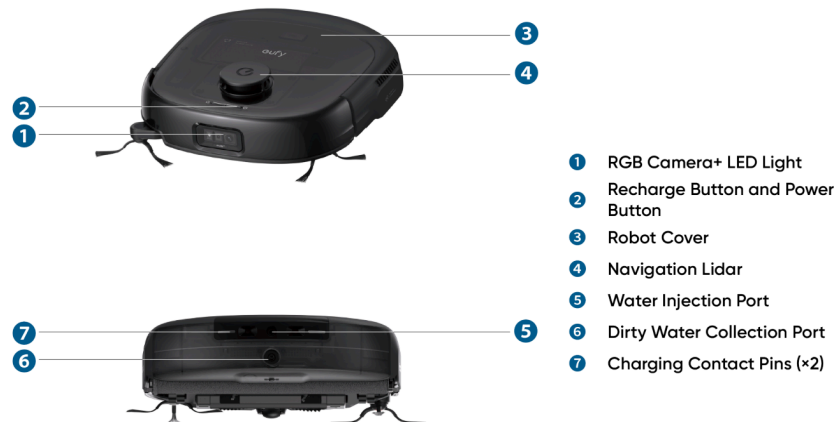


Figure 12: Eufy Itemized Labeling



Figure 13: Eufy Dust Bin Itemized Labeling

Eufy's Human Input Control System

The Eufy app allows the users to issue numerous commands to the robotic vacuum, including redocking the robot back to the charging port, beginning cleaning processes, and switching between different cleaning practices. Like most external mobile apps for high end products, the Eufy app controls the robot via the Wifi and IoT (Internet of Things) allowing information and commands to be shared between the user app and the robot's internal system.

The codes and standards for each aspect of the product have not been documented with the exception of the usage of IoT. According to the Eufy Official Website, these are the codes and standards, focusing on cybersecurity and user privacy, of the all Eufy products listed below. These codes and standards originated from both the U.S. and Europe, to account for the national legality of their product. Considering the smaller scope, this project will use a remote and receiver to prevent violations.

- ISO/IEC 27001:2013
- ISO/IEC 27701:2019
- ETSI EN 303 645
- NISTIR 8425 IoT Security Evaluation
- PSTI Compliance Attestation
- CSA STAR
- ioXt

Section 2.3: Technical Challenges

A technical challenge of this project is reducing the latency between the sensors and actuators to allow the robot to successfully intercept projectiles. To achieve a capture success rate of 75-80% for the trash-catching mode, the Raspberry Pi AI camera must determine the velocity and trajectory based on a live recording of an overhead trash projectile, sending the data to the microcontroller to drive the motors to the corresponding position. Because of this latency, the robot's feature of catching a projectile mid-air has been restrained to only follow a trash object that is manually held above the overhead camera.

In addition, the project faces substantial structural and power management constraints. The 3D-printed PLA base structure must be thin enough to meet weight limits but robust enough to withstand the impact force of a 1 lb object hitting the bin without structural failure. Based on Finite Element Analysis (FEA), the bin requires a precise wall thickness of 0.078 inches to maintain a factor of safety of roughly 3.775. The multifunctional system requires a high power input, which can be fulfilled by the 6S LiPo battery. However, the high power output raises concerns about cable management and precise circuit design to prevent fire hazards or device failure.

Chapter 3: Design Requirements and Concept Development

Section 3.1: Design Requirements

The primary objective of this project is to develop a multifunctional robot capable of transitioning between two states, an autonomous vacuuming mode and a real-time "aim-assist" trash-catching mode. In its vacuuming mode, the robot must be able to navigate indoor household environments and achieve 360° stationary rotation for high maneuverability in tight spaces. The suction system is required to effectively collect common

household debris, such as dust and small waste particles, depositing them into a removable dust bin. To ensure practicality for the user, the robot must support at least 60 minutes of continuous operation on a single charge and allow for seamless mode-switching via remote controller. In its trash-catching mode, the robot's performance requires high-speed perception and mechanical responsiveness to reproduce the following methods of the system: the system must utilize an onboard AI camera to detect projectiles thrown from a distance and move to an interception point with enough speed to catch objects traveling up to 8 mph. Beyond the mechatronic aspects of the design, the collection bin must be able to withstand moderate projectile impacts from trash without deformation or failure while maintaining a reasonable weight and size for practicality.

Section 3.2: Concept Development and Selections

Prime Design Concept

The primary base for this concept was a robotic vacuum with a trash bin attached firmly on top. A bumper and two wheels would be used to navigate and turn in its environment, and the AI camera would be placed near the top of the bin to detect projectile trash. Theoretically, the images fed into the camera would alert the product to move in accordance with the position of the trash to be able to catch the trash as it lands. Figure 14 displays the primary design concept based solely on the initial objective of trash catching and automotive vacuuming.

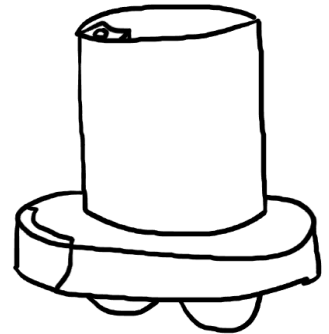


Figure 14. Base Design

Alternate Concepts Considered

Each group member provided several design concepts, attempting to improve on the prime design concept with three or more changes based on the current goals, with more in-depth design consciousness.

Displayed in Figure 15, three concepts were assigned to the same design concept based on the primary design, with minor modifications focusing on user input and navigation. This served as an additional base to some of the group members, as this was the first submission of changes to the primary design before the assigned meeting for concept selection. Three buttons were added to turn on the power and change modes between tracking and cleaning. Lidar is attached to the edge of the bin for sensing the surroundings and mapping. 3 wheels for better balance. Emulating an actual robotic vacuum, the vacuum is placed closer towards the front for further cleanup reach, while the battery and dust bin are located in the back. Alternative concepts for user input and vacuum brushes were built with a USB microphone for voice command inputs rather than buttons, and having five protruding bristles rather than three for more effective area cleanup. In addition, the robot and bin could have been two separate parts. The bin would lock into the robot using an L-shaped twist locking mechanism, which would mean the Raspberry Pi board would have to be implemented at the center of the robot for seamless connection and disconnection of the camera and lidar.

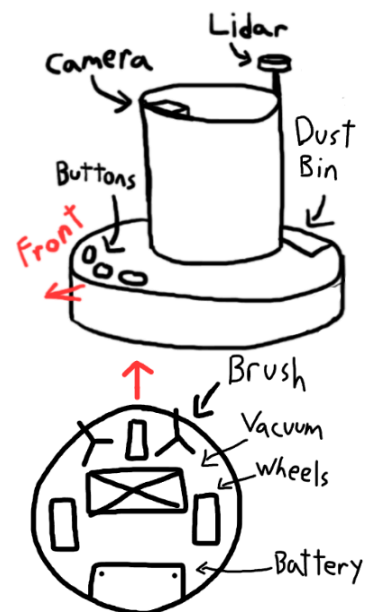


Figure 15. Design Concept 1

Another member provided their insights on improvements to the primary design shown in Figure 16, focusing on sensors. The 3 buttons would be relocated to the side of the robot to introduce a Depth Camera on top of the front side of the robot, to detect the user and complete the aforementioned trash-catching and following functions. The bumper would be replaced by 2 Ultrasonic Sensors placed in front of the robot to prevent collisions. A mechanism would be added to vacuum-seal the trash bag along the inner walls of the bin to expand the bin size.

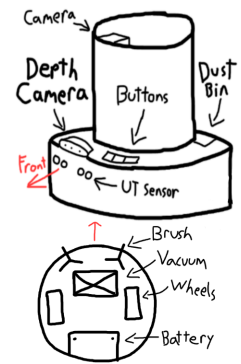


Figure 16. Design Concept 3

Figure 17 built upon and rearranged some components in Figure 15 to focus on repositioning and mobility. There would be two motor wheels and two caster wheels to allow 360° rotation and movement stability, respectively. The same Lidar from Figure 15 would be relocated to the top front side of the robot rather than the bin, for better aesthetics. No additional modifications to the bin were made to provide room for a disposable trash bag. An additional suggestion from the author of this design was to scrap the lidar due to costs and incorporate and modify an existing Roomba's software and design to fit the design goals.

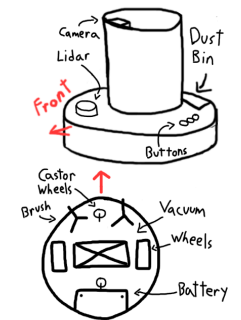


Figure 17. Design Concept 3

Figure 18 was built upon in Figure 15 with various modifications to focus on incorporating a trash bag and user input. The on and off buttons would be replaced with a switch, and the button to switch modes would be a single toggle button. There would also be a sharp cap that could cut a hole through the trash bag to allow the camera to capture its surroundings without obstruction. Alternative designs included combining the trash and dust bins to have a shared storage with a mechanism to open the trash bag through the connection between the two bins, as well as a reposition of the camera on a motorized lid to prevent camera obstruction and exposure of trash odors to surrounding inhabitants.

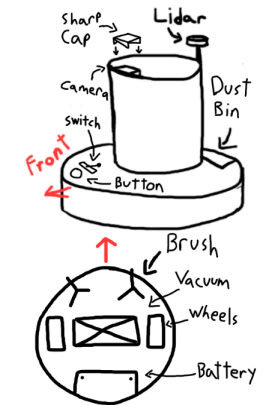


Figure 18. Design Concept 4

Figure 19 displays a design more focused on all-terrain mobility, which is specified as being able to move on land, carpet, or hardwood. This design would use rubber tracks to more effectively climb over rugs or uneven spots on the floor. This would also come with ultrasonic sensors to avoid crashing and an established moving pattern based on said sensors. More innovative was the reshaping of the robotic vacuum portion into a triangular frame, allowing the robot to clean along corners and edges using brushes to push debris towards the vacuum.

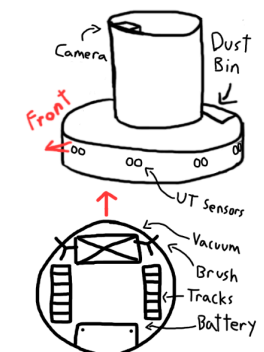
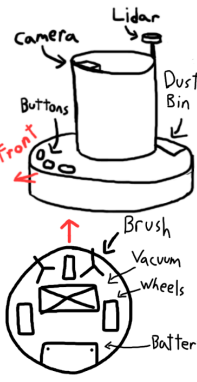
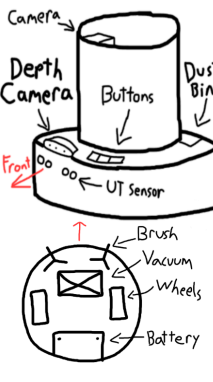
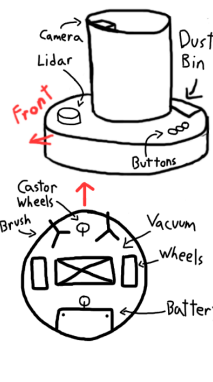
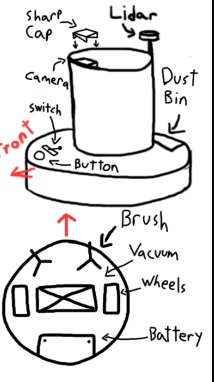
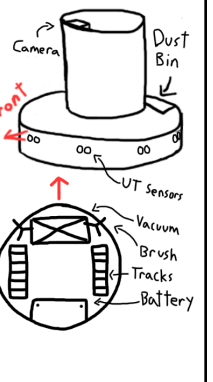


Figure 19. Design Concept 5

Concept Selection

The Pugh Chart uses the correlation of each concept designed by each member to each parameter to judge how effective each design is in meeting the requirements, where -1 is a negative correlation, S is zero correlation, and +1 is a positive correlation. Each score was decided through popular vote.

Table 3: Pugh Chart for Concept Generation

Team Members	Patrick Wong	Gustavo Ornelas	Branden Tran	Kristopher Do	Alejandro Cedeno
Project Idea					
Cost	S	+1	-1	S	+1
Visibility Range	S	-1	-1	S	-1
Cleaning Effectiveness	S	S	S	S	-1
User Friendly	S	S	S	-1	-1
Stability	S	-1	+1	-1	+1
Speed	S	-1	S	-1	-1
Aesthetics	S	+1	+1	S	+1
Software Complexity	S	S	S	S	+1
Development Risk	S	+1	-1	-1	+1
Σ^+	0	+3	+2	0	+4
Σ^-	0	-3	-3	-4	-4
Σ	0	0	-1	-4	0

After evaluating each concept idea based on category fulfillment, with the first concept serving as the baseline for comparison, the team narrowed the options down to concepts 1, 2, and 5. Rather than conducting another

round of voting, it was agreed to move forward with the idea that demonstrated greater familiarity and presence in the market. The final selection was concept 2, originally proposed by teammate Gustavo Ornelas. This concept features a robot chassis as the foundation, equipped with a custom bin mounted on top. The design incorporates two wheels powered by DC motors for mobility, along with two smaller motors driving brushes for sweeping functionality. A built-in vacuum unit provides additional cleaning capability, supported by a dedicated battery section for power management. To enhance navigation and safety, the system integrates two ultrasonic sensors for collision prevention, as well as a depth camera for user tracking and forward mapping. User interaction is managed through three buttons that control power and operational modes. Additionally, a camera positioned near the top of the bin enables object tracking, allowing the robot to detect and catch items thrown toward it. Although this was the finalized concept idea chosen, further improvement in the theoreticals and analysis throughout this and the following semester may warrant future changes.

Section 3.3: Detailed Design and Analysis

Detailed Design

The following images under this category represent the alterations made between 2 revisions made before and after the semester of ME195A. This is to show what changes have been made to the project in terms of Cad model during those times. Refer to Appendix C for detailed engineering drawings on the project. Analysis done on the robot was ran through SOLIDWORKS simulation with an 18 lbf at the epicenter of the robot, this is to mimic a loading of wasted material thrown and held onto by the robot. Refer to Appendix B for further details of analysis.

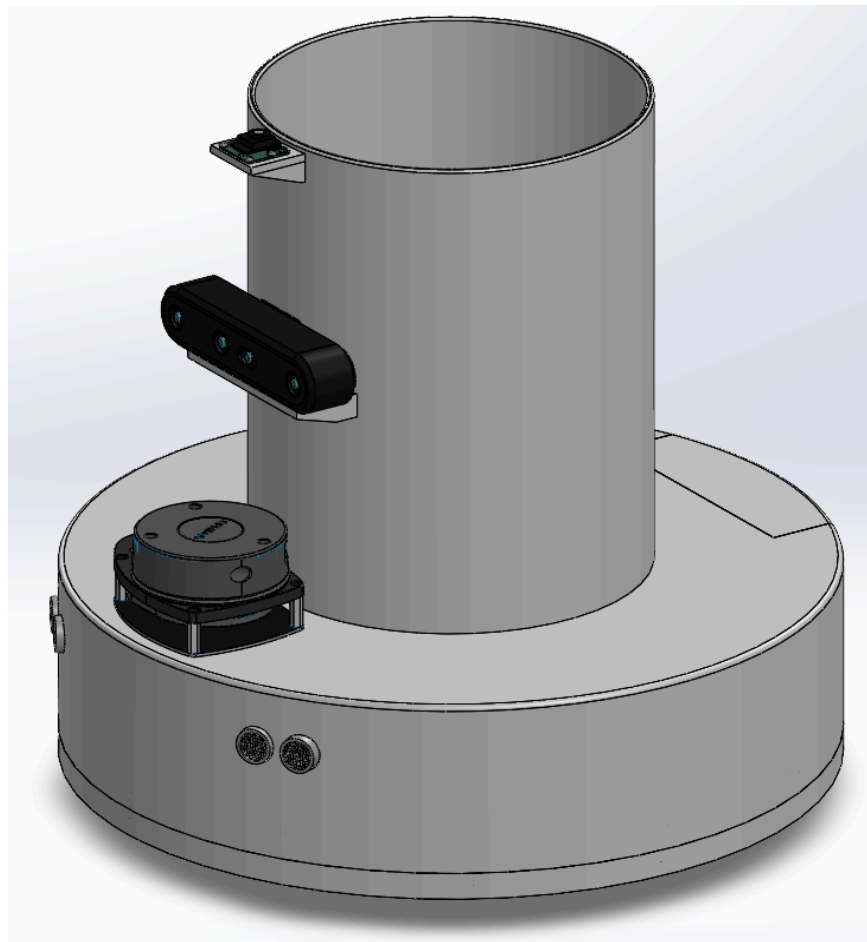


Figure 20: Assembled Cad Model Rev 1

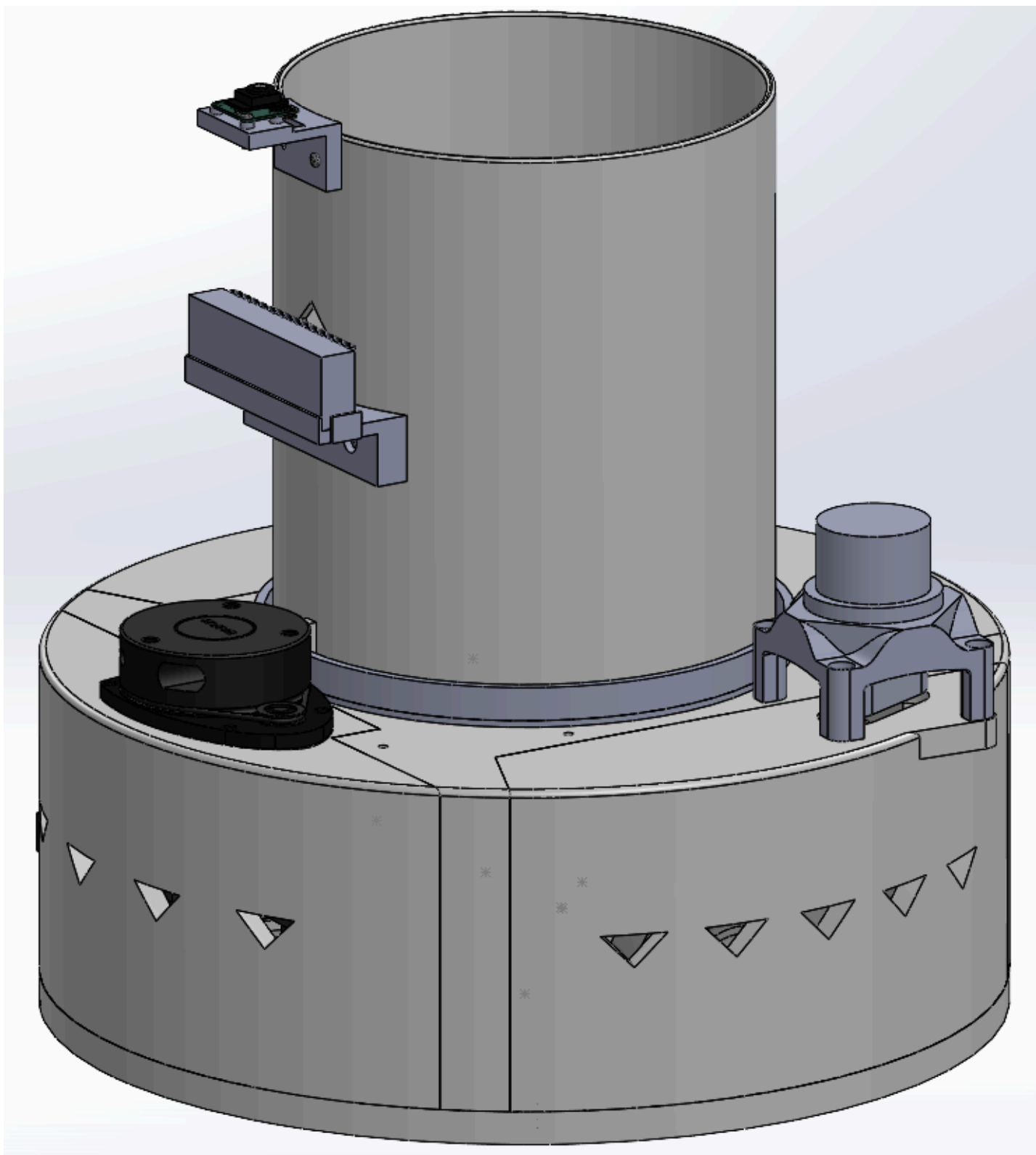


Figure 21: Assembled Cad Model Rev 2

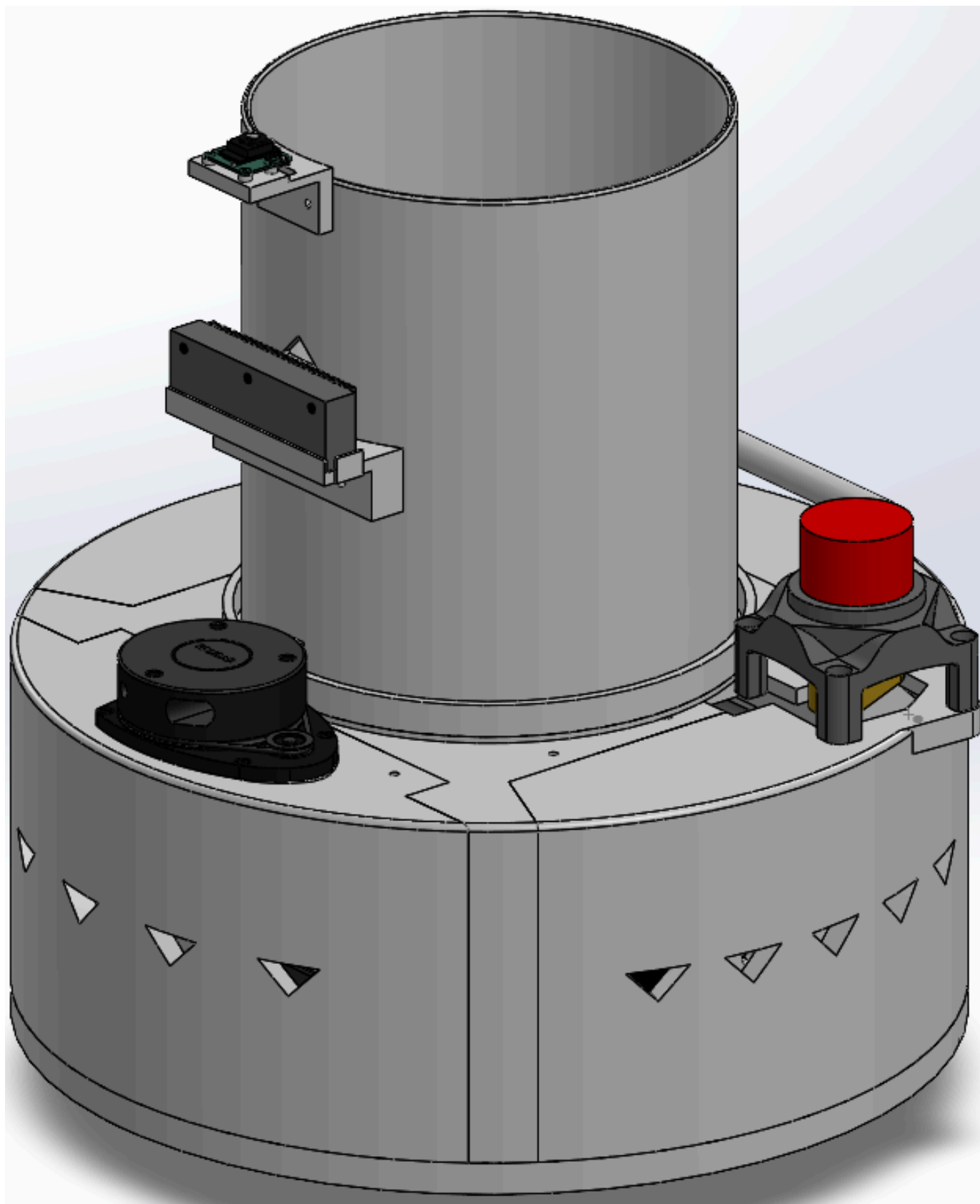


Figure 22: Assembled Cad Model Rev 3

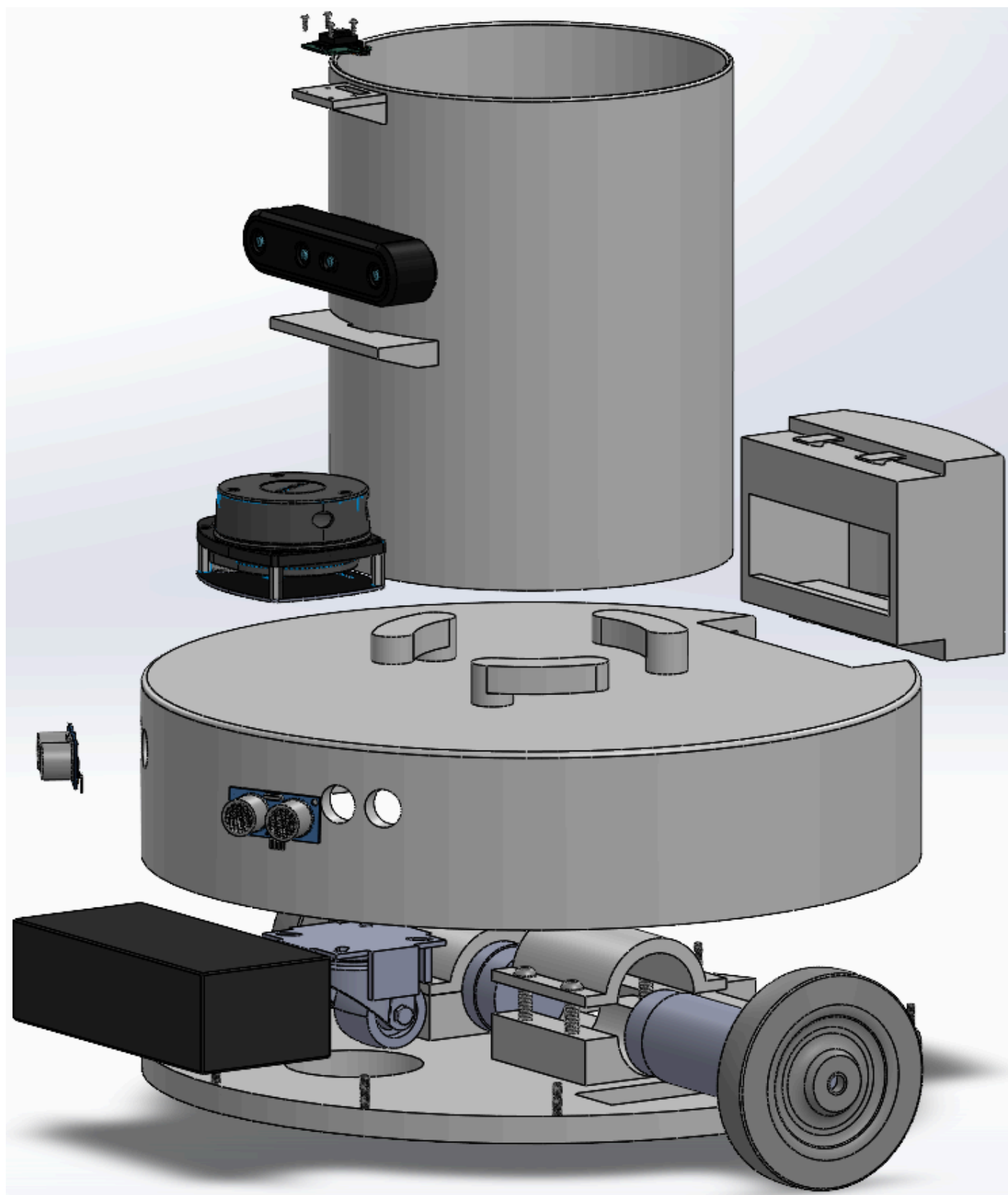


Figure 23: Exploded View CAD Model Rev 1

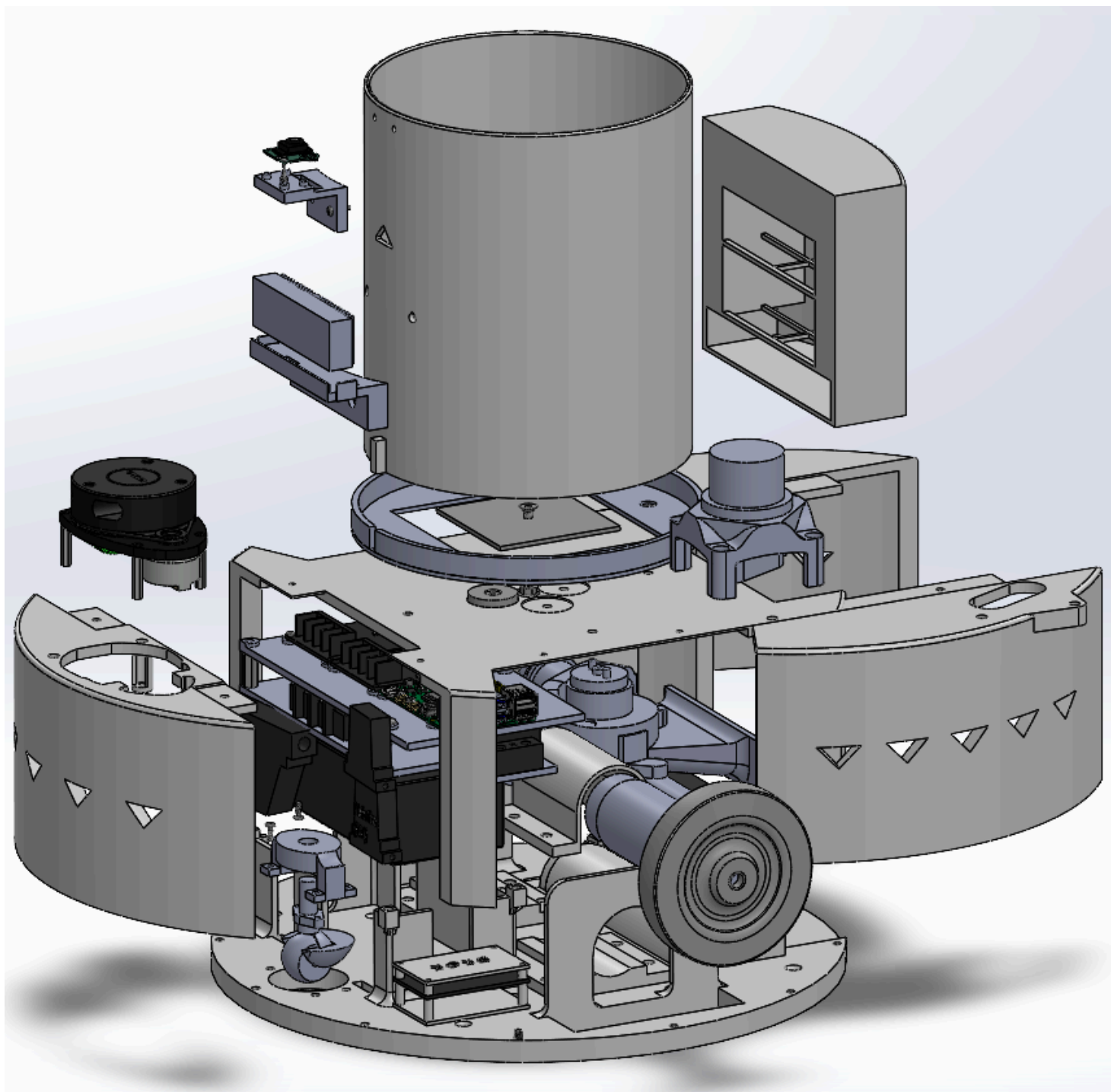


Figure 24: Exploded View CAD Model Rev 2

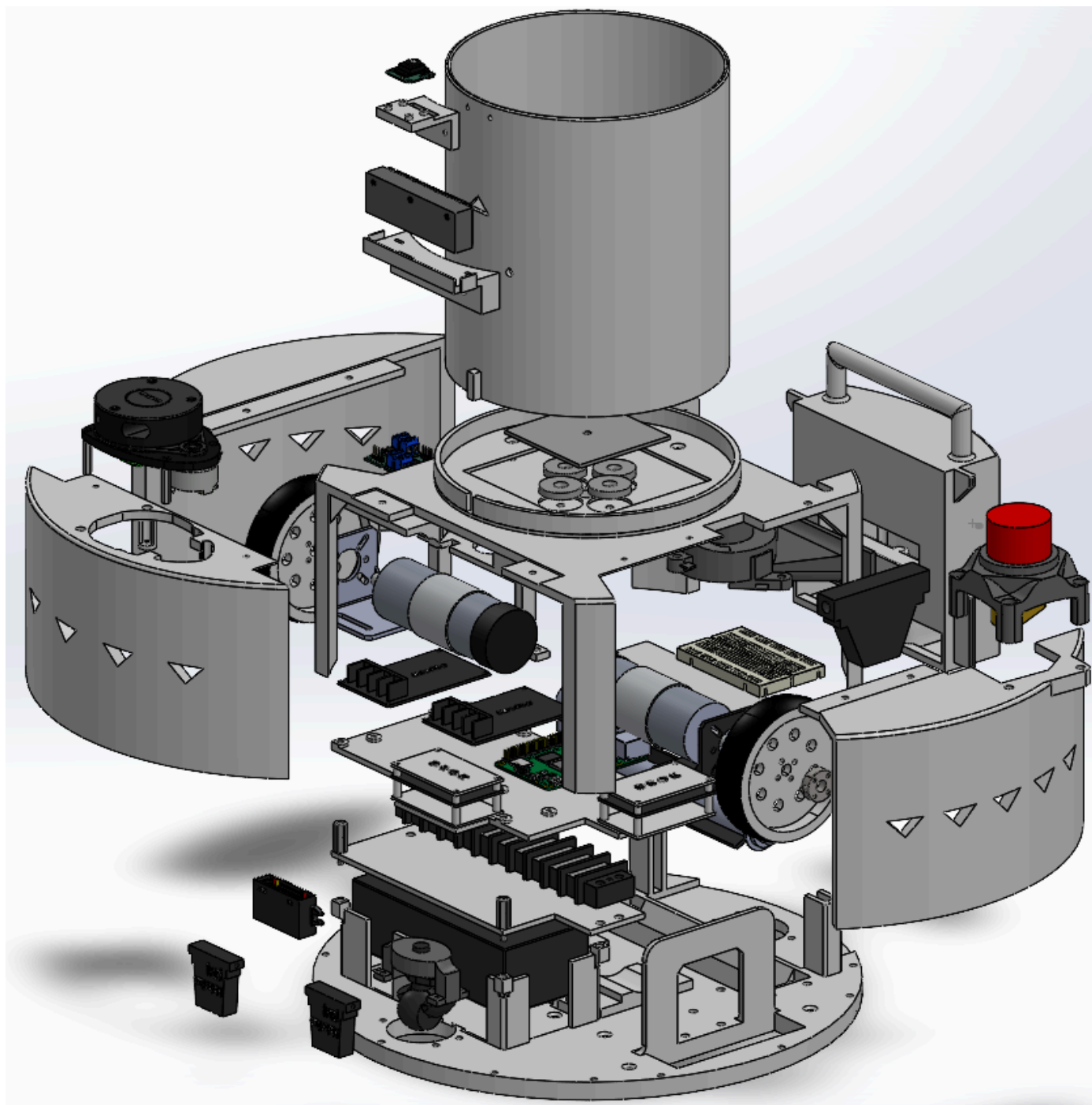


Figure 25: Exploded View CAD Model Rev 3

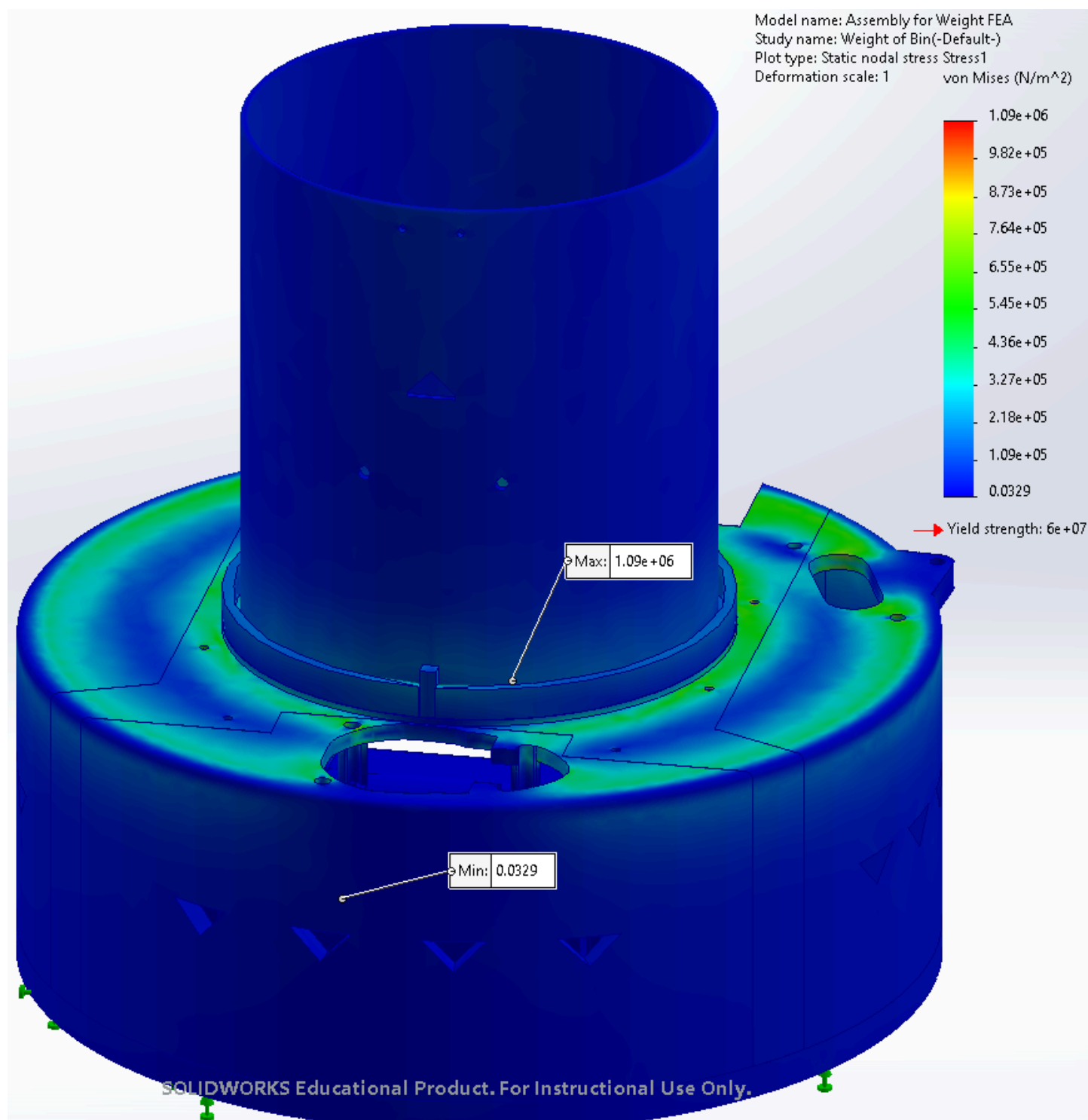


Figure 26: Centered 18 lbf Stress Analysis

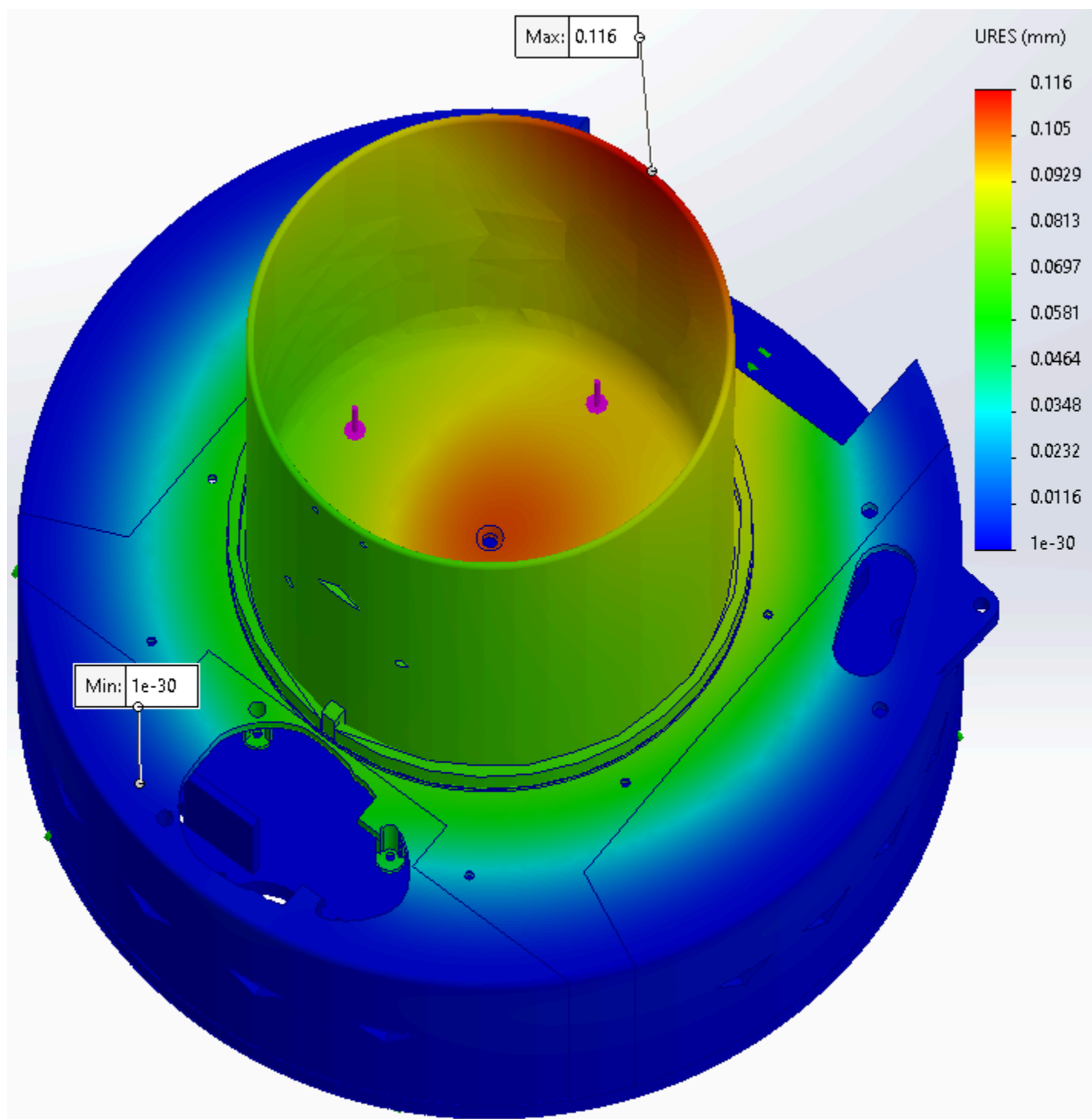


Figure 27: Centered 18 lbf Displacement Analysis

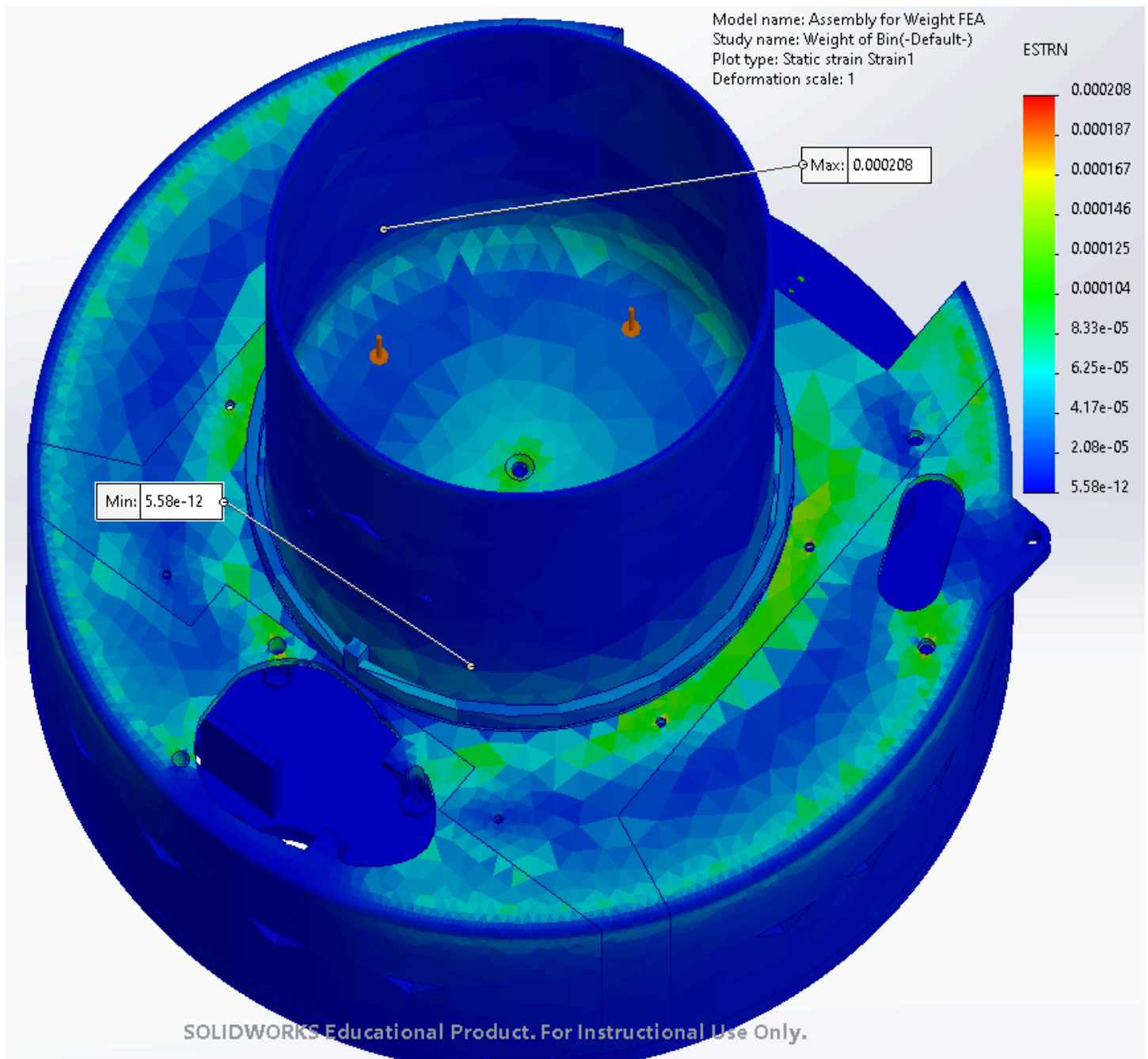


Figure 28: Centered 18 lbf Strain Analysis

Speed Requirements and Motor Selection

The Trash Catching Vacuum Robot is designed to catch an object thrown at 8 mph (3.58 m/s) from a height of 1 m and an angle of 60° from the floor. The robot is assumed to be at full payload capacity with a total weight of 18 lbs (8.16 kg), the height it needs to intercept the object at is 0.5 m, and it is initially situated 0.5 m away from the thrower.

Using (1), the parametric equations for the trajectory of the projectile can be put into a plot, Figure 29 & 30.

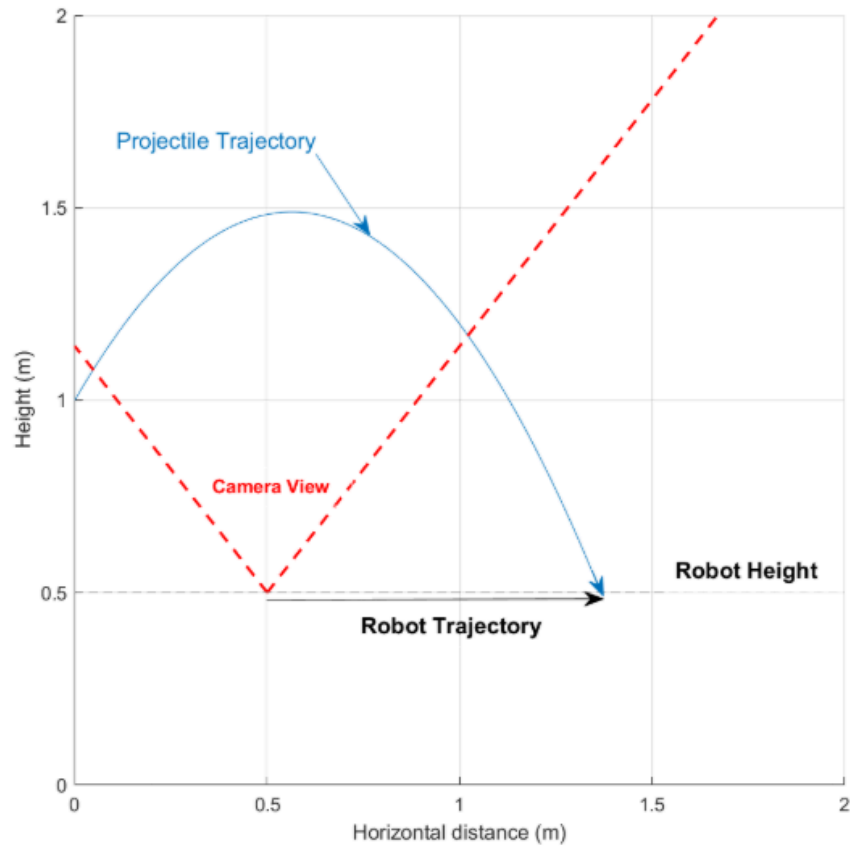


Figure 29: Plot of Projectile and Robot Trajectory

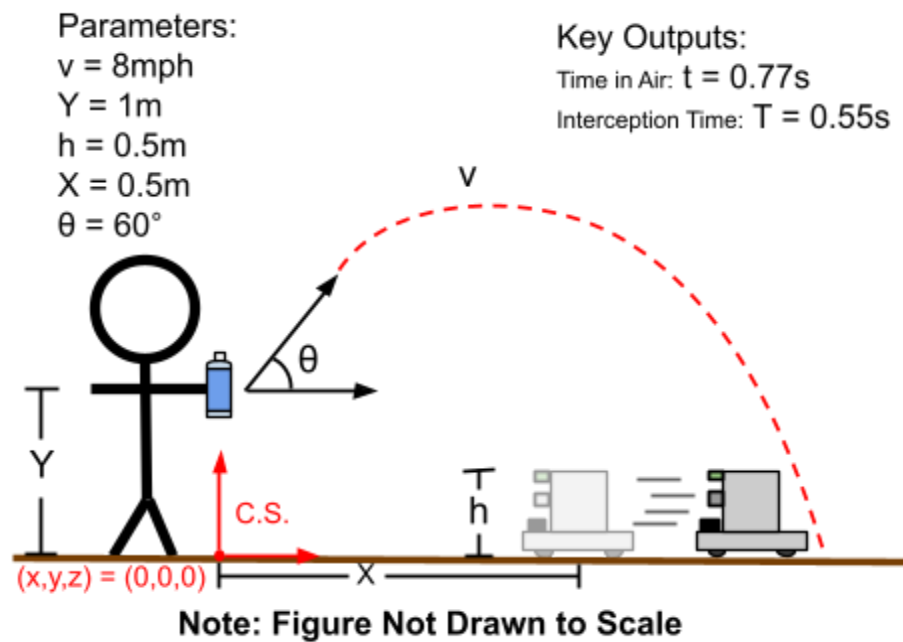


Figure 30: Visual Projectile Motion in Relation to Coordinate System

Expecting that the robot intercepts the object at a height of 0.5m, the horizontal distance that the projectile travels is 1.37m in 0.77 seconds. The robot is expected to start accelerating when the projectile is just over the

center of the robot. To find the time that the object is over the robot, 0.5m is set to equal the parametric equation of x , and isolating for the time variable, it is found that $t = 0.28\text{s}$.

So the distance that the robot needs to travel is $1.37\text{m} - 0.5\text{m}$, or 0.87m , and the time it needs to reach that distance is $0.77\text{s} - 0.28\text{s}$, or 0.49s . Using (1), the acceleration required for this motion is 3.55 m/s^2 .

With (2), the top speed of the robot will be 1.74 m/s , or 3.89 mph .

Motor Requirements

Looking at Figure 4 (FBD of a driven wheel), the frictional force that propels the robot forward using (3) is 29.0 N .

This frictional force applies a torque on two motors, so each motor is actually feeling 14.5 N of frictional force on the 4-inch (0.10 m) diameter wheels. Using (4), the force can be translated as torque and is found to be 0.725 Nm , or 7.39 kg-cm .

Knowing that the top speed of the robot is 1.74 m/s , this linear speed can be translated into angular velocity using (5). The motor will need to have a rated speed of 34.8 rad/s or 332 RPM .

Now that the minimum motor parameters are found, the motor needs to be oversized to account for discrepancies that make the calculations less accurate. The motor datasheet in the appendix states that the selected motor for this project can handle a torque of 15 kg-cm at 820 RPM .

Electrical Power System Design and Verification

The robot is expected to operate with a maximum system voltage of 24 V and 5 V , a combined stall current of 49 A , and a continuous current of 26 A from the two wheel motors. In addition, the vacuum motor requires 12 V and 5 A , while the remaining system devices, including cameras, sensors, and control electronics, draw a total current of 3.9 A as calculated using equation (6). A 22.2 V (6S) LiPo battery is used as the primary power source. With these known electrical requirements established, the remaining unknown parameters can be determined to define an appropriate electrical power system design and to verify that all selected components can safely support the motors and system devices.

Using equation (7), the summed currents from the known electrical requirements yield a total battery current of 60 A . This value represents the maximum current drawn from the battery under worst-case conditions, as it accounts for the combined stall currents of the system. In practice, this current draw will decrease after a short duration, since stall current occurs only briefly, typically for approximately two seconds before the motors return to their continuous (rated) operating current.

Knowing that the system devices operate at 5 V , the minimum required power for the buck converter can be determined using equation (8). With a total device current of 3.9 A and an operating voltage of 5 V , the calculated power requirement is 20 W . This value is then compared to the selected buck converter, which is rated for 60 W ; since 60 W exceeds the required 20 W , the buck converter is deemed suitable for the electrical system. However, the buck converter draws input current from the battery, which must also be accounted for. Using equation (8) with a power of 20 W and an operating voltage of 5 V , the calculated converter input current

is 0.99 A, which is conservatively adjusted to 1.5 A to account for efficiency losses. With this additional current included, equation (7) is used again, resulting in a total battery current of 60 A. This indicates that the battery must be capable of supplying at least 60 A at 22.2 V. Battery verification is then performed using equation (9); given a battery rating of 22.2 V, 10 200 mAh, and 120 C, the maximum continuous current capability is calculated to be 1224 A, which is significantly greater than the required 60 A, confirming the battery is suitable for the robot. Finally, battery performance is evaluated to determine how much of its capability is utilized, since lower utilization reduces stress and improves efficiency. Using equation (10), with a total current draw of 60 A and a battery capacity of 1224 A, the resulting load is 6.5%. This low load percentage indicates minimal voltage sag, cooler operation, and the ability to maintain near-rated motor torque and rotational speed during operation.

One of the most critical components for reliable robot operation is the power distribution hub, terminal block, and kill switch, which must safely handle the system voltage and current supplied by the battery. The selected power distribution hub, terminal block, and killswitch are required to support 22.2 V (preferably 24 V) and the maximum battery current of 60 A as previously calculated. Using equation (11) and the manufacturer's datasheet, the power distribution hub is verified to be rated for 600 V and 75 A. The terminal block is also 600 V and 75 A. And the killswitch is 48 V and 275 A. Since both the voltage and current ratings exceed the system requirements, the selected power distribution hub, terminal block, and kill switch are appropriate for the electrical system. Using equation (12) for the selection of these three parts, it will also be greater than the continuous current since it was already greater than the stall current. In addition, fuses are incorporated as a safety measure to protect critical components of the electrical system. In this design, fuses are placed at the battery output and at the motor drivers connected to the wheel motors. Proper fuse selection is performed using equation (13), which applies the continuous current rather than the stall current, as slow-blow fuses allow brief overcurrent conditions to accommodate motor stall events. Using a battery with a continuous current of 32.5 A, the calculated fuse rating is 40.63 A before blowing. The same equation is applied to the motor driver fuses using their continuous current rating of 13 A, resulting in a fuse rating of 16.25 A. It is important to note that fast-acting fuses would require higher current ratings, as they trip immediately once the rated current is exceeded. Additionally, the selected motor drivers must be capable of handling both the continuous and stall currents of the motors to ensure safe and reliable operation.

Impact Force on Bin's Wall and Thickness Required

Based on the initial parameters set to find the speed requirements and motor selection of the robot. It will be applied here in determining the impact force and thickness required when it comes to designing the custom bin. The allowable stress must be determined first by using the researched tensile stress of 50 MPa with a safety factor of 4. It is 4 because that is the engineering number for continuously used products to ensure the longevity of an item or object. Applying equation (14) will give a result of 12.5 MPa for the allowable stress. This will ensure that any calculations made after will meet the safety requirements of 4, meaning the design will be made strong enough to withstand expected forces.

Now going back to the set parameters from the speed requirements and motor selection, the person's tossing speed is thrown at an angle of 60° , which impacts the wall of the bin at a 22.1° angle to the wall. The object thrown for this scenario will be an apple that weighs 200g. Knowing the rigidity of an apple, the parameter set for the contact time of an apple to the wall was set to be 10ms. Using equation (15) helps get the resulting velocity x coordinate system towards the bin's wall, the result being 1.35 m/s. The velocity y coordinate system parallel to the bin's wall ended up being 3.32 m/s. This is later then implemented to equation (16) to determine

the momentum in x of the object, result was 0.27 kg(m/s) and momentum in y is 0.664 kg(m/s) This was then incorporated into equation (17) to determine the impact force applied perpendicular to the wall and when calculated, gave an impact force x of 27 N and an impact force y of 66.4 N.

Hoop's stress application is used to determine the thickness required, specifically the thin-wall theorem. Knowing the diameter of the bin is 7.25 in, and the general contact area of the object striking the walls of the bin is 1 cm². Equation (18) can be applied to find the localized pressure applied to the contact area, which just so happens to be 270 kPa for that area of contact. This is then applied to the Hoop's equation (19) to find the required thickness to handle the pressure. That thickness is 0.078 inches according to the calculations. Running the forces applied to the thickness under the impact forces calculated will result in the FOS being 3.775.

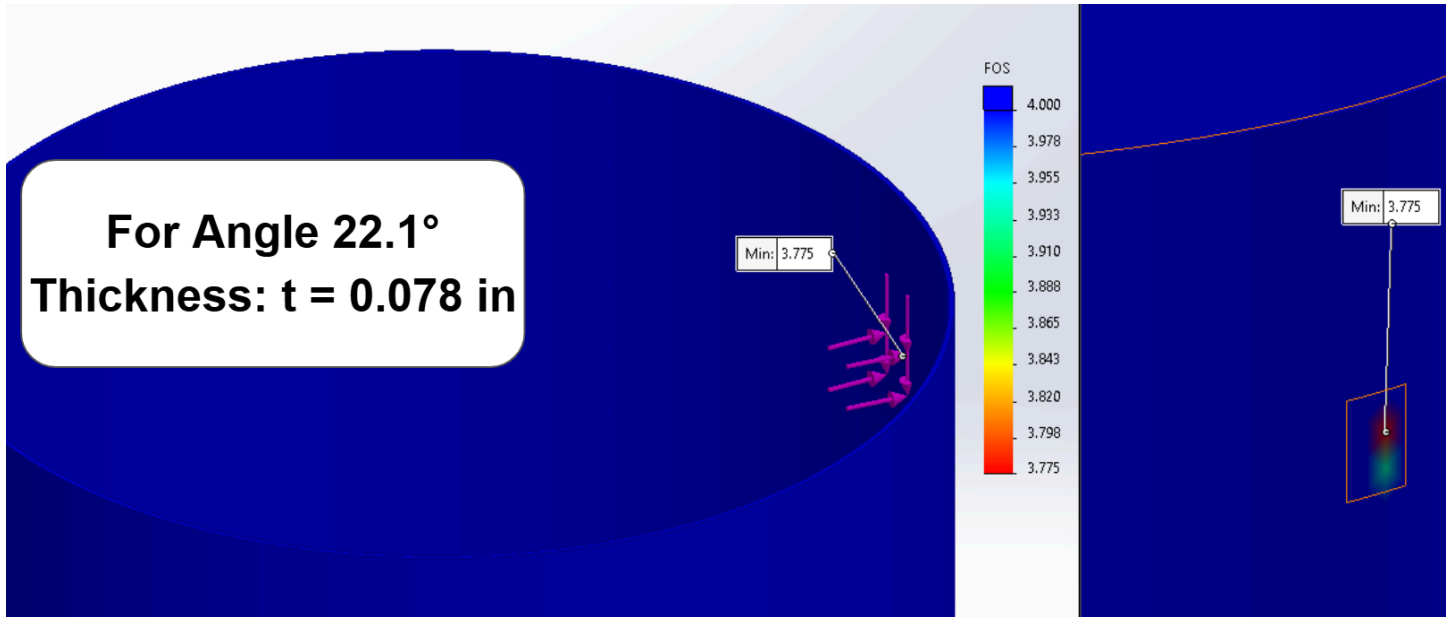


Figure 31: FEA Analysis on Bin for Angle at 22.1°

Section 3.4: Final Design relation to Engineering Codes & Standards

As the project shifts from design to building and wiring for the robot, Engineering codes and standards will be referenced and applied to ensure that each system is considered safe and reliable, and that the systems meet professional engineering expectations. In this project, codes will be viewed as mandatory safety regulations that must be followed, and standards will be regarded as the technical blueprint on how to design, fabricate, and document the robot appropriately. Codes and standards will assist in the development of the wiring layout, selecting components, and building the overall mechanical system.

For the electrical system, the project will follow the National Fire Protection Association's National Electrical (NFPA NEC) Code, which shows safe practices for wiring, grounding, and circuit protection. This will help determine the correct wire size, insulation, and ideal fuses for the robot's components, such as the 24-volt DC motors, power distribution hub, buck converters, and power supply that will be used to charge the 6S Li-Po battery. Moreover, safety practices enforced by the Occupational Safety and Health Administration (OSHA) for electrical safety in the workplace and the Electrical Equipment Maintenance Code (EEMC) will also be applied to make sure that all electrical work is performed safely and that the robot remains functional for a long time.

These will help prevent potential hazards such as short circuits, overheating, and improper grounding, especially when handling the robot's high-current battery system.

For the mechanical part of the project, the American Society of Mechanical Engineers (ASME) and the American Society for Testing and Materials (ASTM) standards will be used for geometric dimensions and tolerancing in the computer-aided design model. This will ensure that parts fit correctly and align during assembly. ASME standards will also help with choosing specific bolts, screws, and hardware to mount the motors, brackets, and panels. The ASTM material standards will be used for selecting and testing the plastic components. This will help confirm that the frame and enclosures meet the necessary strength and durability requirements. Furthermore, the International Organization for Standardization (ISO) standards will be used for testing tolerances for manufacturing and planning for mechanical safety; specifically identifying and minimizing risks, such as pinch points, sharp edges, and areas with rotating parts, such as motors and vacuum brushes. Recommendations from Underwriters Laboratories (UL) and the American National Standards Institute (ANSI) will be used for control components, including relays, switches, and plastic enclosures. This will ensure flame resistance and adequate working electrical spacing. Even though formal UL certification is not needed, the associated principles of safety will be applied to enhance the quality and protection of the wiring system. Regarding the wiring system with moving elements, ANSI and the Robotic Industries Association (RIA) will be consulted, particularly for safety integration, including a main power switch, emergency shutdown, and protection of the moving elements. Finally, the drawings will comply with the Institute of Electrical and Electronics Engineers (IEEE) standards to maintain the use of consistent electrical symbols and documentation quality.

By applying these codes and standards throughout the design and construction process, the robot's electrical and mechanical systems will be built safely, efficiently, and aligned with real-world engineering practices of codes and standards. This approach will help ensure that the system of the robot, including its 24-volt motors, LiDAR, Raspberry Pi 5, and AI camera, operates reliably while minimizing electrical and mechanical risks.

Chapter 4: Circuitry, Microcontroller, and System Interface

Section 4.1: Circuitry

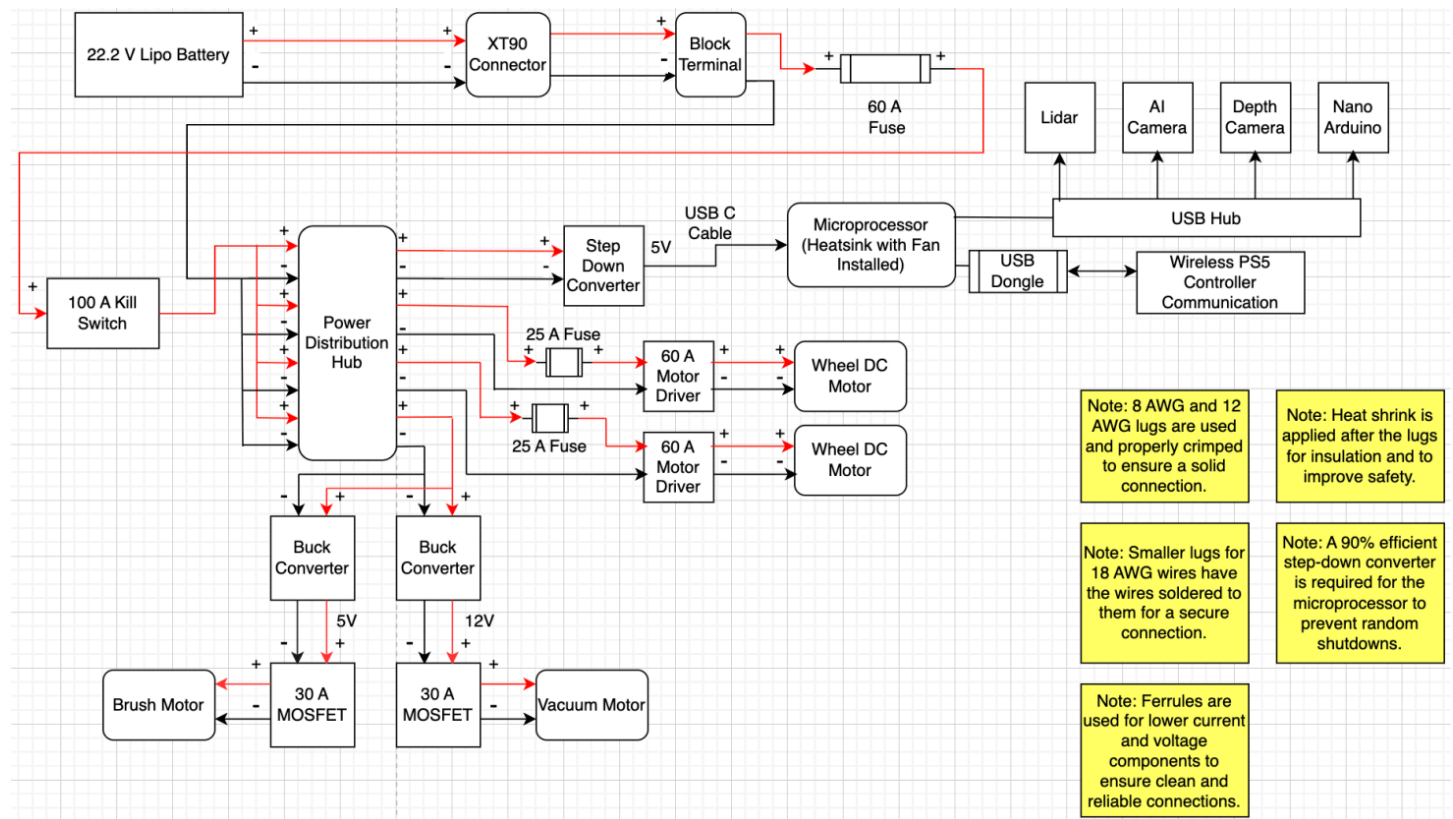


Figure 32: Electrical Circuit Layout

1. Primary Power Source and Protection:

As shown in Figure 32, a 22.2 V LiPo battery serves as the primary power source for the robot. The battery is connected through an XT90 connector, which was selected due to the high current expected from the system. The battery wiring uses 8 AWG cable, which is more compatible with an XT90 connector than an XT60 connector. Using an XT60 connector with this wire gauge and current level could result in poor electrical contact or overheating. The output wires from the XT90 connector are soldered to the connector terminals to ensure a secure and reliable connection. A two-pole terminal block is then used to connect the XT90 output wires to the rest of the circuit. This simplifies assembly and allows the fuse to be installed without permanently soldering all connections. A 60 A fuse is used in the main power line because the estimated maximum current draw of the robot is approximately 55 A under worst-case conditions, such as motor stall. The fuse protects the system by interrupting the positive power path, while the ground connection remains uninterrupted. To safely shut down the robot, a 100 A kill switch is included in the circuit. This allows the entire system to be powered off without disconnecting the battery directly, preventing improper shutdown of the Raspberry Pi 5.

2. Power Distribution System

After the main protection components, power is routed to a power distribution hub, which distributes voltage to the rest of the robot's electrical components. The hub allows multiple subsystems to be powered from the same battery source while keeping the wiring organized. All connections are arranged in parallel, ensuring that each subsystem receives the same supply voltage. Ground connections follow the same configuration. Because the battery voltage is 22.2 V, certain components require buck converters to step the voltage down to appropriate operating levels. These converters provide regulated voltages for sensitive electronics and other subsystems.

3. Motor driver and Actuation System

The robot uses two dc wheel motors for the motion. Each motor is controlled by a 60 A motor driver, which is capable of handling the peak current requirements of the motors. The motor drivers have a continuous current rating of 20, which is sufficient because the motors themselves have a continuous current rating of 13 A. For protection, 25 A fuses are installed before each motor driver. The DC motors have a stall current of approximately 24.5 A, so the fuse rating is slightly above this value to prevent unnecessary tripping while still protecting the system during stall conditions. The vacuum motor operates with a lower-rated motor driver. Since the system voltage is 22.2 V, a buck converter reduces the voltage to 9 V to match the operating requirements of the vacuum motor driver.

4. Control System and Sensors

The Raspberry Pi 5 acts as the main controller (brain) of the robot, processing sensor data and controlling the system. Because the Raspberry Pi requires 5 V, a buck converter steps the battery voltage down from 22.2 V to 5 V. Power is delivered through a USB-C cable, which allows the Raspberry Pi to be powered safely through its standard power port. Several sensors are connected to the Raspberry Pi, including a LiDAR sensor, depth camera, and AI camera. To reduce the current load on the Raspberry Pi's USB ports, a USB hub is used. The LiDAR and depth camera connect through the hub, while the AI camera connects directly to the Raspberry Pi. Finally, wiring lengths are kept as short as possible to minimize electrical resistance and voltage drop, helping maintain stable system performance.

Section 4.2: Microcontroller and Electronic Component Features

On the control side, the system is structured so that the Raspberry Pi 5 microprocessors and the Arduino Nano microcontroller perform different roles. Two Raspberry Pi 5 microprocessors handle the computational tasks associated with perception and high-level decision making. These tasks include acquiring image and depth data from the cameras, processing LiDAR data for partial mapping, performing object detection, estimating object trajectories, and determining whether the robot should continue vacuuming or switch to trash-catching mode.

The Arduino Nano microcontroller is connected to the Raspberry Pi through a USB connection that allows serial communication between the devices. While the Raspberry Pi performs high-level processing, the Arduino Nano is responsible for low-level actuator control. Specifically, it adjusts the speed of the DC wheel motors by receiving motion commands from the Raspberry Pi and converting them into PWM signals for the motor drivers. Encoder feedback is used to implement closed-loop speed control, ensuring accurate and stable motor

performance. The microcontroller is used for these actuator-related tasks because it provides consistent timing and reliable hardware control, which is critical for motor control and real-time responses. By separating the computational workload between the Raspberry Pi and the Arduino Nano, the system can maintain stable motor control even when higher-level processes such as AI detection or sensor processing temporarily increase the computational load.

In addition to the control hardware, several electronic components are used to support the robot's operation. The AI camera is used in trash-catching mode to detect overhead trash and provide positional information that helps the robot reposition itself to intercept the object. The depth camera and LiDAR sensor are used during autonomous vacuuming mode to detect obstacles and assist with navigation within the room. Together, these sensors allow the robot to operate safely and efficiently in both modes.

Section 4.3: System Integration and Data Acquisition

In this project, the robot software is implemented using ROS2, which allows different components of the system to communicate through multiple nodes. ROS2 was installed on both the development computer and the Raspberry Pi 5, with SSH (secure shell) enabled on the Raspberry Pi non-remotely initially to allow wireless access from the computer, from now on. This setup eliminates the need to repeatedly connect the Raspberry Pi directly with cables such as a micro-HDMI, keyboard, and mouse. Instead, the code and files are developed on the computer and then transferred to the Raspberry Pi through the ROS2 environment, simplifying development and testing. Below in Figure 33 is the initial setup for the Raspberry Pi 5 to install ros2 and enable ssh. Where now it can be accessed through the username and IP address of the wifi network.



Figure 33: Setup for the Raspberry Pi 5 for ROS2 installation and enabling of ssh

Nonetheless, in this architecture, individual nodes (code) are responsible for specific tasks such as sensor data processing, robot state publishing, motor control, and perception. These nodes exchange information through ROS2 topics, enabling the sensors, control algorithms, and actuators to operate together as a coordinated system. To support development and testing, a simulation environment was created using Gazebo. The robot model used in the simulation was built using URDF and Xacro files, where the physical structure of the robot is defined through links and joints. These files specify the robot's chassis, wheel assemblies, and other components, as well as their dimensions, inertial properties, and collision models. The wheel joints were defined as continuous joints, allowing them to rotate freely during simulation. Additional macros were implemented to simplify the calculation of inertial properties for different robot components. A snippet from the robot's Xacro model file used for simulation is shown below. This section defines the chassis link and its fixed joint connection to the base frame. The same robot model is used in Gazebo and RViz, allowing the simulated robot to behave consistently with the physical system during real world testing.

```
<!-- chassis link -->
<joint name="chassis_joint" type="fixed">
  <parent link="base_link"/>
  <child link="chassis"/>
  <origin xyz="-0.1 0 0.1188"/>
</joint>

<link name="chassis">
  <visual>
    <origin xyz="0 0 0"/>
    <geometry>
      <cylinder radius="0.1778" length="0.136"/>
    </geometry>
  </visual>
</link>
```

Figure 34: Snippet of the robot Xacro model used to build the robot in Gazebo simulation

In Figure 35, the chassis of the robot is defined as a cylindrical link connected to the base frame using a fixed joint. The geometric dimensions describe the physical structure of the robot, while the joint values establish how the component is positioned within the robot's coordinate frame. Nonetheless, the robot model is launched in Gazebo through a ROS2 launch file which uploads the Rviz file, which then loads the robot description and initializes the simulation environment. The simulation also integrates `ros2_control`, which allows controllers to interact with the simulated robot hardware. In this system, a joint state broadcaster publishes the state of each joint, while a differential drive controller which needs to be loaded, and then activated converts velocity commands into wheel rotation commands. These controllers allow the robot to move within the Gazebo environment in a manner that closely represents how the real robot will behave. Visualization and debugging of the robot state are performed using RViz2 command, which provides a graphical interface for observing the robot model, coordinate frames, and sensor data. The robot description and joint states are published through the `robot_state_publisher` node, allowing RViz to display the full kinematic structure of the robot. Note that `ros2_control` and `robot_state_publisher` nodes are files that have been created, and coded to be used for the robot. This visualization helps verify that all joints, links, and coordinate frames are properly defined within the URDF model. Note that the image below in Figure 36 has been readjusted to have the links (wheels) and caster wheel in the appropriate place as you will see below.

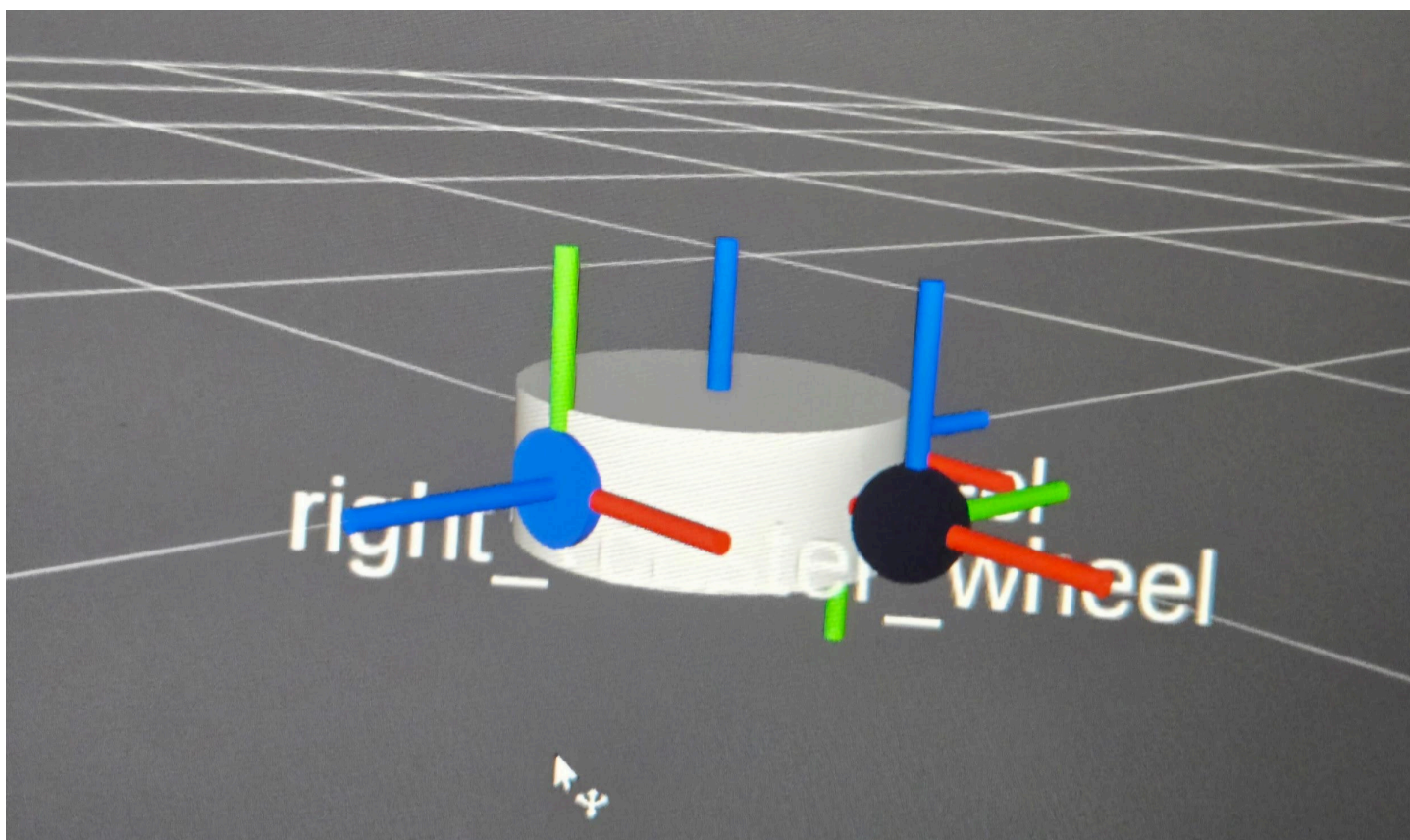


Figure 35: Rviz2 model (has been adjusted below for gazebo)

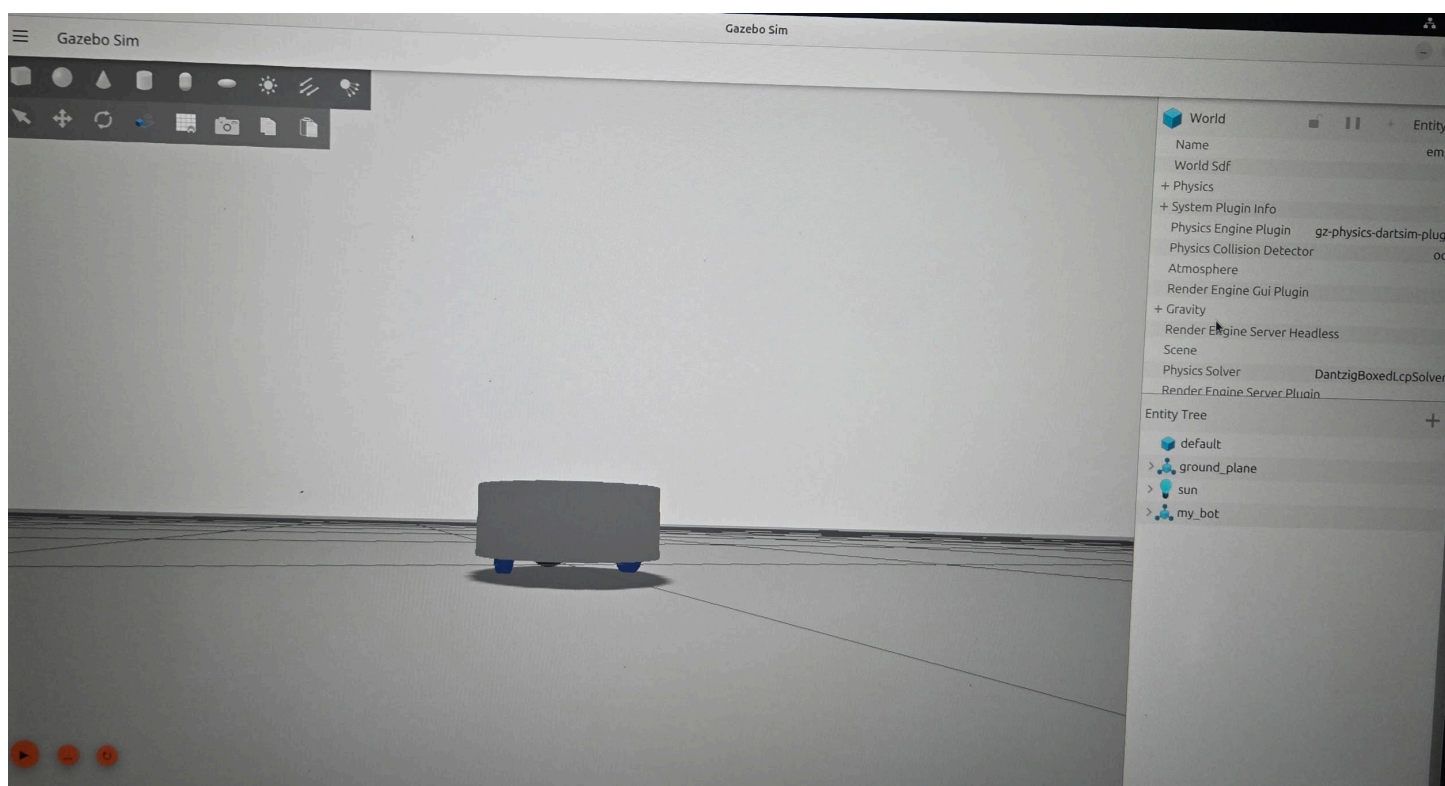


Figure 36: Gazebo with the updated rviz2 model for real simulation

Sensor integration is also incorporated into the ROS2 framework. The LiDAR sensor and depth camera will be represented as links attached in the xacro file where the model is defined for testing to the robot model, allowing them to publish data relative to the robot's coordinate frame. This enables the robot to process environmental data for navigation and obstacle avoidance. Nonetheless, using Gazebo and RViz together allows the robot's behavior, motion control, and sensor integration to be tested and verified in simulation before deploying the software on physical hardware. This simulation-based workflow significantly improves development efficiency by allowing software debugging and controller validation without risking damage to the robot's physical components.

The Raspberry AI camera serves an important job inside its trash-catching mode. It will give data on what the object is to avoid catching a structurally damaging object, such as a brick. This will have to be trained through an AI model. Its second job is to locate the center of the thrown object with respect to its camera coordinates. With this data, the velocity of the thrown object can be calculated and given to the microcontroller to find out where the object is going to land. The microcontroller will then tell the robot to move in the direction of the velocity of the object at a certain speed until the object gets out of frame. As more hands-on testing into the workings of the AI camera with the Raspberry Pi 5 develops, the implementation of the AI camera into the ROS2 framework and Gazebo simulation will be feasible.

Chapter 5: Prototyping and Fabrication

Section 5.1: Materials and Processes

The majority of the project's build material will be in 3D printed PLA. This method was chosen due to some components requiring intricate dimensioning in the project, such as the placement of the drive motors or the vacuum features. Using a 3D printer gives an edge in precision in tight spaces. Most of the 3D prints will be carried out through Makerspace in the Davidson College of Engineering, and smaller features will be printed in-house through a project member's personal 3D printer. The 3D print material was chosen to be PLA because it is the cheapest filament. Other filaments, such as the heat-resistant ABS or the flexible Thermoplastic Polyurethane (TPU), were deemed unnecessary because the most important structural factor in the project is rigidity to withstand impacts of thrown objects or bumping into walls, and PLA does that just fine. Please refer to Appendix A for more details on PLA. To reduce wastage of filament during the manufacturing phase, the mounting features of smaller components will be tested with the personal 3D printer instead of having to reprint a large part in the case where a feature is misdimensioned. This aligns with the ABET SO4, which describes the student's ability to recognize ethical and professional responsibilities in engineering as wastage of materials was reduced.

Section 5.2: Assembly Procedures

Assembly Summary

- Build Material: PLA
- Build Process: 3D Printing via Makerspace and personal printer
- Fastening Types:
 - Bolts
 - Nuts
 - Heat-set Inserts

- Tools:
 - Soldering Iron
 - Scraper
 - Flush Cutters
 - Type 2 Phillips Screw Driver
 - Allen Keys

The fastening features in this project will involve bolts, heat-set inserts, and hex nuts. Before beginning assembly of the 3D-printed chassis, the overhang support features that are created during the 3d printing process have to be removed, which will leave rough surfaces that will have to be cleaned or sanded down. The next step is to place heat-set inserts in areas that are often fastened and unfastened together, or areas where putting a hex nut is inaccessible or unattractive. The heat-set insert kit will be ordered online, and the soldering iron to place the inserts in will be accessed through Makerspace.

Assembly will first be carried out by fastening the parts that are required to be split into multiple prints with the use of bolts and inserts that were in place from the previous step. From here, the electronics and physical mechanisms can be put in place. Finally, loose wires from electronic components will be taped together with electrical tape to avoid being tangled with the drive system components.

Section 5.3: Challenges During Fabrication/Build

A big constraint for 3D printers is a limited build size, and the largest accessible build area is 350 mm x 350 mm x 340mm (13.8in x 13.8in x 13.4in). Because the project is larger than the 3D printer's build area, the project will have to be printed in parts and fixed together. In many prints, supports were needed because there were many overhangs that the 3D printer could not directly print. These supports became a challenge as they add an extra layer of PLA, and are not part of the design, which is too hard to remove, so the result is longer than designed in CAD. To fix this, the overhang supports need to be eliminated in that area.

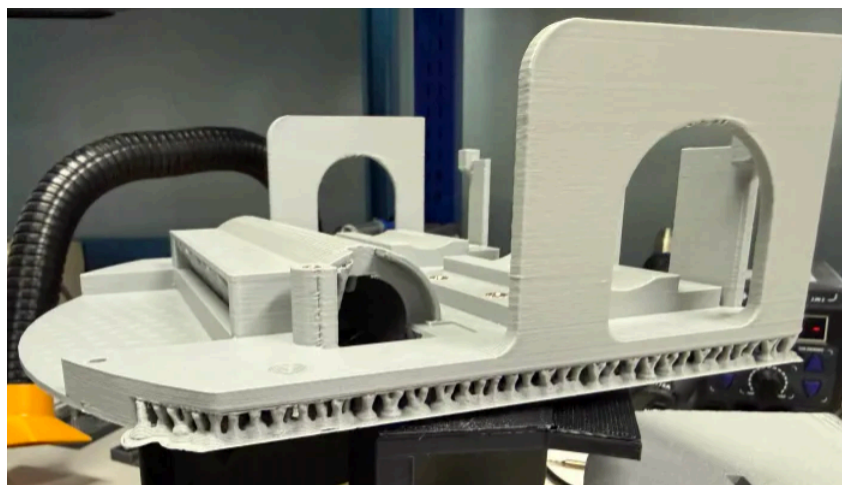


Figure 37: Overhang supports that are difficult to remove



Figure 38: Supports adding extra hidden length, causing misalignment in the final assembly.

Section 5.4: Cost Effectiveness of Prototype

The BOM lists all components required for the prototype, including the DC motors, drivers, the 6S LiPo battery, LiDAR unit, AI and depth camera, Raspberry Pi, microcontroller, chassis materials, connectors, and other electrical safety components. Each item in the BOM is listed with its quantity, price, and vendor, which allows the team to track costs and verify that the project remains within the initially desired budget. A cost breakdown shows that the most expensive components were the battery, motors, sensors, and cameras, while the mechanical materials, wiring, fuses, and other minor components represented a smaller portion of the total cost.

Cost analysis is also utilized for evaluating design decisions, as well as future optimization possibilities. For example, if the sensing suite has been determined to be over-specified in terms of its performance gains, future development cycles could look to either lower-cost alternatives or consider a modular configuration of sensors that can be scaled-up or down depending on required price points. Likewise, if additional structural components are found to be costing more than previously expected, it may be possible to adjust the type of materials used or change the way they are manufactured in order to lower costs while still meeting the requirements of safety and reliability. Collectively, the BOM (Bill of Materials) supports the overall conclusion that the selected design is an acceptable financial undertaking for a student capstone project, while identifying areas in which the majority of investments are being made.

Table 4: Bill of Materials (BOM)

Components	Unit Cost \$	Quantity	Ship+tax+total
24 Volt Planetary DC Wheel Motor	\$54.50	2	\$178.21
DROK DC Buck Converter Adjustable Voltage Regulator	\$15.99	1	\$17.49
18 AWG Stranded Tinned Copper Wire	\$7.68	1	\$7.63
Ampper Battery Disconnect Switch 12V -	\$12.14	1	\$13.28

48V - Kill Switch			
Raspberry Pi 5 8GB (Microcontroller)	\$80.00	1	\$88.00
A1M8 Lidar	\$99.00	1	\$108.28
Raspberry Pi AI Camera	\$77.00	1	\$90.02
Cooling fan/Heatsink for Raspberry Pi	\$10.99	1	\$13.00
HCL-HVC 22.8V-10200mAh 120C G10 Protection Plates	\$119.95	1	\$132.95
12 AWG Silicon Copper Wires (3 ft)	\$7.98	1	\$8.73
8 AWG Fuse Holder	\$6.49	1	\$7.15
12 AWG Fuse	\$5.29	1	\$5.79
6 PCS Dual High-Power MOSFET Trigger Switch Drive Module	\$5.29	1	\$6.93
8 AWG Silicone Wire (3 ft)	\$11.98	1	\$13.10
Depth Camera	\$149	1	\$169.00
GLOSO TBE Double Row Stainless Terminal Blocks 10 poles	\$15.95	1	\$16.35
Cytron 20Amp Bi-Directional 6V-30V DC Motor Driver Speed Controller 60A Peak	\$28.50	2	\$62.34
Nano Module CH340 5V/16M USB-C Micro-Controller Board for Arduino Nano V3.0	\$12	1	\$13.56
10 pack Terminal Block White Screw Terminal Connector Dual Row	\$5.99	1	\$6.55
OVONIC X1 Pro Dual RC Lipo Battery Balance Charger	\$84.79	1	\$92.74
5 Pairs Amass XT90S XT90-S XT90 Connector Anti-Spark	\$11.99	1	\$13.11
Constant Current CC CV Buck Converter Module DC 6-40V to 1.2-36V 20A 300W	\$9.99	1	\$10.06
XHF 10PCS 8 AWG-3/8 Heavy Duty Battery Lugs with 3:1 Heat Shrink Tubing	\$5.21	1	\$5.70
GLOSO TBE Double Row Stainless	\$10.95	1	\$11.98

Terminal Blocks, Marine Grade, 4-14 AWG Cu, 75A/600V - 1 Pack, 2 Poles			
Threaded Inserts 400Pcs (Ktehloy)	\$15.98	1	\$17.42
16 Pcs Blade Fuses Maxi APX Replacement Fuses	\$8.99	1	\$9.83
WGGE Automatic Heavy Duty Wire Stripper/Wire Cutter 8-18 AWG Solid and Stranded Electrical Wire	\$14.99	1	\$16.40
Lexar E-Series 64GB Micro SD Card, microSDXC UHS-I Flash Memory Card with Adapter, 100MB/s, C10, U3, A1	\$18.99	1	\$21.30
XT60 to XT90 Adapter Cable	\$10.11	1	\$10.93
TETRIX Wheels and Hub	\$38.85	1	\$41.99
Magnets (x8)	\$15.17	1	\$16.40
IVIN USB 3.0 Hub	\$8.59	1	\$9.29
650 pcs Heat Shrinks	\$9.20	1	\$9.94
Udemy	\$27.75	1	\$30.00
Micro HDMI to HDMI Cable	\$9.38	1	\$10.15
Flex Cable for Pi Camera 100mm	\$3	1	\$3.26
Flex Cable for Pi Camera 300mm	\$4.02	1	\$4.35
Hopper Heat Sink for Raspberry Pi	\$5.05	1	\$5.45
Female to XT90 Male Plug Extension	\$13.15	1	\$14.21
SELTERM 8AWG Copper Wire Lugs	\$11.55	2	\$24.98
USB C Cable 3ft	\$5.15	1	\$5.57
iCrimp Battery Terminal Crimper	\$27.26	1	\$29.47
Motor Mount Braket	\$11.80	1	\$12.79
Arducam canera 160	\$50.85	1	\$55.00

Dielectric Greaser	\$4.30	1	\$4.65
Permatex Thread Locker	\$6.52	1	\$7.05
ASUS BT500 Dongle	\$21.37	1	\$23.10
Shark® Robot Front Wheel Kit	\$19.92	1	\$21.53
Plusrock Waterproof Converter	\$17.55	1	\$18.98
MOSFET Trigger Switch Drive Module	\$7.10	1	\$7.69
Male to Female Wires	\$7.62	1	\$8.24
Rasp Pi USB Keyboard	\$17.15	1	\$18.54
Breadboard Power Kit	\$6.05	1	\$6.54
Total:			\$1557.00

Additionally, the selection of these components was made taking into account their functionality, quality, electrical ratings, and cost effectiveness. The DC motors for the wheels were decided upon since they would deliver the required torque and RPM to the robot without overstressing the power supply, while offering a price that, though low, was not a compromise on quality when compared to other vendors. The DROK buck converter was chosen because it could supply the necessary current and voltage to the Raspberry Pi 5 and other low-powered devices without a problem, but at a lower cost since its maximum power rating was not high. The Raspberry Pi 5 was selected because it was powerful enough to perform AI tasks, process camera images, and handle data from several sensors all at once, while basic microcontrollers could not take such high-level processing; besides, it was much less expensive than the \$200 or more alternative microprocessors, which would have put the project over budget. The battery kill switch was chosen due to its low price, well-known manufacturer, handling ability of the battery's maximum current and voltage, and being on sale during the Black Friday event. RPLiDAR A1M8 was the selection made because it guarantees reliable and accurate mapping performance at a price that is not prohibitive; more expensive LiDAR sensors do offer longer range, but these capabilities are not needed for this robot that mainly works indoors. The Raspberry Pi AI camera, despite its higher price, was chosen because of its great picture quality, precise results, and smooth integration; thus, it is amazing for the purpose of distinguishing and recognizing the trash types. Along with the heatsink, a fan has been incorporated to guarantee proper cooling of the Raspberry Pi 5, which takes care of various sensor and camera inputs. The depth camera was the one that was picked because it offers performance that is on par with cameras in the \$300–\$500 range while being very cheap at the same time. Cytron motor drivers were selected for controlling because most of the alternatives were unable to deliver the high current needed for the 24 V planetary DC motors, even if they were rated for the voltage; other sources that could do so were priced above the motors, thus making them unreasonably expensive. A nanomodule motor driver was chosen for the vacuum motor because it can handle the required voltage and current, it's easy to use, and it's reasonably priced. XT90-S connectors were the ones chosen due to their anti-spark feature, which limits inrush current when connecting high-current components; therefore, the risk of electrical damage and safety hazards was reduced.

while also being the most affordable option available. The vacuum motor and additional buck converter were purchased using a gift card, which did not have any effect on the project budget. Note that a nano Arduino microcontroller was bought to interchange information with the Raspberry Pi 5 through serial communication using a usb power, which both of these things were given for free at Makerspace. Additionally, the same brand of terminal blocks was chosen due to the fact that they can handle the system's voltage and current safely, support multiple wire gauges, and are available under a cost-saving promotion. Wire sizes of 8 AWG, 12 AWG, and 18 AWG were chosen according to the electrical code standards that apply to current requirements; short three-foot lengths were used to keep the cost down and fit the robot's small size. Heavy-duty battery lugs were selected to guarantee a reliable mechanical and electrical connection to the kill switch, as ferrules by themselves would not exert enough force; the lugs chosen were the right size and were available within the project's time frame. A high-performance LiPo battery was chosen despite its higher price because it offers a large current capacity, long operation time, and dependable performance, together with a compatible high-power charger. Even if the electrical components, such as sensors and cameras, were the main budget item, further cost-cutting would reduce the system's performance, accuracy, and reliability. In the end, \$500 of the budget was covered by external funding, plus support from the family's \$500+ out-of-pocket cost that was shared among the team.

Section 5.5: Safety, Environment, and Engineering Codes

Safety was an important factor considered throughout the design of the Trash-Catching Vacuum Robot. The robot contains moving mechanical components, electrical systems, and a high-current Li-Po battery, which require careful design to reduce potential hazards. To address these concerns, the robot includes several safety features, such as a main power switch and an emergency shutoff that allows the system to be quickly powered down if unexpected behavior occurs. Proper wiring practices are also used, including appropriate wire gauges, fuses, and secure connections to minimize the risk of overheating, electrical shorts, or component failure. The design also considers thermal safety. Components such as the Raspberry Pi, motor drivers, and power electronics generate heat during operation. To prevent overheating, heatsinks and cooling fans are included, and ventilation openings are incorporated into the robot shell to allow airflow and maintain safe operating temperatures.

Environmental considerations were also taken into account during the design process. The robot shell is manufactured using additive manufacturing with PLA material, which helps reduce material waste compared to many traditional manufacturing processes. In addition, some components of the vacuum system are reused from an existing robotic vacuum, which helps reduce electronic waste and lowers the environmental impact of the project. Engineering codes and standards referenced earlier in the design process also support these safety and environmental goals. Guidelines from organizations such as NFPA, OSHA, ASME, ASTM, ISO, ANSI, UL, and IEEE help ensure that the electrical and mechanical systems follow accepted engineering practices. By considering these standards, the robot is designed to operate safely, reliably, and in accordance with professional engineering expectations.

Section 5.6: Final Overview and Modifications from Initial Design

The final iteration of the design will be described in a few systems: drive, microcontroller, sensors, and chassis. The drive system is composed of two DC motors and drivers rated for 840 RPM at 1.42 Nm of torque and a castor wheel. Four-inch wheels will be connected to the motor shaft. The microcontroller for this project will be the Raspberry Pi 5 with 8 Gb of RAM. This brand was chosen because of the accessibility to many open source

projects and code that will aid in the development of the programs. It was deemed that 8 Gb of memory was needed, as it is typical for ROS programs to take that much RAM. The Raspberry Pi will be connected to three sensors, that being the Raspberry AI camera, the A1M8 Lidar, and the OAK-D depth camera. The initial design also included two ultrasonic sensors, which were removed in the final design as the sensors were believed to lack complex room mapping capabilities, which are accounted for using the lidar. And finally, the chassis will be 3D printed in PLA, consisting of a max diameter of 14 inches, a bin diameter of 7.25 inches, and a total height of 14.5 inches. The final chassis design also included ventilation holes to improve airflow and cooling that is needed for the microcontroller.



Figure 39: Fully Assembled Prototype

Chapter 6: Testing and Results Analysis

Section 6.1: Dust Bin Experimentation & Analysis

Methodology

To evaluate the performance of the vacuum system, a computational fluid dynamics (CFD) analysis was conducted using SOLIDWORKS Flow Simulation. The purpose of this testing was to analyze airflow behavior, suction effectiveness, and vortex formation inside the bin under realistic operating conditions. The simulation was set up as an internal flow analysis using air as the working fluid. Standard atmospheric conditions were assumed, with air density approximately 1.2 kg/m^3 . A pressure-driven flow approach was used to replicate the behavior of a vacuum system. An environmental pressure of approximately $101,325 \text{ Pa}$ was applied at the inlet. At the outlet, a reduced static pressure condition was applied to simulate suction, defined as atmospheric pressure minus 1200 Pa . This created a pressure differential of approximately 1200 Pa across the system, which drives the airflow through the dust bin. The computational domain enclosed the entire internal volume of the dust bin, including the funnel region, dome geometry, and suction outlet. A global mesh was generated automatically, and the solver iterated until convergence was achieved based on goal stabilization. To quantify system performance, several engineering goals were defined, including maximum dynamic pressure, average static pressure, average velocity, and volume flow rate. These goals allowed for evaluation of both airflow energy and motion throughout the system.

Two geometries were evaluated: A smoother bin and an updated bin with a dome covering the caster wheel opening

Name	Current Value	Progress	Criterion	Averaged Value
GG Maximum Dynamic Pressure 2	1251.12 Pa	Achieved (IT = 64)	37.8949 Pa	1245 Pa
GG Minimum Dynamic Pressure 1	0 Pa	Achieved (IT = 51)	0 Pa	0 Pa
SG Average Static Pressure 2	101063 Pa	Achieved (IT = 66)	8.12752 Pa	101066 Pa
SG Average Velocity 1	20.5179 m/s	Achieved (IT = 57)	0.593903 m/s	20.4163 m/s
SG Volume Flow Rate 3	0.0857656 m^3/s	Achieved (IT = 58)	0.00252658 m^3/s	0.0853063 m^3/s

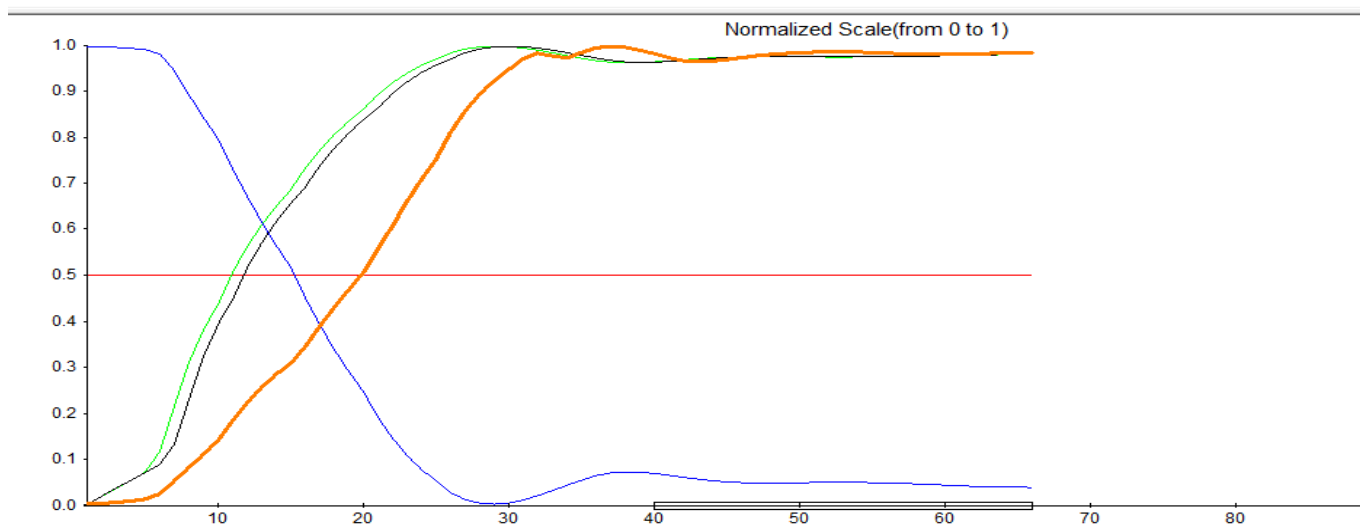


Figure 40: Convergence plot for bin design showing stabilization of velocity, pressure, and flow rate values.

Name	Current Value	Progress	Criterion	Averaged Value
GG Maximum Dynamic Pressure 2	1569.95 Pa	Achieved (IT = 70)	50.205 Pa	1613.37 Pa
GG Minimum Dynamic Pressure 1	0 Pa	Achieved (IT = 45)	0 Pa	0 Pa
SG Average Static Pressure 2	101053 Pa	Achieved (IT = 63)	9.02724 Pa	101048 Pa
SG Average Velocity 1	17.9774 m/s	Achieved (IT = 67)	0.550756 m/s	18.2243 m/s
SG Volume Flow Rate 3	0.0497708 m ³ /s	Achieved (IT = 51)	0.00148821 m ³ /s	0.0503459 m ³ /s

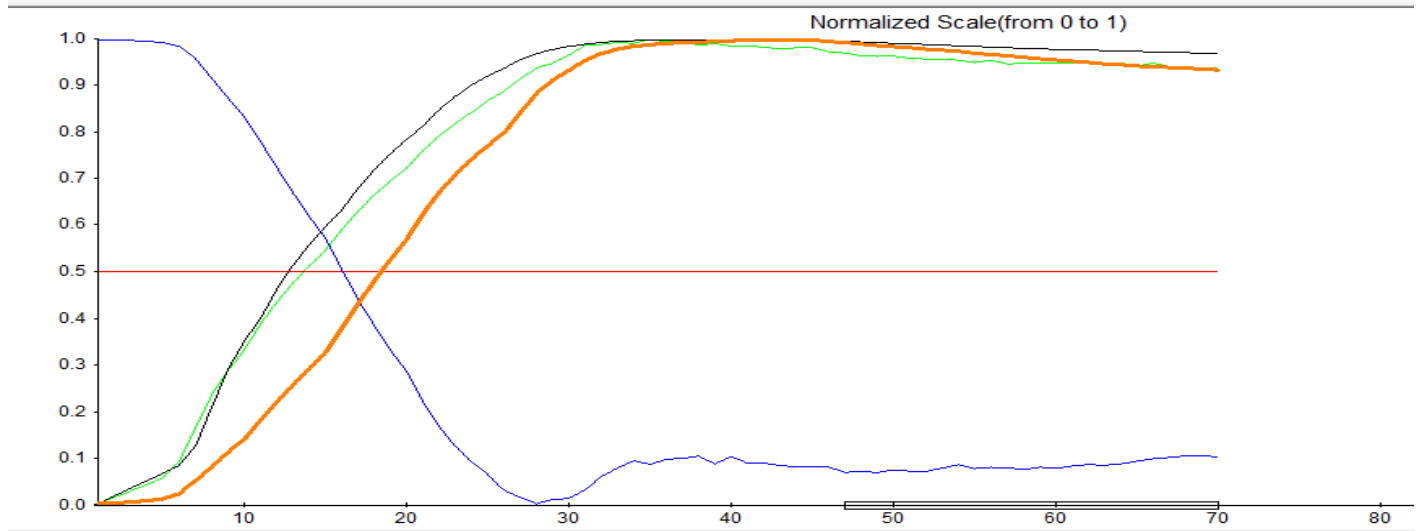


Figure 41: Convergence plot for dome design showing convergence of simulation parameters.

These figures confirm that both simulations reached steady-state conditions and that the results are reliable for analysis. The simulation results were obtained after convergence of all defined goals under a pressure difference of approximately 1200 Pa.

Data Collection & Analysis

Smoother Bin Design Results:

- Average Velocity: ~20.5 m/s
- Volume Flow Rate: ~0.0857 m³/s (~181 CFM)
- Maximum Dynamic Pressure: ~1251 Pa
- Average Static Pressure: ~101066 Pa

Dome Design Results:

- Average Velocity: ~17.98 m/s
- Volume Flow Rate: ~0.0497 m³/s
- Maximum Dynamic Pressure: ~1569.95 Pa
- Average Static Pressure: ~101000 Pa

Dynamic pressure is defined as:

$$q = \left(\frac{1}{2}\right) \rho V^2 \quad (20)$$

and represents the kinetic energy of the airflow, which is directly related to suction effectiveness.

Where ρ is the air density and V is the airflow velocity. This equation relates airflow velocity to kinetic energy and provides an estimate of suction strength.

The dynamic pressure calculated using this equation is based on average velocity values and represents an approximate average condition. In contrast, the maximum dynamic pressure values reported in the simulation are obtained from localized high-velocity regions within the flow field and therefore provide a more accurate representation of peak performance.

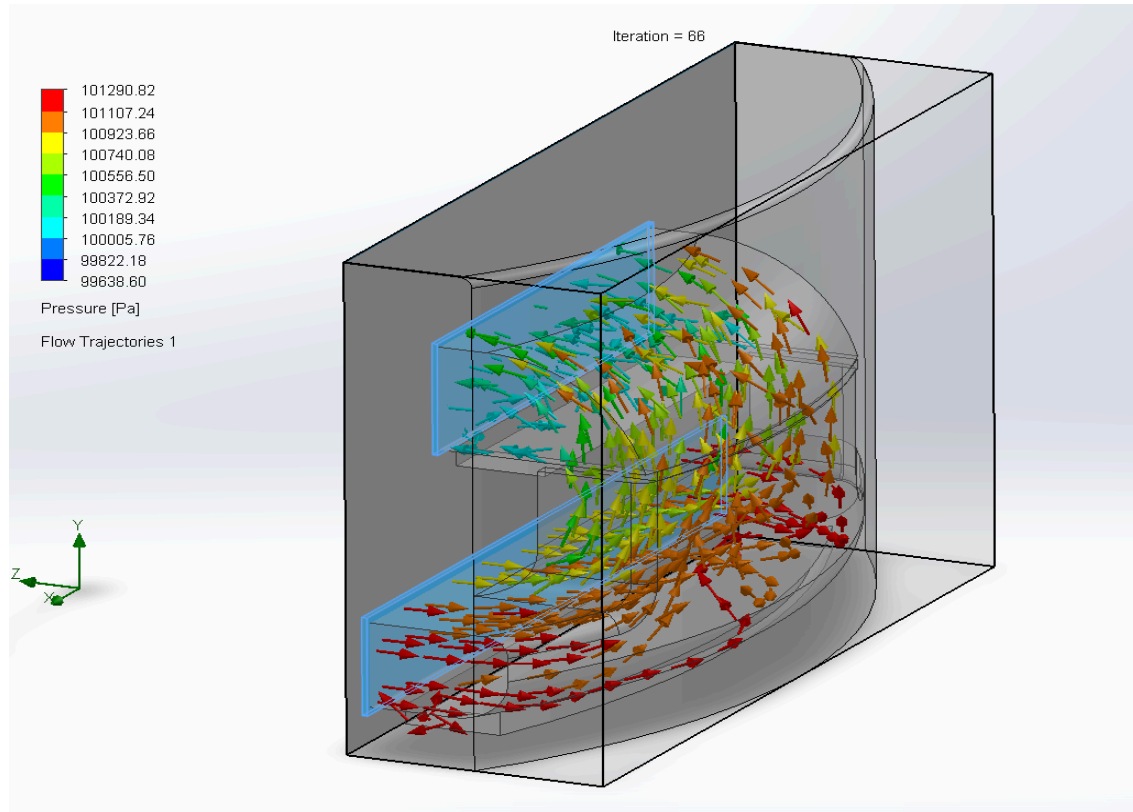


Figure 42: Flow trajectories for smoother dust bin showing smoother airflow and reduced vortex intensity.

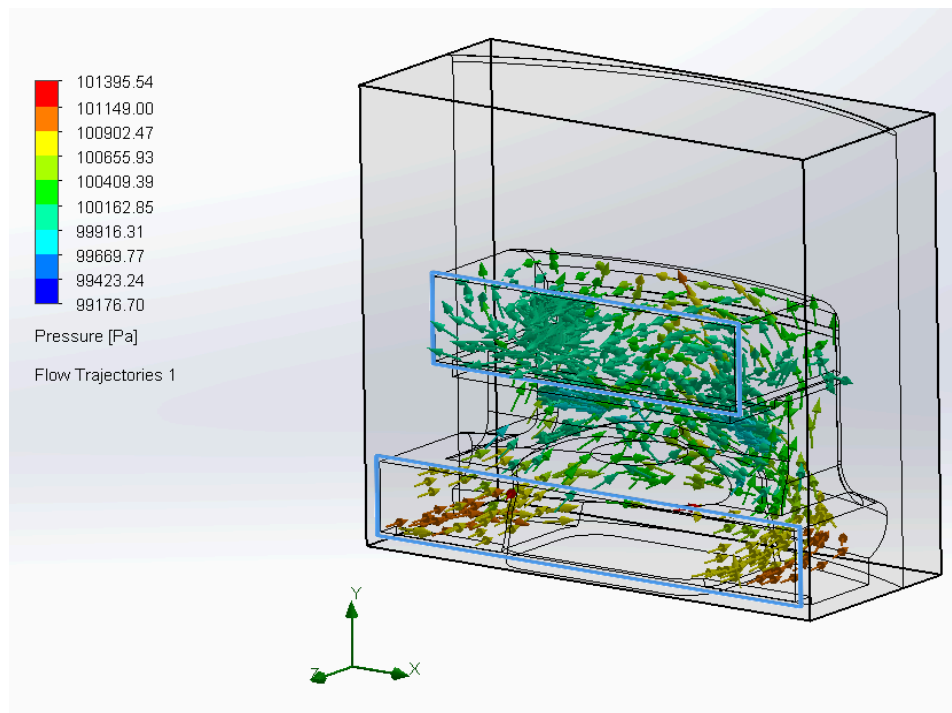


Figure 43: Flow Trajectories for Dome Dust Bin Design

As shown in Figure 42, the smoother design guides airflow more smoothly toward the outlet, reducing turbulence and minimizing recirculation within the bin. In contrast, Figure 43 shows the updated dome design, where airflow is further controlled and directed due to the elimination of the bottom opening. The dome prevents unwanted airflow leakage and forces the air to follow a more efficient path toward the suction outlet. As a result, airflow remains more organized and consistent throughout the bin. The simulation data was analyzed using built-in computational tools within SOLIDWORKS Flow Simulation. Goal plots were used to verify convergence and numerical stability, while flow trajectories were used to visualize airflow behavior, velocity distribution, and pressure variation. These tools enabled both quantitative comparison and qualitative analysis of flow patterns between designs.

Discussion and Conclusion

The results demonstrate that internal geometry has a significant impact on airflow behavior, even under identical pressure conditions. The applied pressure difference of approximately 1200 Pa represents a realistic operating condition for a small vacuum system, ensuring that the results reflect real-world performance. The funnel design provides a balance between airflow velocity and flow control. Its curved geometry reduces abrupt directional changes, minimizing turbulence and allowing airflow to remain more organized. As shown in Figure 42, the airflow follows smoother trajectories, which improves energy utilization and debris transport. The smoother design also produces a relatively high dynamic pressure, indicating effective concentration of airflow energy. The updated dome design introduces additional resistance, which results in a decrease in average velocity and volume flow rate. However, it significantly improves airflow control by eliminating leakage through the bottom opening and guiding airflow along a more efficient path. As shown in Figure 43, the airflow is more directed toward the suction outlet, with reduced recirculation and improved consistency. Despite the reduction in velocity, the dome design produces a higher maximum dynamic pressure, increasing from approximately 1250 Pa to approximately 1570 Pa. This indicates that airflow energy is more effectively

concentrated within the system. As a result, the ability of the system to transport debris is improved, even with a lower average velocity.

When compared to the design criteria, both designs generate sufficient airflow for debris pickup. However, the dome design better satisfies the requirement for controlled airflow and efficient debris routing. This is especially important for integration with rotating brushes, which rely on consistent airflow to transport debris into the bin. The results satisfy the design requirements for the system, which include generating sufficient airflow for debris pickup while maintaining controlled flow behavior. The dome design meets these criteria by providing adequate velocity (~ 18 m/s) while reducing recirculation and improving airflow direction. This ensures that debris can be effectively lifted and guided into the collection bin. The testing results confirm that internal geometry plays a critical role in airflow performance. While the smoother design provides good airflow characteristics, the dome design further improves performance by reducing energy losses and increasing effective dynamic pressure. These results demonstrate that maximizing velocity alone does not guarantee better performance. Instead, optimizing airflow direction and minimizing energy losses leads to a more efficient and practical design. Therefore, the updated dome dust bin is selected as the final design for the vacuum robot system.

Section 6.2: Trash Detecting AI Model

Methodology

The current camera used a demo image recognition model, detecting people and miscellaneous objects. The AI model for actual trash detection that will be applied to the camera was developed and tested under Google Colab. From prior experience, Google Colab proved to be a readily accessible software and qualified to train unique image recognition models through the existing "You Only Look Once" version 8 (YOLOv8) model, developed by software company Ultralytics. In addition, the AI model used a preexisting dataset, Trash Annotations in Context (TACO) database, a compilation of images of trash and their respective labels and segmented outlines. Several examples of the images extracted from the official website are shown in Figure 44. The program will reconfigure the dataset for all labels to be combined into one label, "trash", replace segmentation outlines with bounding boxes to reduce computational load, and divide the data into training and test data.



Figure 44: TACO Dataset Example Images

The four main models tested on TACO Dataset were labeled “run_20_epochs” “run_50_epochs” “run_yolov8s” “run_highres”. For the purposes of this report, the four models will be referred to as Model 20E, Model 50E, Model 8S, and Model High-Res, respectively. Model 20E ran YoloV8n (Nano), a version of YoloV8 that prioritized processing speed rather than accuracy and precision to prevent hardware issues. This model went through 20 epochs of training with image quality set to 640 by 640 pixels, with padding for rectangular images. Model 50E increased the epoch count to 50 from Model 50E. Model 8S replaced YoloV8n to YoloV8s from Model 50E to increase the accuracy and precision of the model at the cost of processing speed. Model High-Res increased the image resolution to 800 pixels from Model 8S.

The full program written in Google Collab established a workspace folder in Google Drive, downloaded and reconfigured the TACO Dataset, trained and tested several variations of models through YOLOv8, compared the models, and saved the models and overall data onto Google Drive.

Data Collection and Analysis

The program compared the image recognition capabilities of each model. Figure 45, displays the original test image and the model’s annotated images side by side.

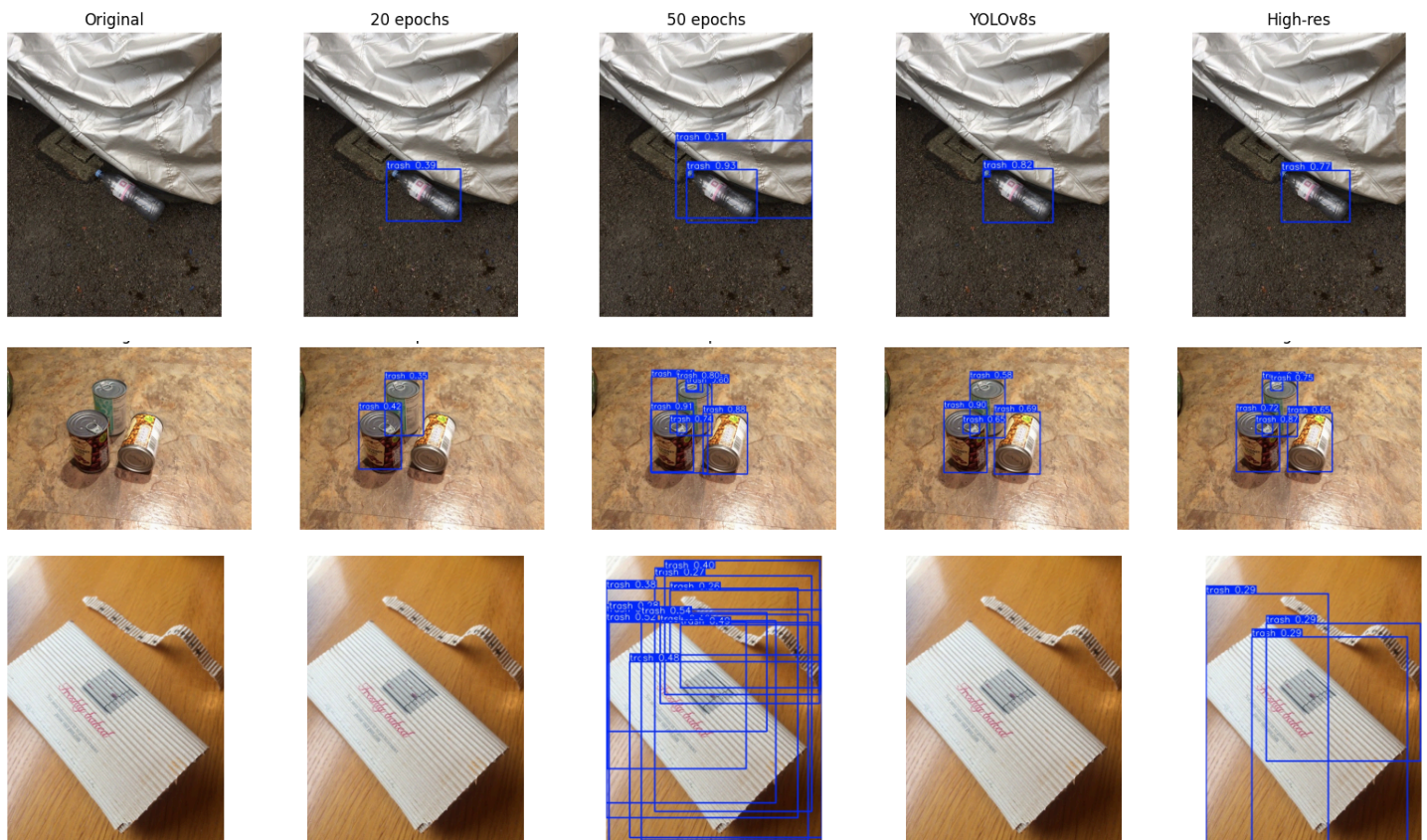


Figure 45: Sample Comparisons of Test Images and Bounding Boxes

In addition the program was able to compare the tests and confusion matrices of each model, providing the table with the given parameters shown in Table 5. In the context of the table, precision is the ratio between true positives to all positive predictions, while recall is the ratio between true positives to the sum of true positives

and false negatives. In general, higher precision implied lower false positives and recall implied lower false negatives. The mean average precision at 50% (mAP50) Intersection over Union (IoU) determined how well the boxes were drawn or if the object could be found based on the test labels and outline, with 50% of the box overlapping the test outlines. The mAP50-95 followed stricter criteria from 50% to 95% of the box.

Table 5: AI Model Parameter Comparison

	Precision	Recall	mAP50	mAP50-95
20E	0.434213	0.311924	0.298823	0.200703
50E	0.742216	0.311924	0.542862	0.381990
8S	0.683970	0.473926	0.525129	0.367631
High-Res	0.736570	0.512270	0.588373	0.432992

Discussion

Based on the tabular data, the values for precision, recall, mAP50, and mAP50-95 were all lower than 0.9, providing a sub-par performance. Precision is relatively high compared to the other parameters in contrast to recall, suggesting lower false positives with higher false negatives. This can be observed in Figure 45, displaying cases of unboxed pieces of trash. The lower value could be attributed to an imbalance of trash orientation of the training images, consisting of either a variety of trash scattered with low resolution or clear close-up of trash components, causing certain types of trash to identify trash from one frame over another. The mAP50 and mAP50-90 are low as well due to the false negatives and positives mentioned in the precision and recall, as well as irregular orientation as shown in the last row of Figure 45. Overall, Model High-Res had the greatest performance, dominating three out of four parameters against other models. Model 50E has potential as well, with the highest precision parameter, but couldn't be used effectively as the AI camera component itself had poor lens quality and registration of objects, causing a delay in the robot's reaction time of maximum 2 seconds of object recognition.

Recommendations for AI Improvements

Model High-Res has the highest potential for improvements. Decreasing the imbalance of the trash configuration of each image requires manual checking and is inefficient in the timeframe of the project. Increasing epoch count and introducing spatial drop-out may improve the parameters. Spatial drop-out involves removing a specific percentage of feature maps, including filters identifying shade, details, etc., rather than a node from typical drop-out to prevent overfitting. In addition, extracting the coordinates from the boxes may allow the camera and robot design to work with the AI model.

Section 6.3: Trash Detection and Approach System Using Camera

Trash Detection and Approach System Using Camera

The camera subsystem was tested under controlled conditions to verify the full perception to action pipeline. Testing was performed after all the nodes were initialized in the raspberry pi 5 and the robot successfully transitioned into trash_catch mode from initial idle to nav (navigation) using slam to the one desired. In this case trash_catch mode using the robot_mode_manager node.

System Configuration

The system consisted of an upward-facing USB camera (v4l2_camera), a YOLOv8-based detector (trash_detector.py), a trash approach controller (trash_approach.py), and a robot mode manager (robot_mode_manager.py). The main parameters used during testing were:

- Confidence Threshold: 0.35
- Image Size: 416
- Centre Tolerance: 0.06
- Linear Gain (kp_linear): 0.5
- Angular Gain (kp_angular): 2.0

Bottle detection was evaluated using COCO class 39. Detection messages were continuously published to /trash_detections, confirming real-time perception and tracking.

Detection Output From Testing:

- Class ID: '39'
- Confidence: ~ 0.60
- Bounding Box Center:
 - $x \approx 0.713$
 - $y \approx 0.623$
- Bounding Box Size:
 - $size_x \approx 0.181$
 - $size_y \approx 0.740$

As the bottle was moved within the camera's field of view, the bounding box position updated continuously, confirming accurate tracking.

```

bbox:
  center:
    position:
      x: 0.7131258010864258
      y: 0.6227258046468099
    theta: 0.0
  size_x: 0.1809236526489258
  size_y: 0.740102047390408
  id: ''
---
header:
  stamp:
    sec: 1775512888
    nanosec: 956817395
  frame_id: camera_link_optical
detections:
- header:
  stamp:
    sec: 0
    nanosec: 0
  frame_id: ''
  results:
  - hypothesis:
    class_id: '39'
    score: 0.6001554131507874
    pose:
      pose:
        position:
          x: 0.0
          y: 0.0
          z: 0.0
        orientation:
          x: 0.0
          y: 0.0
          z: 0.0
          w: 1.0
      covariance:
        - 0.0
        - 0.0

```

Figure 46: Output Values From Camera After Bottle Detection Sent to trash_detections

Velocity commands observed on /cmd_vel showed correct controller behavior. Note that when it's at zero, no object has been detected.

Non-zero outputs (object off-center from camera):

- Linear Velocity (x): ~ -0.038 to -0.084 m/s
- Angular Velocity (z): ~ -0.294 to -0.426 rad/s

Zero outputs occurred when:

- The object was within center tolerance (± 0.06)
- Detections were lost (1.5 s timeout)
- The robot was not in trash_catch mode

Note that when linear and angular velocity is 0.0, that means the object is aligned in the center of the camera, hence the robot doesn't need to move anymore to be aligned with the trash.

```
linear:
  x: 0.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.0
---
linear:
  x: -0.038349448310004364
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.2945514678955079
---
linear:
  x: -0.061362902323404966
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: -0.4262516021728515
---
linear:
  x: -0.08417277865939671
  y: 0.0
```

Figure 47: Velocity Commands Sent to Topic /cmd_vel

Control Approach

The trash approach controller computes motion commands based on the difference between the detected object position and a target position in the image. Angular velocity is driven by horizontal error, while linear velocity is driven by vertical error:

- $\text{angular.z} \propto (\text{target_x} - \text{detected_x})$
- $\text{linear.x} \propto (\text{target_y} - \text{detected_y})$

This proportional control allows the robot to continuously adjust its motion and align with the detected object.

System Behavior and Limitations

The system operated as expected, with smooth mode transitions (idle to nav to trash_catch), reliable detection, continuous tracking, and stable control response. The robot also correctly stopped when detections were lost.

One limitation is that the system was assumed that the target position is the center of the image:

- $\text{target_image_x} = 0.5$
- $\text{target_image_y} = 0.5$

Since the camera is mounted near the edge rather than directly above the bin, this introduces alignment error.

```

trash_approach = Node(
    package='my_bot',
    executable='trash_approach.py',
    name='trash_approach',
    output='screen',
    parameters=[
        'centre_tolerance': 0.06,
        'kp_angular': 2.0,
        'kp_linear': 0.5,
        'max_linear': 0.35,
        'max_angular': 1.5,
        'invert_y': False,
        'lost_timeout': 1.5,
        'target_image_x': 0.5,
        'target_image_y': 0.5,
    ],
)

```

Figure 48: Parameters of AI Camera From Initial Assumptions

Calibration and Conclusion

To correct this, the system must be calibrated by placing a bottle at the bin center, recording the corresponding bounding box values, and updating the target image parameters. Overall, the AI camera subsystem successfully demonstrated full integration of perception, decision-making, and control. The system was able to detect, track, and respond to objects in real time, with calibration remaining as the final step to improve accuracy.

The results show that the AI camera subsystem is functioning correctly, with successful integration between detection, control, and mode management. The system was able to reliably detect the bottle and publish bounding box data in real time, with confidence values around 0.60 indicating acceptable detection performance under the test conditions. The velocity outputs confirm that the controller is properly converting visual information into motion. Non-zero linear and angular velocities were generated when the object was off-center, and these values scaled with the positional error, which is consistent with proportional control. Zero outputs occurred under expected conditions, such as when the object was centered, lost, or when the robot was not in trash_catch mode, confirming correct system logic and safe behavior. A key limitation identified is the camera placement. Since the camera is mounted near the edge of the robot, assuming the target position is at the image center introduces alignment error. This means the robot may appear aligned in the image but not be physically

aligned with the bin, making calibration necessary. Detection performance was generally reliable, but it can be affected by lighting, object orientation, and model limitations. Overall, the system behavior was stable and predictable, and the remaining issues are mainly related to calibration and environmental conditions rather than flaws in the system design. And future implementation of a ps4 controller to switch modes using the L1 button will occur as well.

The AI camera subsystem successfully met its primary objective of demonstrating a complete perception-to-action pipeline. The system consistently detected the target object, tracked its position, and generated appropriate motion commands in real time. Integration between all components operated reliably, and the robot responded correctly under both active and inactive conditions. The controller showed stable and predictable behavior, with motion outputs scaling appropriately based on object position. In addition, safety features such as stopping when detections were lost or when alignment was achieved confirm that the system is suitable for real time operation. The main limitation identified is the need for calibration due to the current camera placement as well as its quality, which affects alignment accuracy and object detection. Overall, the subsystem can be considered successful, with performance somewhat meeting expectations but can definitely be secured and improved with the right necessary AI camera component and additional components such as an AI accelerator for the microcontroller.

Lidar Data And Slam Mapping

The LiDAR subsystem was tested to verify that valid scan data was being published and could be used by SLAM Toolbox for real-time mapping. Testing was performed using the robot's physical LiDAR sensor with the slam.launch.py file and the mapper_params_online_async.yaml configuration.

System Configuration

The SLAM setup used slam_toolbox in online asynchronous mapping mode. The main configuration included:

- Odom Frame: odom
- Map Frame: map
- Base Frame: base_footprint
- Scan Topic: /scan
- Resolution: 0.05 m
- Max Laser Range: 8.0 m
- Mapping Mode: mapping
- Loop Closing: enabled

This configuration allowed the LiDAR scans to be processed continuously for live map generation.


```

debug_logging: false
throttle_scans: 1
transform_publish_period: 0.02
map_update_interval: 5.0
resolution: 0.05
max_laser_range: 8.0
minimum_time_interval: 0.5
transform_timeout: 0.2
tf_buffer_duration: 30.0
stack_size_to_use: 40000000
enable_interactive_mode: true

```

Figure 49 : Slam Configuration for LiDAR

LiDAR Data Results

LiDAR data published on the /scan topic confirmed that the sensor was operating and returning valid range measurements. Representative scan data showed:

- Frame ID: lidar_frame
- Angle Range: approximately -3.14 to 3.14 rad
- Angle Increment: 0.00593 rad
- Scan Time: 0.134 s
- Range Min: 0.15 m
- Range Max: 12.0 m

The scan data contained a mixture of finite range readings and inf values. The finite values indicate detected nearby surfaces, while inf values represent directions where no obstacle was detected within sensor range. Example measured distances were approximately 0.11 m to 0.14 m in several scan regions, showing that the LiDAR was successfully sensing surrounding objects at close range.

The intensity values were also populated, with many readings near 47.0, confirming that the LiDAR was returning reflection data along with distance measurements.

```

header:
  stamp:
    sec: 1775712809
    nanosec: 26748882
  frame_id: lidar_frame
angle_min: -3.1407299041748047
angle_max: 3.136415481567383
angle_increment: 0.00592742720618844
time_increment: 0.00012649904238060117
scan_time: 0.1339624971151352
range_min: 0.15000000596046448
range_max: 12.0
ranges:
- .inf
- 0.14000000059604645
- 0.13899999856948853
- 0.13899999856948853
- 0.13899999856948853
- 0.13899999856948853
- 0.13899999856948853
- 0.13899999856948853

```

Figure 50: LiDAR Scan Data #1

```

- .inf
- 0.11800000071525574
- 0.11800000071525574
- 0.11800000071525574
- 0.11800000071525574
- 0.11800000071525574
- 0.11800000071525574
- 0.11800000071525574
- 0.119000000274181366
- 0.119000000274181366
- 0.119000000274181366
- 0.11999999731779099
- 0.11999999731779099
- 0.11999999731779099
- 0.11999999731779099
- 0.12099999934434891
- 0.12099999934434891
- '...'
intensities:
- 0.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0
- 47.0

```

Figure 51: LiDAR Scan Data #2

Discussion of Results

The LiDAR results show that the sensor was functioning properly and publishing consistent scan data for use in SLAM. The angular coverage and scan timing indicate that the sensor was providing a full 2D sweep suitable for mapping. The presence of both valid ranges and infinite readings is expected in normal operation, since some directions contain nearby objects while others do not contain detectable obstacles within the measurement range. Because SLAM Toolbox was configured correctly to subscribe to /scan and use the odom, map, and base_footprint frames, the LiDAR data was suitable for real-time mapping. Overall, the results confirm that the LiDAR subsystem was operating as intended and provided the necessary input for the mapping process.

Chapter 7: Conclusions and Future Work

Section 7.1: Design, Calculations, and Simulations Assessment

Shown throughout chapters 2 through 6, the team applies engineering principles that they have learned throughout their mechanical engineering career of courses taken from previous years before and during senior project.

The design of the robot was inspired through competitors in the market that have similar dimensions amongst each other, thus, the initial design specifications that was implemented onto the robot until further changes that deemed itself essential for the robot. Calculations that were made in regards of the speed, thickness, and power were driving factors to the material and components used for the project which go hand to hand with electrical calculations on what was needed to ensure the project ran safely within the constraints it was in.

Electrical calculations helped finalize what purchases were mandatory to meet the functionality requirements for the project, this is to ensure components don't overload in current and voltage in such that testing can be done multiple times during prolonged trials of prototyping. Which was a major concern during the manufacturing phase of the project, but was soon resolved with the help of purchasing buck converters and voltage regulators to prevent burnout.

Simulations that tested the structural integrity of the robot reflected on the design of the robot which proved to be an effective way of visualizing the flaws in the design. Causing major developments in the design to cater the stress analysis done on the robot to secure and finalize the design of the whole structure of the robot. Especially during testing when the robot was under continuous use which required a strong frame for the robot to withstand itself from breakage.

Overall, the team has successfully applied engineering principles to the project to validate it's design and electrical build.

Section 7.2: Experiment Results to Design Specification Comparison

As detailed in Section 1.2, the major design specifications are those for mobility, trash-catching, vacuuming performance, user interaction, and general constraints.

In general constraints and UI specifications, it was listed that the size of the robot should be 12.75" tall and 14" diameter and the control mode is switched remotely between vacuuming and trash-catching. In the final design, the diameter stayed the same but the height grew to 14.32 inches. In this section, the weight was also listed to have a 12 lbs unloaded weight and 6 lbs of payload before derating the speed of the motors. In the final assembly, the base weight was 16.8 lbs and any weight added derated the max speed of the robot. The control mode switching was achieved through a PS5 controller and a bluetooth dongle.

For mobility specifications, it was stated in specifications that the robot should have 360° rotation, move 1 ft/s in vacuuming mode, move 3.9 mph in trash-catching mode, and navigate around household environments while avoiding obstacles. The end-prototype was able to satisfy three of the four specifications. The only specification it did not meet was the speed at trash catching mode. Under no load, the robot was only able to reach a max speed of 3.35 mph. This does not mean that the robot will fail at trash-catching because it is limited by the motors. It means that the throw speed of the projectile will have to be reduced. The motors are capable of catching slower objects.

The specifications for vacuuming performance details that the vacuum system should be able to collect dirt particles, have a dust bin volume of 300 mL, and operate continuously for 60 minutes. The end result was able to meet those specifications with the exception of the dust bin volume. The volume of the dustbin was only about 50 mL as it was severely limited by the amount of space allotted for the vacuum system. However, it was able to lift large particles such as dirt. The vacuum system for this project served mainly as a proof of concept as more time was needed to refine the packaging of the entire robot to give more space for the vacuum. Continuous run-time was greatly exceeded however, reaching a run-time of four hours. This meant that the battery was oversized for the project. A smaller battery would've increased the usable space in the project.

Lastly, the trash-catching specifications stated to maintain structural integrity when catching a one-pound object, have a success rate of 75% on catching, and recognize an object and move towards it within 0.25 seconds. The end product did not succeed in trash-catching of a projectile, but it was able to react and move toward an object held above it. The reaction time of the trash-catching system was estimated to be within two to four seconds. This was not sufficient enough as to catch an object in the air, the reaction time should be within the milliseconds to catch anything. However, it was able to maintain structural integrity after having an orange, one-pound object, tossed onto it.

Section 7.3: Team Collaboration Reflection

The team's initial collaboration started off strong, but slowly became hindered due to personal lives and schedules that wouldn't align amongst every member. Despite this being the case, the team held strong to ensure the project meets its expected demands and functionality. By splitting the team into separate groups, it proved to be efficient as tasks were handled simultaneously by each group member without holding each other back. The only time when collaboration from all teams was necessary, was during prototyping and testing phase where manufacturing the robot mid way, allowed all teams to understand the flaws within the robot and make room for improvements without needing to fully build the prototype. This saved a lot of time for both ends of the party, allowing deadlines to be met and individual members to manage their time easier. What could be made for improvements would be communication, as response timing or lack of clarification, paused work or caused misleading information which have delayed progress within the project. However, the team has made note of these errors and made their efforts to resolve this issue throughout the year during the making of the project.

Section 7.4: Cost Analysis Evaluation

Based on the bills of materials of the project shown in subsection 5.4, the team has concluded that some purchases may not have been necessary or deemed the quality of certain products not reflecting their cost as there were performance issues for the robot. When it comes to competing against cleaning robots of today's market in comparison to the project, pricing may be an issue as most high end cleaning robots tend to average between \$800-\$1500. The price of the project may have fit within the price range, however, the effectiveness of the components within the robot, proves to be a problem on concluding whether or not the project may be considered a rival against other cleaning robots. Despite the unique function the project may have, if it were to have the capabilities to catch trash, that feature alone has costed an estimate of ~\$500 due to the needed components to drive the robot fast enough to detect objects during flight of less than a second and the speed to be within the thrown objects landing by calculating its trajectory. If such component was removed, the size and overall cost of the robot would have dropped significantly as a bin would not be necessary anymore nor the need of a depth camera as a lidar is able to scan the room with complete freedom of 360°. The validation for spending on such components is deemed reasonable as a powerful robot is needed to ensure such a functionality can exist in coinexistence of regular cleaning robot functions such as a roomba.

Section 7.5: Project Limitations

The most definite deciding factor of the project was the budget as it is a self funded project and supported project from the mechanical engineering department with a funding of \$500. This has affected what parts were chosen in this project which has deemed itself an affecting factor towards the robot's functionality. Necessary improvements for parts of the project such as the AI camera and Microcontroller is needed as the robot tends to internally crash itself as multiple softwares and hardwares make their attempt to coincide with eachother to run the robot towards it's desired functions. Weight was another factor that has affected the design of the robot as it needed to be light enough to catch objects. This meant the material of the robot had to be lightweight but also durable enough to withstand impacts of tossed objects that it would need to catch. PLA, was the main candidate as it was the most accessible material for SJSU students, provided by MakerSpace which mainly uses PLA for their 3D printers.

Section 7.6: Potential Improvements & Future Work

Improvements that can be done on the project to guarantee its success as a product would have to be a redesign on the overall robot for aesthetic, ergonomics, and size purposes. Visually it is not bad looking as it appears to be plain, similarly as any other cleaning robot in today's market. However, the ergonomics of the robot is unfriendly as removable parts such as the dustbin and bin tend to be a struggle to not only just remove, but to align and slot them back in as well. Gaining access to charging the robot and the internal components require full disassembly of the robot rather than subsections, proving it to be difficult to work with when it comes to replacing any parts or analyzing the circuitry of the robot. Dust bin holds very little volume of debris as it lacked the space to contain any once brushed/vacuumed in. Wiring of the robot could use proper slottings such as embedding cable clips within the design of the robot's chassis or using cable ties to prevent messy wires. The number of electronics overburdened the microcontroller as well as the weight of the robot, causing expected calculations to not be met for the robot's full potential functionality. Components that are a part of the robot that needs to be replaced would be the microcontroller as it's CPU and Ram Usage was too much for it to handle when it comes to running the multiple elements needed to meet the project's objectives. AI camera needs to be replaced as it's low performance quality has hindered the project greatly as improving the robot even through

code would not have been enough for it to perform as well as the team wants it to be. The robot's code can be improved to increase the effectiveness and efficiency of the robot's room scanning, object identification, and speed control for it's current state is close to those in the market, but not enough to be considered a competition. If there was a chance for the team to ever continue this project, these factors will be worked on to ensure the project isn't just a prototype, but a product that can be potentially sold to the many consumers out there in a need of a clean space with ease of access of a nearby disposal unit.

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Appendices

Appendix A: Detailed Calculations

1. Electrical Circuit Analysis

DC Motors Require 24 Volts and 12 Volts

Voltage Required to Make Devices Work = 5 V

Minimum Battery Voltage Needed is 6S (22.2 Volts)

Current of Wheel Motor: $I_{\text{Stall}} = 24.5 \text{ A}$

$$I_{\text{Rated}} = 13 \text{ A}$$

Current of Vacuum Motor: $I_{\text{Stall}} = 5 \text{ A}$

$$I_{\text{Rated}} = 4 \text{ A}$$

Two Wheel Motors: $I_{\text{Stall}} = 49 \text{ A}$

$$I_{\text{Rated}} = 26 \text{ A}$$

Current of Devices:

Raspberry Pi 5 = 1.5 A

A1M8 LiDAR = 1 A

Raspberry AI Pi Camera = 0.1 A

OAK D Lite Camera = 0.9A

Motor Driver = 0.4 A

$$I_{\text{devices}} = I_1 + I_2 + I_3 + I_4 + I_5 = 3.9 \text{ A} \quad (6)$$

Voltage Required to Make Devices Work = 5 V

Minimum Power Required For Buck Converter

$$P = IV \quad (8)$$

$$= (5\text{V})(4\text{A}) = 20 \text{ W}$$

Chosen Buck Converter Support 60 W > 20 W ✓

Current Draw by Buck Convert Input

$$I = \frac{P}{V_{\text{battery}}} = \frac{20 \text{ W}}{22.2 \text{ V}} = 0.99 \text{ A} \approx 1.5 \text{ A (Due to Efficiency Losses)}$$

Total Current Draw By The Battery (22.2 V) Battery:

$$I_{\text{Battery}} = I_{\text{Buck Converter}} + I_{\text{Wheel Motors}} + I_{\text{Vacuum Motor}} \quad (7)$$

$$I_{\text{Battery}} = 1.5 \text{ A} + 49 \text{ A} + 5 \text{ A} = 60 \text{ A (Due to Voltage Fluctuations)}$$

Hence, Battery Must Handle At Least 60 A at 22.2 V

Verification : HCL-HVC (22.2V, 10200mAh, 120C)

$$A_{\text{Continuously}} = Ah \times C \quad (9)$$

$$A_{\text{Continuously}} = (10.2 \text{ A})(120 \text{ C}) = 1224 \text{ A} > 60 \text{ A} \quad \checkmark$$

$$\begin{aligned} \text{Battery Load} &= \frac{I_{\text{Draw continuous}}}{A_{\text{battery continuous}}} \times 100 \quad (10) \\ &= \frac{60 \text{ A}}{1224 \text{ A}} \times 100 = 6.5 \% \end{aligned}$$

Power Distribution Hub (PDH), Terminal Block (TB), and Kill Switch (KS) Must Handle 22.2 V (Preferably 24 V) and Max Current That The Battery Supplies.

$$\begin{aligned} I_{PDH, TB, KS} &> 2 \times I_{\text{Continuous Wheel Motors}} + I_{\text{Continuous Vacuum Motor}} + I_{\text{Devices}} \quad (11) \\ I_{PDH, TB, KS} &> 2 \times 13 \text{ A} + 4 \text{ A} + 2.5 \text{ A} \\ I_{PDH, TB, KS} &> 32.5 \text{ A} \end{aligned}$$

$$\begin{aligned} I_{PDH, TB, KS} &> 2 \times I_{\text{Stall Wheel Motors}} + I_{\text{Stall Vacuum Motor}} + I_{\text{Stall Devices}} \quad (12) \\ I_{PDH, TB, KS} &> 2 \times 24.5 \text{ A} + 5 \text{ A} + 5 \text{ A} \\ I_{PDH, TB, KS} &> 60 \text{ A} \end{aligned}$$

Fuse Selection for Battery and Motor Drivers

$$\begin{aligned} \text{Fuse}_{\text{Battery, motor driver}} &= 1.25 \times I_{\text{Continuous}} \quad (13) \\ &= (1.25)(32.5 \text{ A}) = 40.63 \text{ A} \end{aligned}$$

$$\text{Fuse}_{\text{Battery, motor driver}} \rightarrow (1.25)(13\text{A}) = 16.25 \text{ A}$$

Notes:

These fuses will need to handle higher current if it's a fast-acting switch.

Motor drivers must supply 13 A continuously for Wheel DC Motors, and Max current it handles must be greater than 24.5A for safety.

1.2 Speed Requirements and Motor Selection

Parametric equations representing the trajectory of the projectile and robot:

$$[x, y]_p = [(3.58 \text{ m/s})\cos(60^\circ)t, 0.5 \text{ m} + (3.58 \text{ m/s})\sin(60^\circ)t + 0.5(-9.8 \text{ m/s}^2)t^2]$$

$$[x, y]_R = [0.5 \text{ m} + 0.5(3.55 \text{ m/s}^2) * t^2, 0.5]$$

Acceleration required from the robot

$$\begin{aligned} x_f &= x_i + v_i t + \frac{1}{2}at^2 \quad (1) \\ 0.87 \text{ m} &= 0.5a * (0.49 \text{ s})^2 \rightarrow a = 3.55 \text{ m/s}^2 \end{aligned}$$

Top speed of the robot:

$$\begin{aligned} v_f &= v_i + at \quad (2) \\ v_f &= 0 + (3.55 \text{ m/s}^2)(0.49 \text{ s}) \rightarrow v_f = 1.74 \text{ m/s or } 3.89 \text{ mph} \end{aligned}$$

Frictional force applied to driven wheels:

$$F = ma \quad (3)$$

$$F_{fr} = (8.16 \text{ kg})(3.55 \text{ m/s}^2) \rightarrow F_{fr} = 29.0 \text{ N}$$

Torque applied to the motor:

$$\tau = Fd \quad (4)$$

$$\tau = (14.5 \text{ N})(1/2 * 0.10 \text{ m}) \rightarrow \tau = 0.725 \text{ Nm or } 7.39 \text{ kg} * \text{ cm}$$

Linear velocity to angular velocity:

$$\omega = v/r \quad (5)$$

$$\omega = \frac{1.74 \text{ m/s}}{0.05 \text{ m}} \rightarrow \omega = 34.8 \text{ rad/s or } 332 \text{ RPM}$$

1.3 Impact Forces on Bin's Wall and Thickness Required

Parameters

Mass: $m = 200 \text{ g} = 0.2 \text{ kg}$, Soaring Speed: $v = 8 \text{ mph} = 3.58 \text{ m/s}$, Impact Angle: $\theta = 22.1^\circ$,

Contact Time: $t_c = 0.01 \text{ s} = 10 \text{ ms}$, Contact Area: $A = 1 \text{ cm}^2 = 0.0001 \text{ m}^2$,

Bin Outer Diameter: $D_o = 7.25 \text{ in}$, Safety Factor: $SF = 4$, Ultimate Tensile Strength: $\sigma_u = 50 \text{ MPa}$

Allowable Stress

$$\text{Allowable Stress: } \sigma_{allowable} = \frac{\sigma_u}{SF} \quad (14)$$

$$\sigma_{allowable} = \frac{50 \text{ MPa}}{4} = 12.5 \text{ MPa}$$

Velocity of Object to Bin Wall

$$\text{Normal to Wall Velocity: } v_n = v \cos(\phi) \quad (15)$$

$$v_n = v \cos(\phi) = (3.58 \text{ m/s}) \cos(90^\circ - 22.1^\circ) = 1.35 \text{ m/s}$$

$$\text{Tangent to Wall Velocity: } v_t = v \sin(\phi)$$

$$v_t = v \sin(\phi) = (3.58 \text{ m/s}) \sin(90^\circ - 22.1^\circ) = 3.32 \text{ m/s}$$

Momentum

$$\text{Momentum: } \Delta p = mv \quad (16)$$

$$\Delta p_x = mv_n = (0.2 \text{ kg})(1.35 \text{ m/s}) = 0.27 \text{ kg} \frac{\text{m}}{\text{s}}$$

$$\Delta p_y = mv_t = (0.2 \text{ kg})(3.32 \text{ m/s}) = 0.664 \text{ kg} \frac{\text{m}}{\text{s}}$$

Impact Forces

$$\text{Impulse Momentum: } \Delta p = Ft_c \quad (17)$$

$$F_x = \frac{\Delta p_x}{t_c} = \frac{0.27 \text{ kg} \frac{\text{m}}{\text{s}}}{0.01 \text{ s}} = 27 \text{ N}$$

$$F_y = \frac{\Delta p_y}{t_c} = \frac{0.664 \text{ kg} \frac{\text{m}}{\text{s}}}{0.01 \text{ s}} = 66.4 \text{ N}$$

Localized Pressure

$$\text{Pressure: } P = \frac{F}{A} \quad (18)$$

$$P = \frac{F}{A} = \frac{27}{0.0001\text{m}^2} = 270 \text{ kPa}$$

Thickness Required

$$\text{Hoop's Stress Thin Wall: } \sigma_{\theta} = \frac{P * r}{t} \quad (19)$$

$$t = \frac{P * \frac{D_o}{2}}{\sigma_{allowable}} = \frac{270 \text{ kPa} * \frac{7.25 \text{ in}}{2}}{12.5 \text{ MPa}} = 0.078 \text{ in}$$

Dynamic Pressure

$$q = \left(\frac{1}{2}\right) \rho V^2 \quad (20)$$

Appendix B: Detailed Simulation Results & Computer Programs

The following simulations shown in this appendix are tested on a bin wall thickness of 0.25 in. The simulated results go in order of Factor of Safety, Displacement, and Vonmises respectively in order. The mesh creation done was set to peak Fine before running solidworks simulation. The impact angle range goes in order from 90°, 60°, 45°, and 30°. The contact area of force applied is 1 cm² to the surface of the interior bin. The respective forces applied to the following impact angles are as follows:

90°: $F_x = 71.6 \text{ N}$ $F_y = 0 \text{ N}$

60°: $F_x = 62 \text{ N}$ $F_y = 35.8 \text{ N}$

45°: $F_x = 50.6 \text{ N}$ $F_y = 50.6 \text{ N}$

30°: $F_x = 35.8 \text{ N}$ $F_y = 62 \text{ N}$

```

%% Impact Forces on Bin Wall and Required Thickness
% All parameters are adjustable
clc; clear;
% Parameters
% Mass and velocity
m = 0.2;           % mass (kg)
v_mph = 8;         % speed (mph)
v = v_mph * 0.44704; % speed (m/s)
theta_deg = 22.1;  % impact angle (degrees)
phi = 90 - theta_deg; % angle relative to wall (degrees)
tc = 0.01;         % contact time (s)
A = 1e-4;          % contact area (m^2) (1 cm^2)
sigma_u = 50e6;    % ultimate tensile strength (Pa)
SF = 4;            % safety factor
D_in = 7.25;       % bin outer diameter (in)
D = D_in * 0.0254; % diameter (m)

```

```

sigma_allow = sigma_u / SF; % allowable stress (Pa)
vn = v * cosd(phi);         % normal velocity (m/s)
vt = v * sind(phi);         % tangential velocity (m/s)
dp_n = m * vn;              % normal momentum (kg·m/s)
dp_t = m * vt;              % tangential momentum (kg·m/s)
Fn = dp_n / tc;              % normal impact force (N)
Ft = dp_t / tc;              % tangential impact force (N)
P = Fn / A;                  % localized pressure (Pa)
r = D / 2;                   % mean radius (m)
t = (P * r) / sigma_allow;   % required thickness (m)
t_in = t / 0.0254;           % thickness (in)
fprintf('--- Impact Analysis Results ---\n');
fprintf('Speed: %.2f m/s\n', v);
fprintf('Normal velocity: %.2f m/s\n', vn);
fprintf('Tangential velocity: %.2f m/s\n\n', vt);
fprintf('Normal force: %.2f N\n', Fn);
fprintf('Tangential force: %.2f N\n\n', Ft);
fprintf('Localized pressure: %.2f kPa\n', P / 1e3);
fprintf('Allowable stress: %.2f MPa\n\n', sigma_allow / 1e6);
fprintf('Required wall thickness:\n');
fprintf('  %.4f m\n', t);
fprintf('  %.3f in\n', t_in);

```

Figure B-1: Matlab Code for Applying Parameters for Bin's Thickness

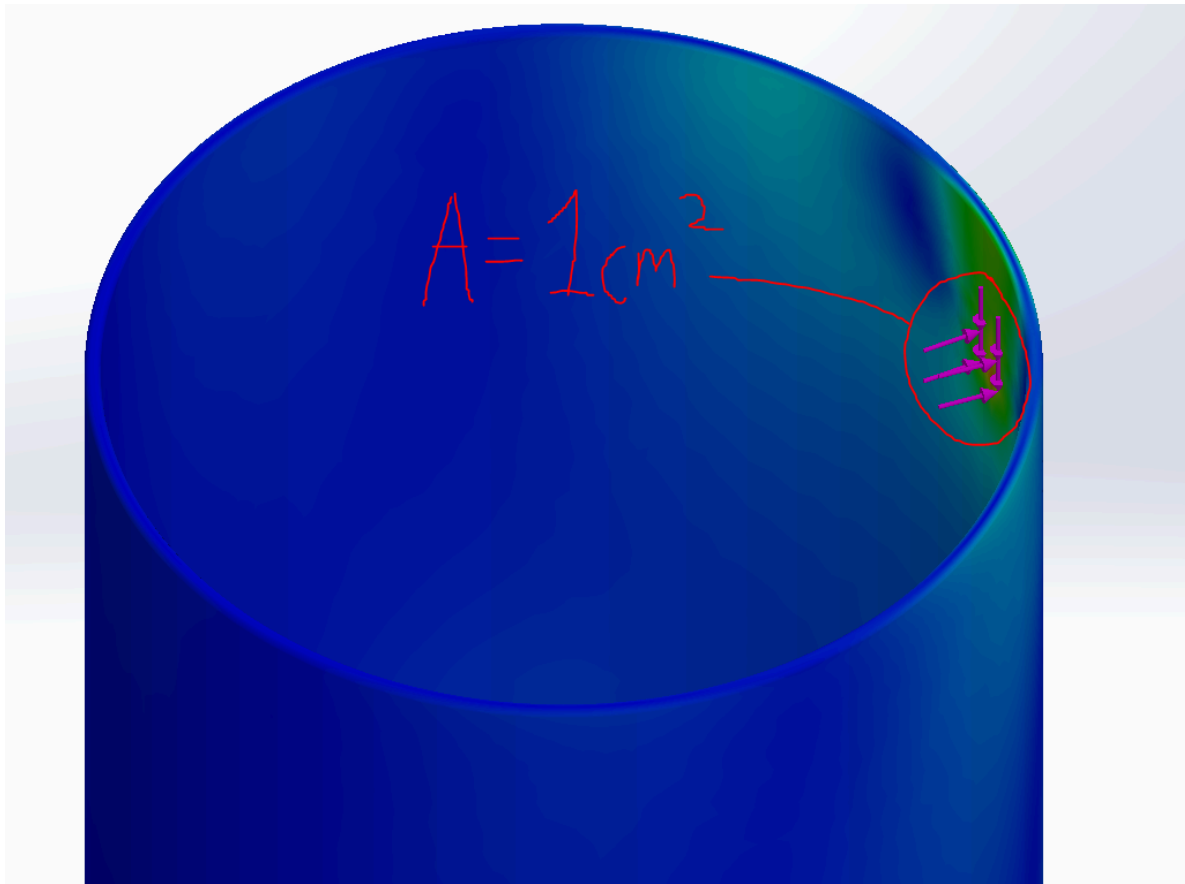


Figure B-2: Visual Parameters for Force Application

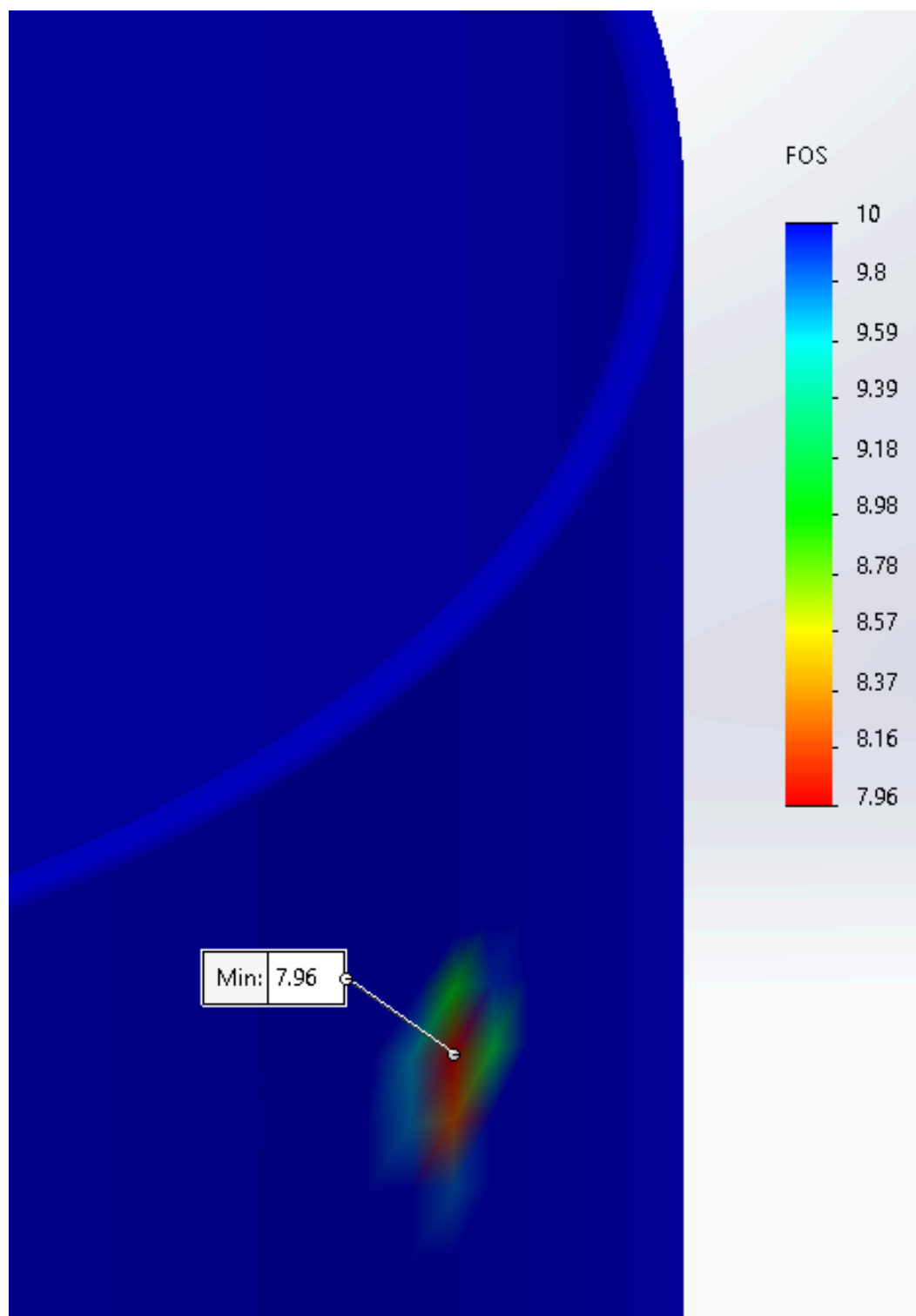


Figure B-3: FOS of Bin at Impact Angle 90°

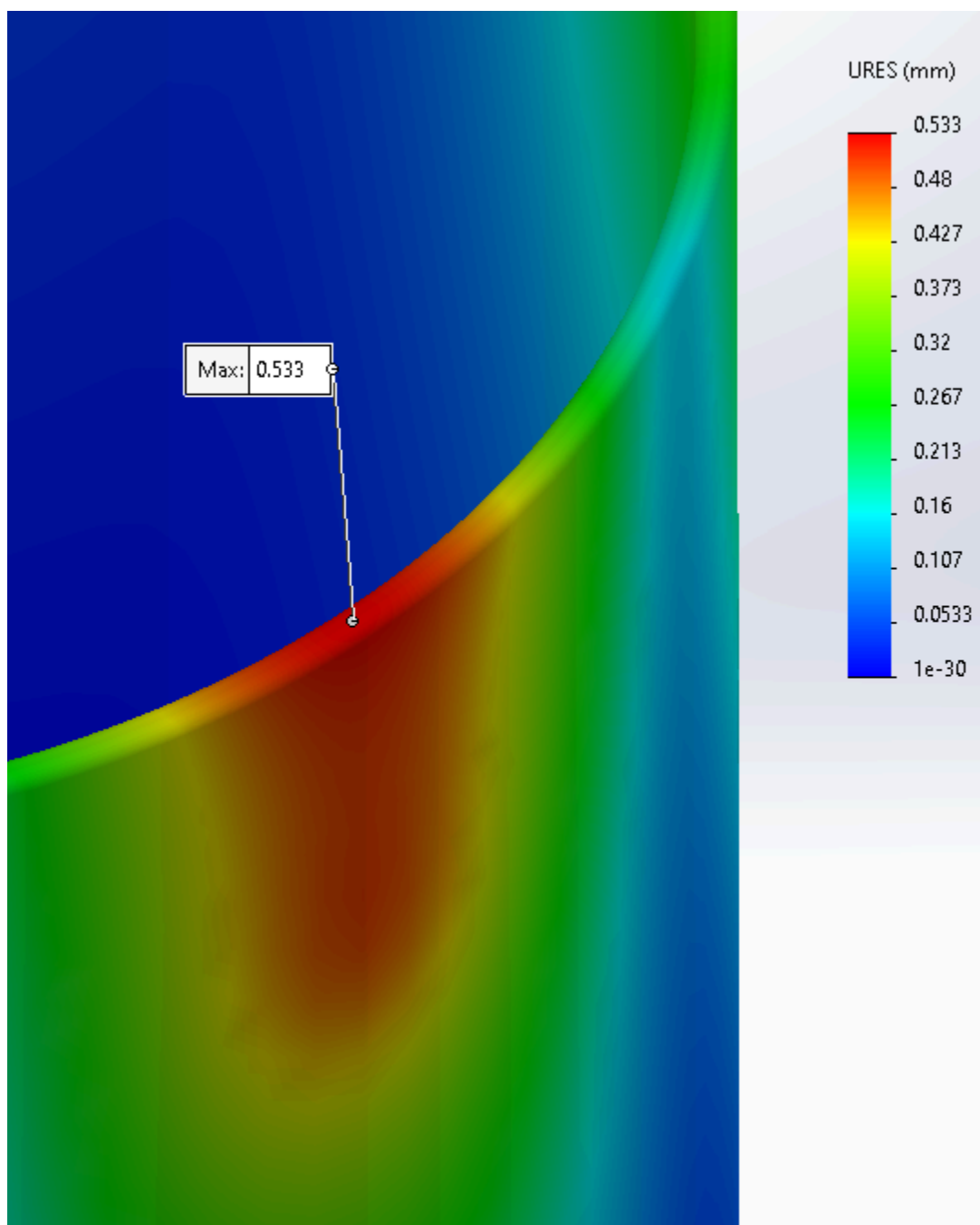


Figure B-4: Displacement of Bin at Impact Angle 90°

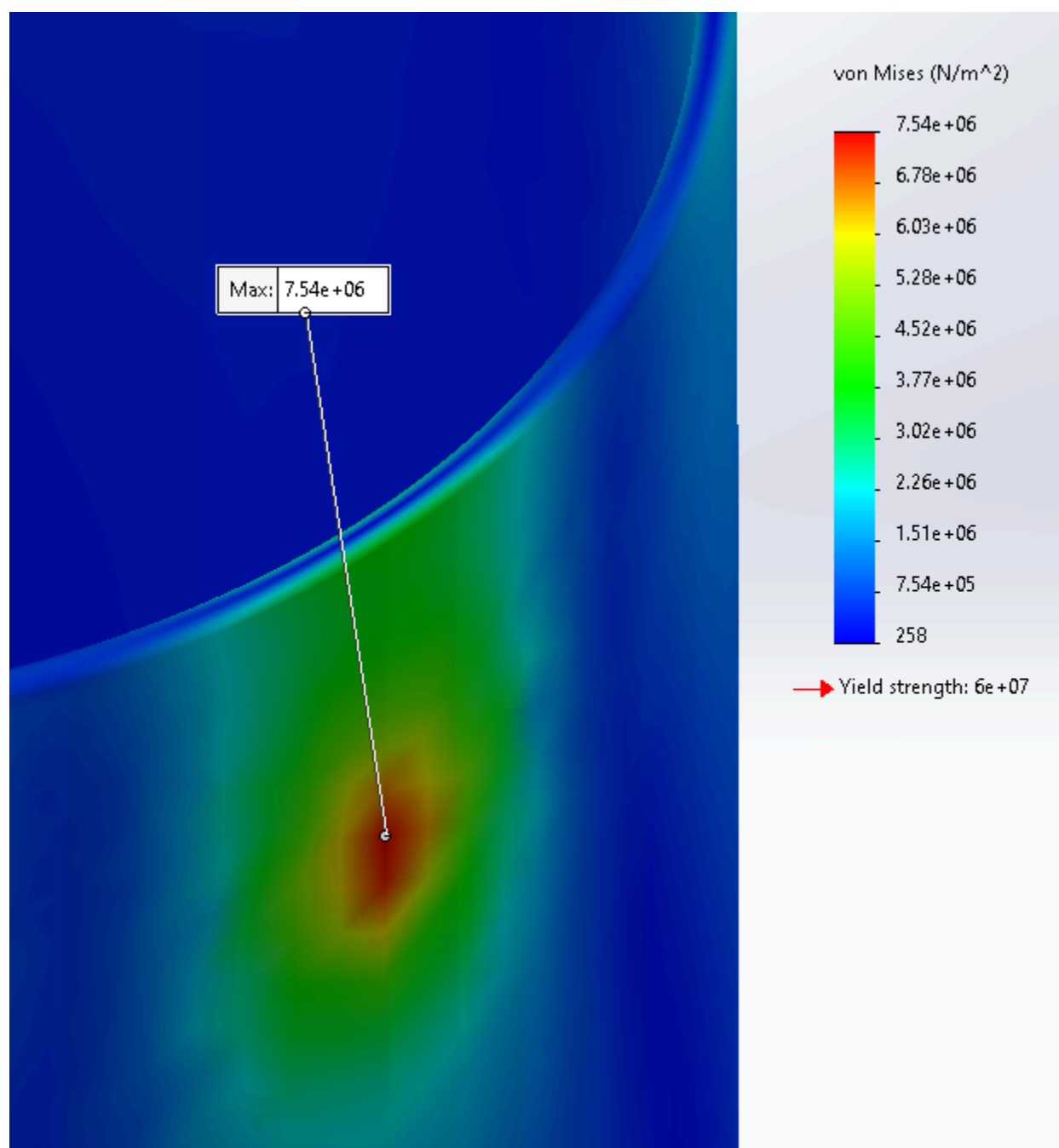


Figure B-5: Von Mises of Bin at Impact Angle 90°

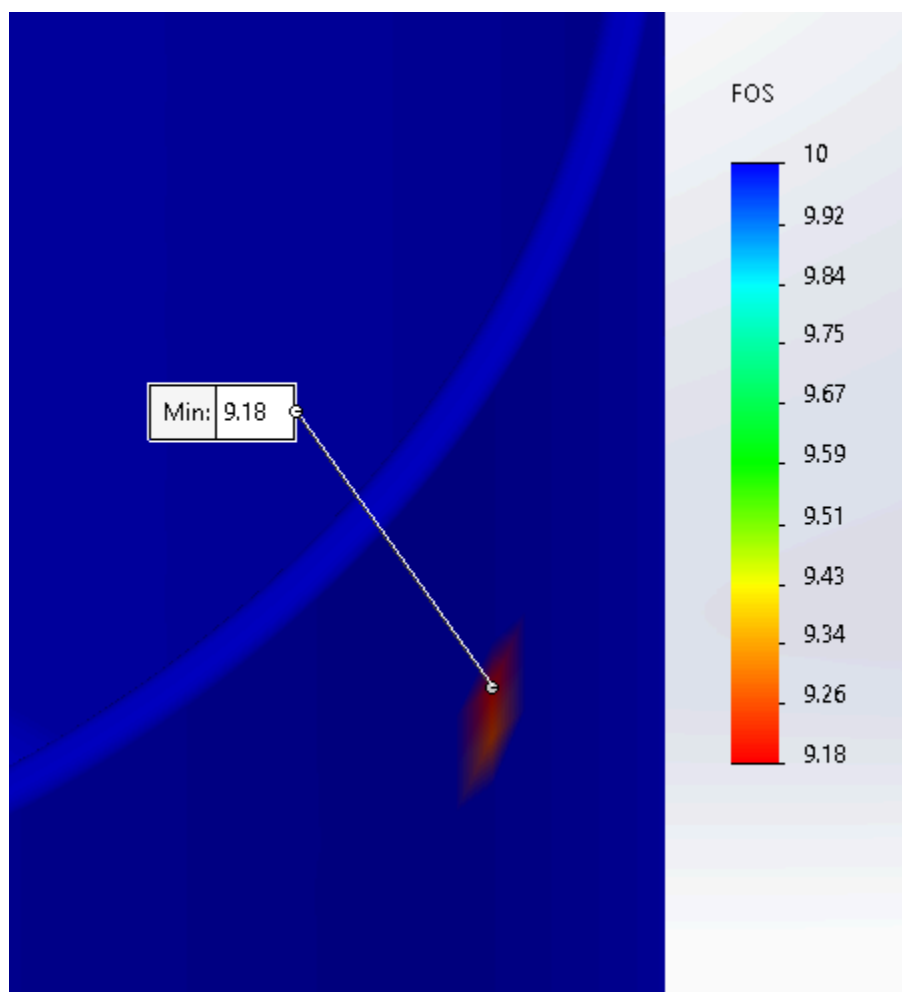


Figure B-6: FOS of Bin at Impact Angle 60°

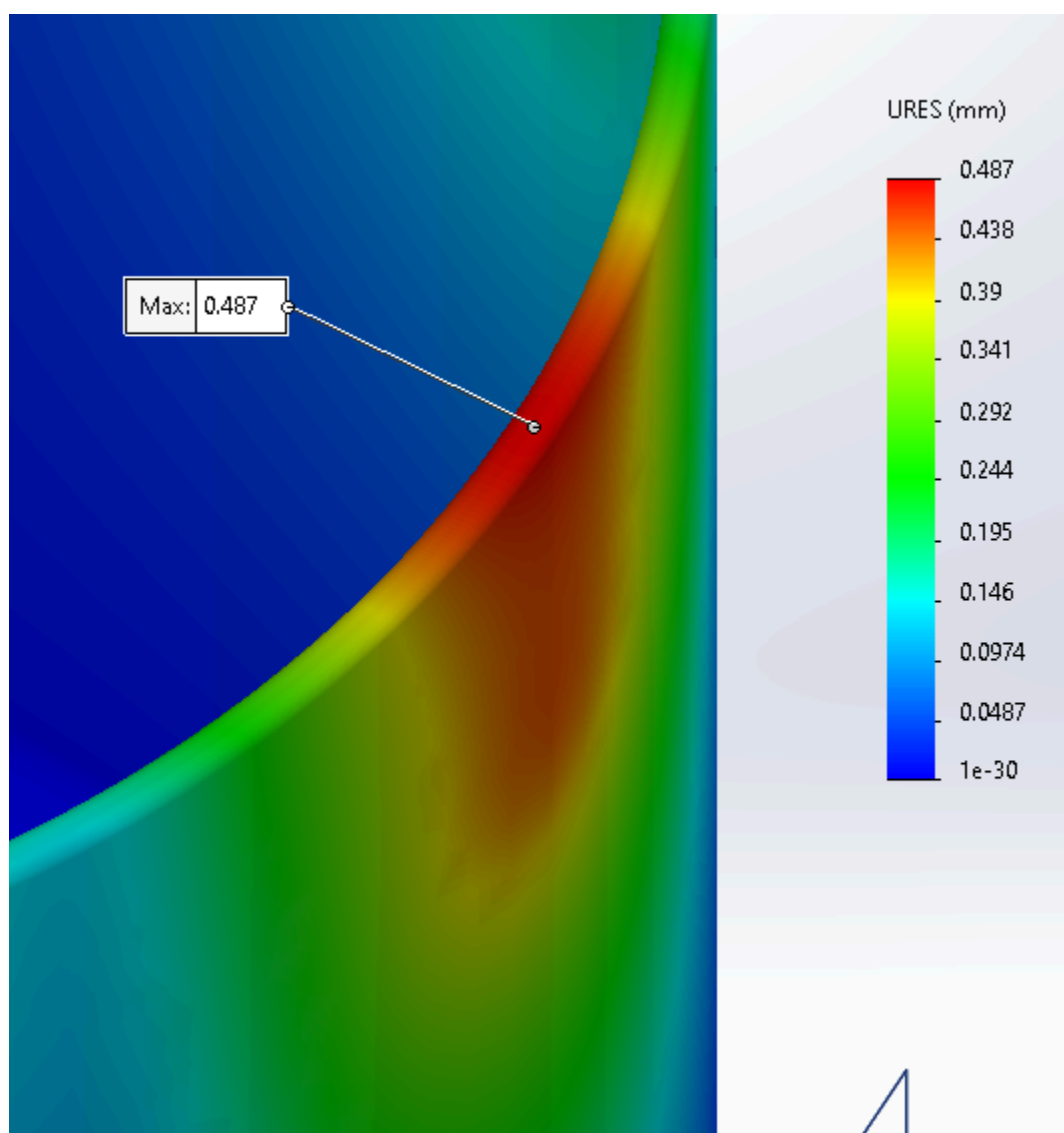


Figure B-7: Displacement of Bin at Impact Angle 60°

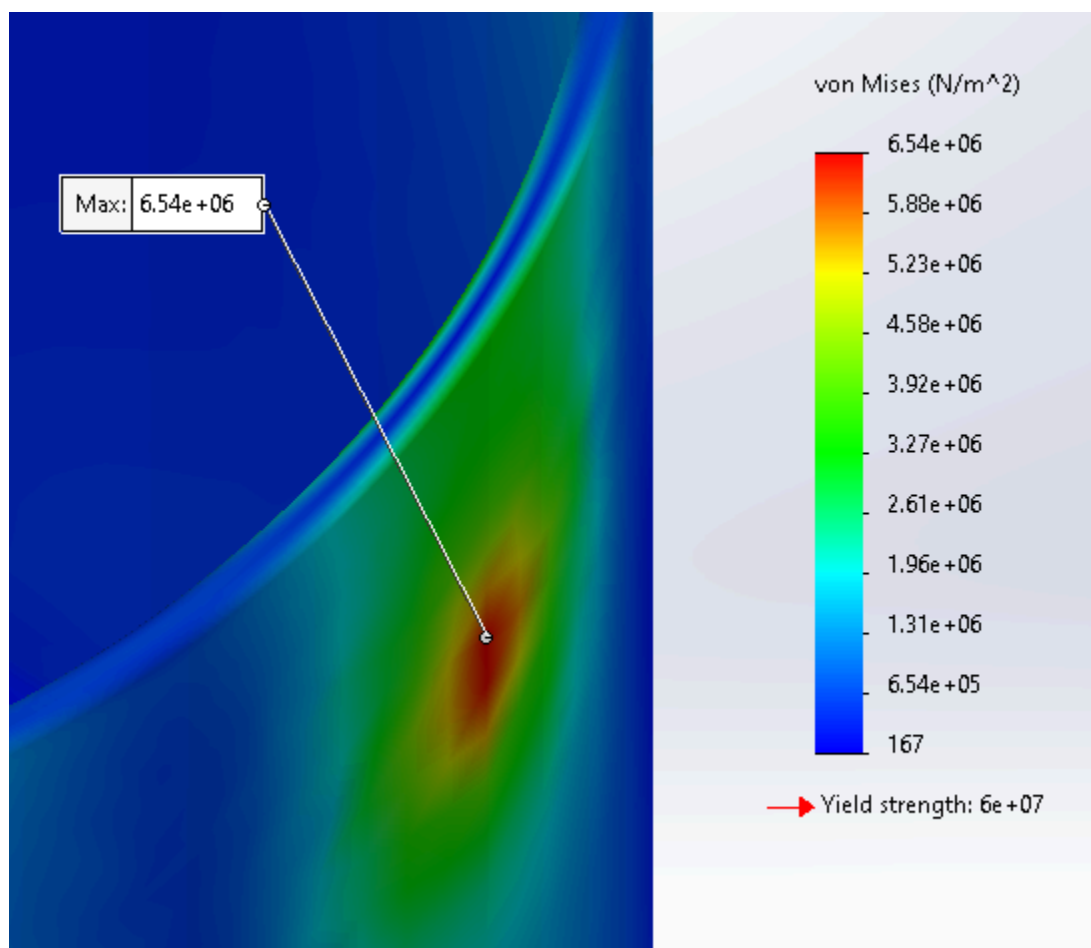


Figure B-8: Von Mises of Bin at Impact Angle 60°

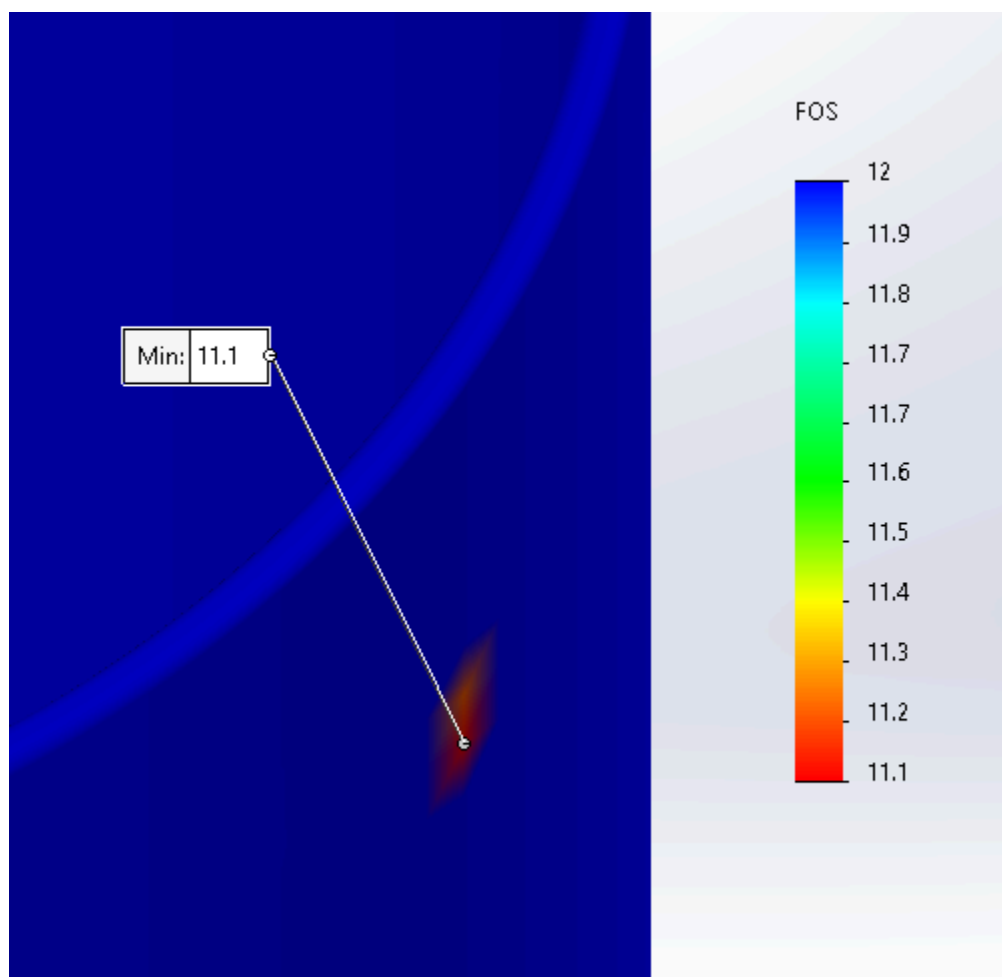


Figure B-9: FOS of Bin at Impact Angle 45°

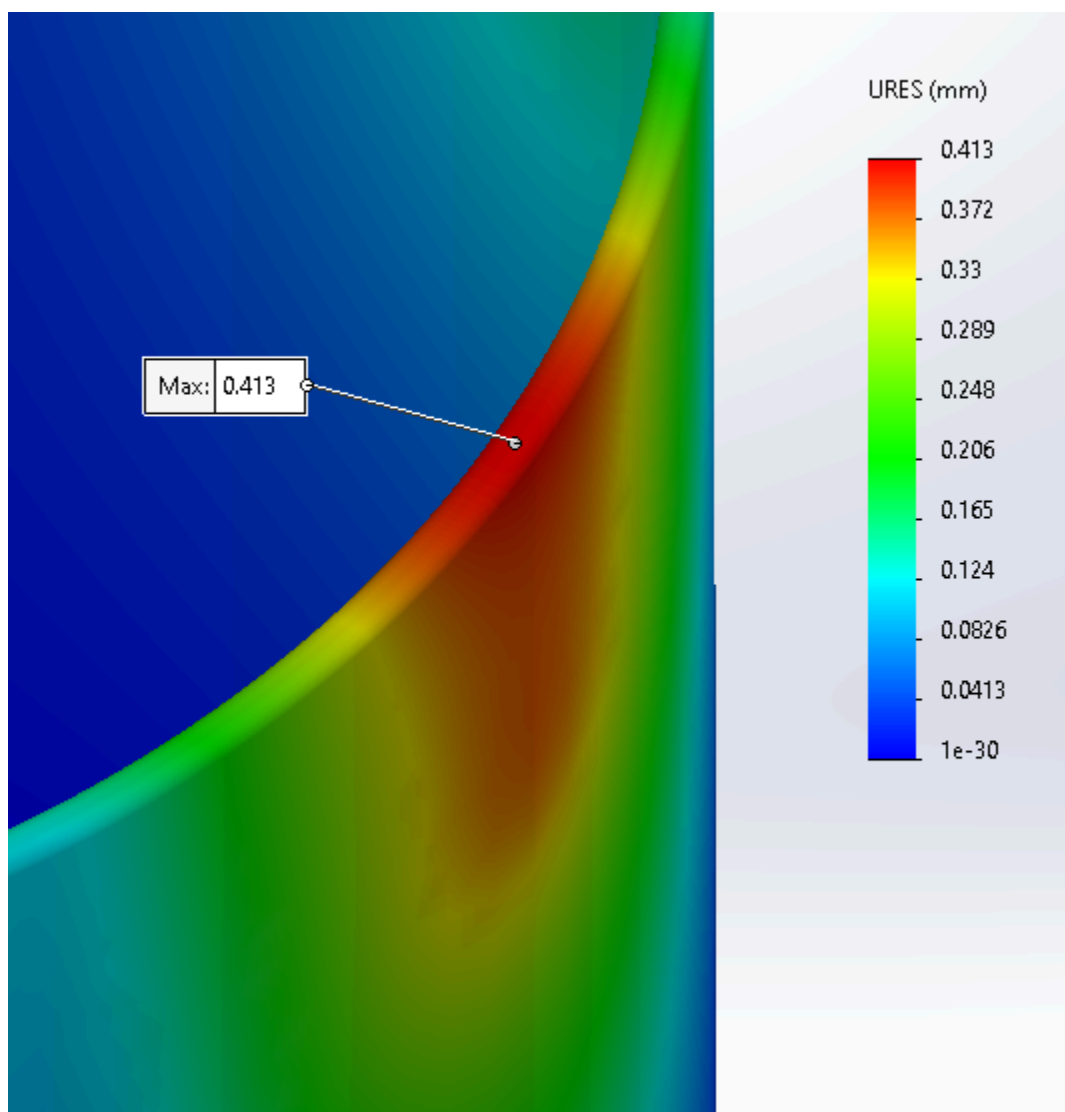


Figure B-10: Displacement of Bin at Impact Angle 45°

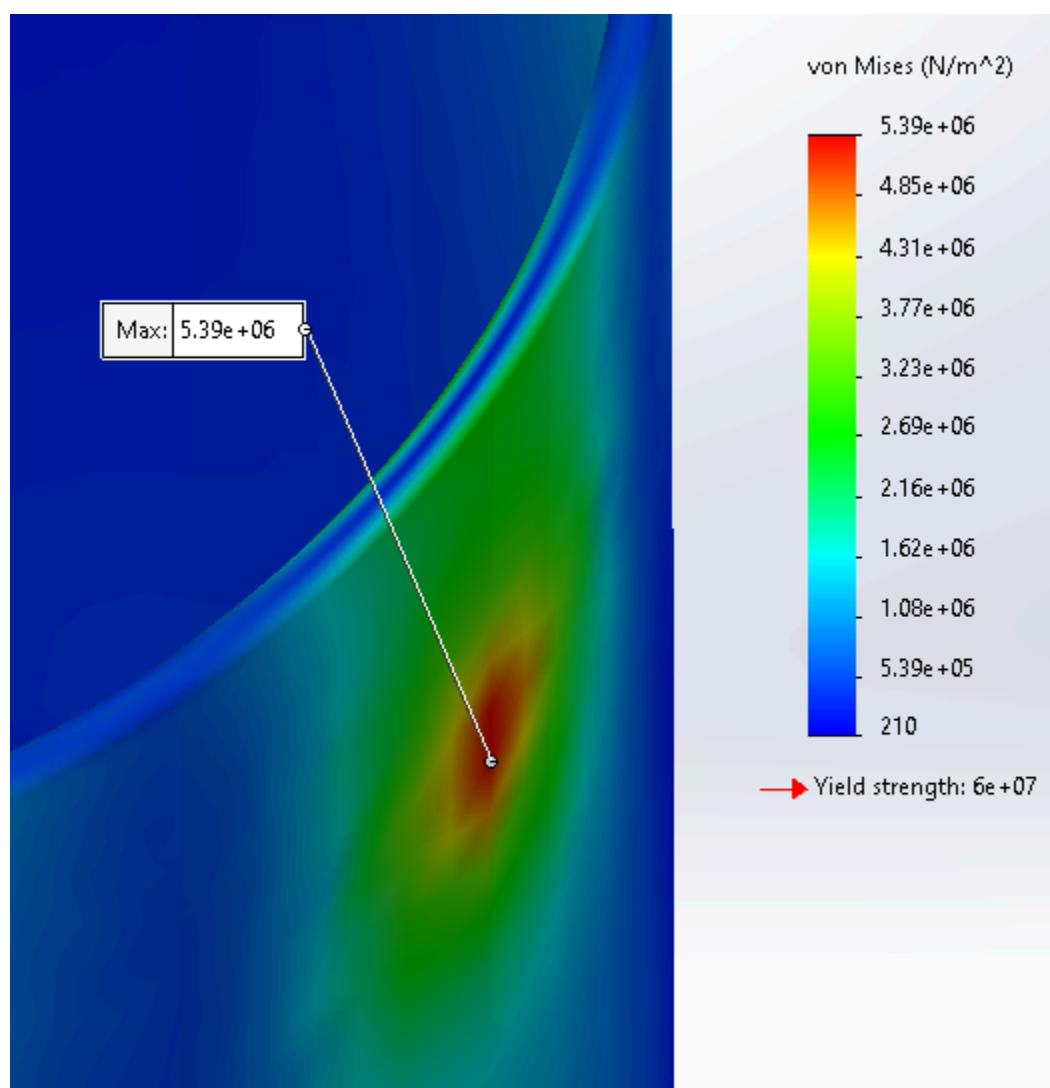


Figure B-11: Von Mises of Bin at Impact Angle 45°

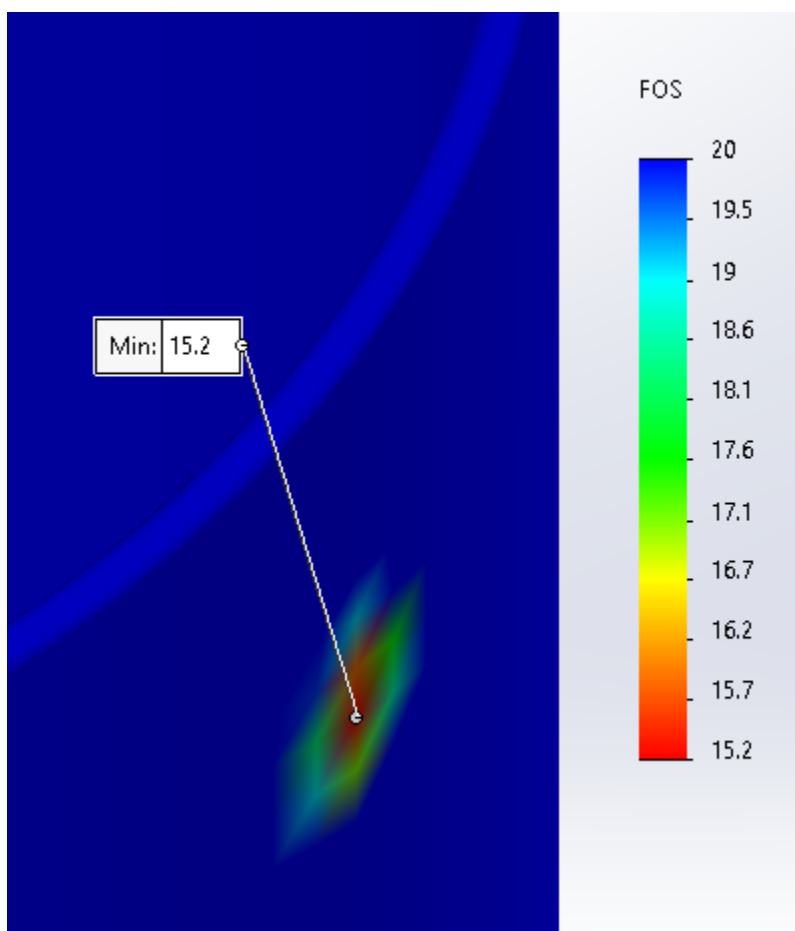


Figure B-12: FOS of Bin at Impact Angle 30°

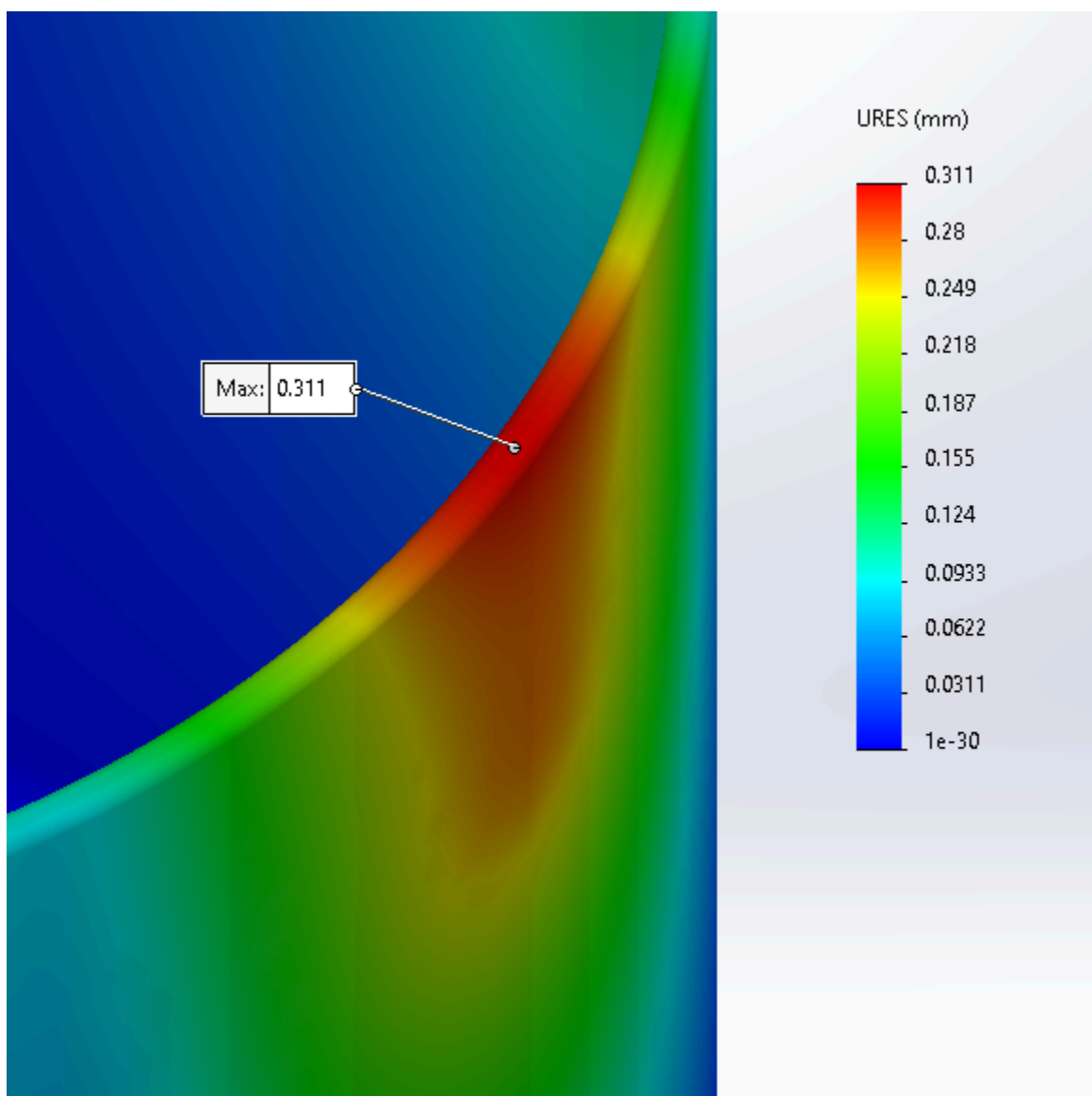


Figure B-13: Displacement of Bin at Impact Angle 30°

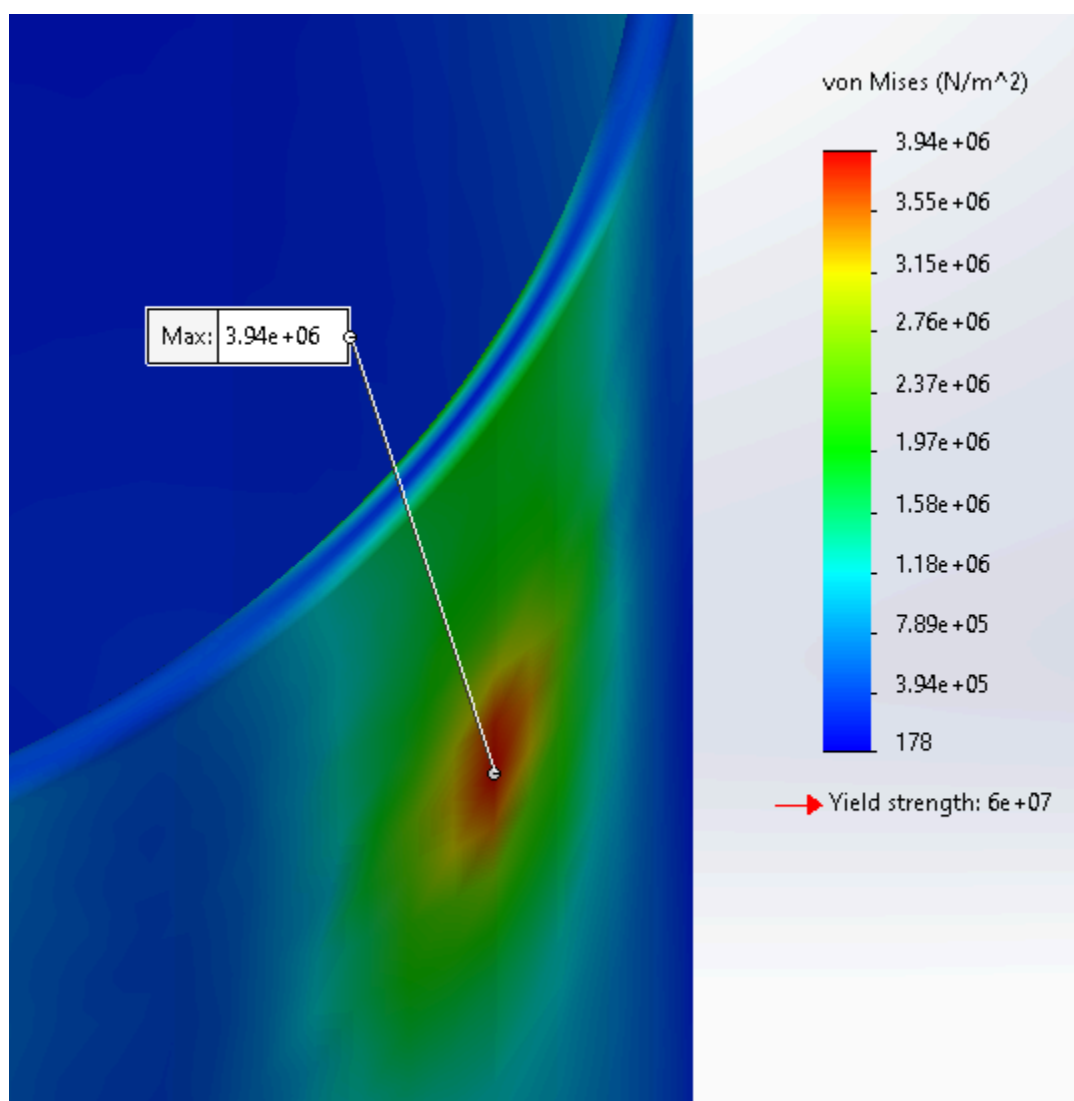


Figure B-14: Von Mises of Bin at Impact Angle 30°

Appendix C: Detailed Design Drawings

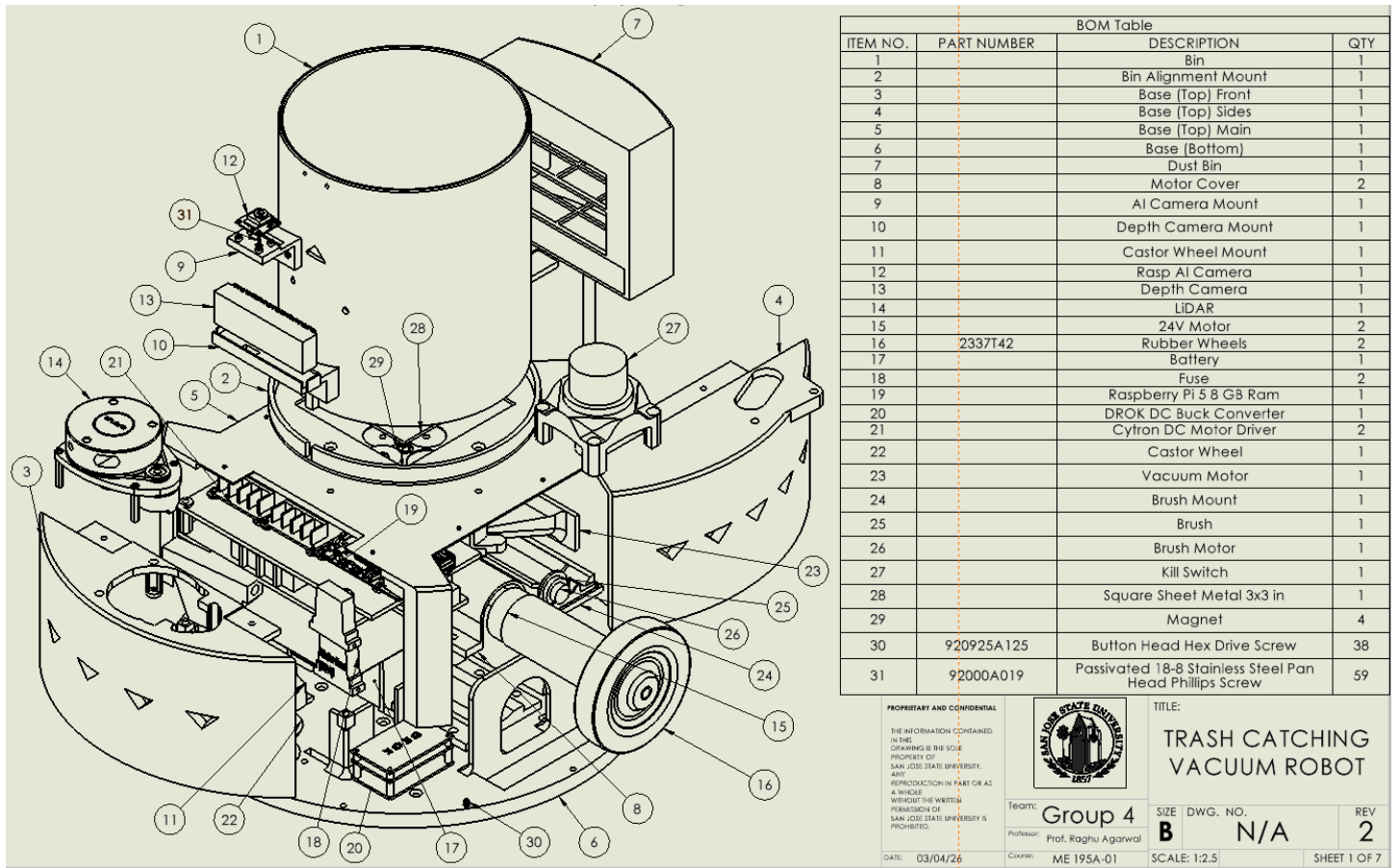


Figure C-1: CAD Drawing of Assembly (w/ Components)

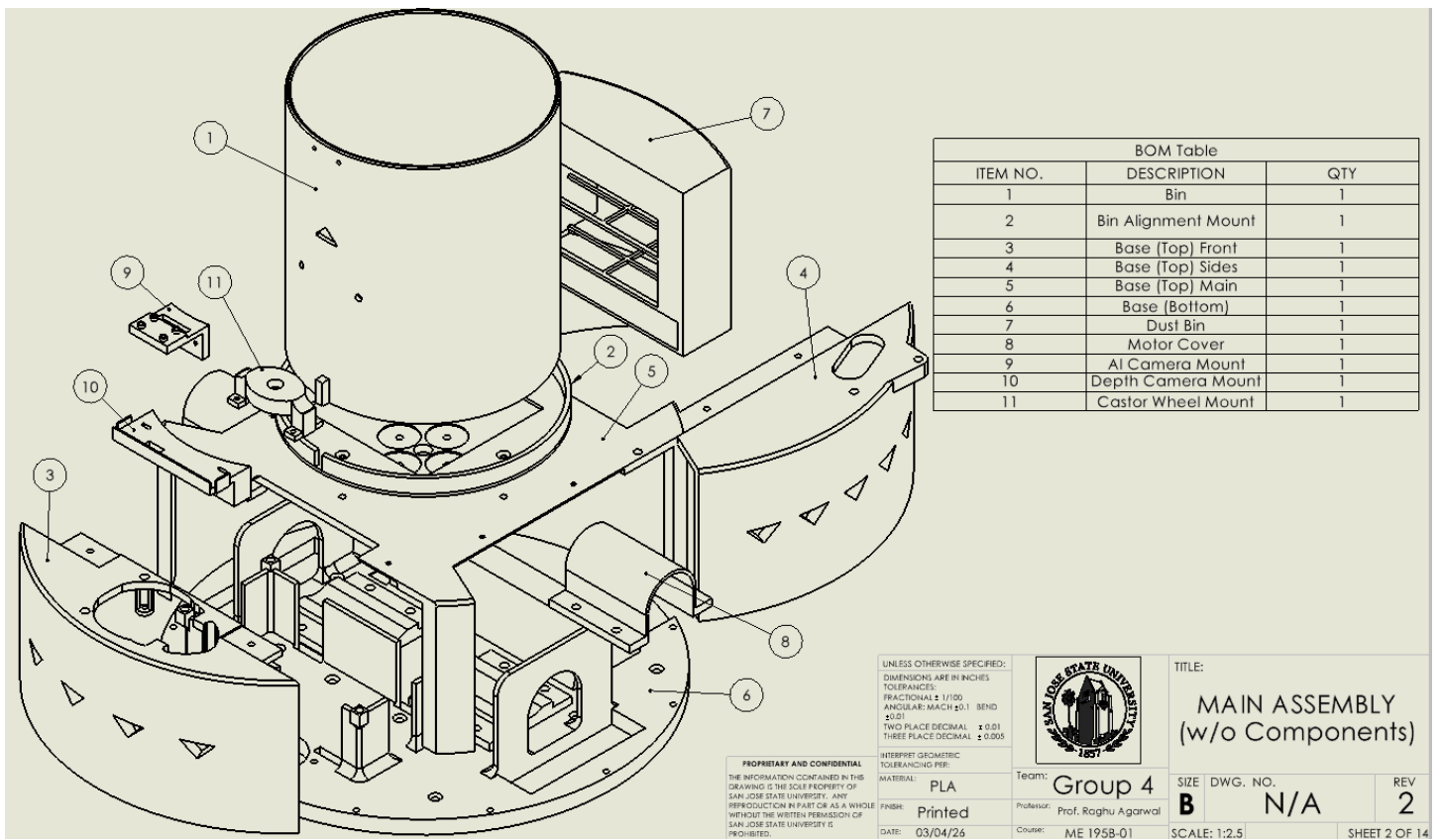


Figure C-2: CAD Drawing of Assembly (w/o Components)

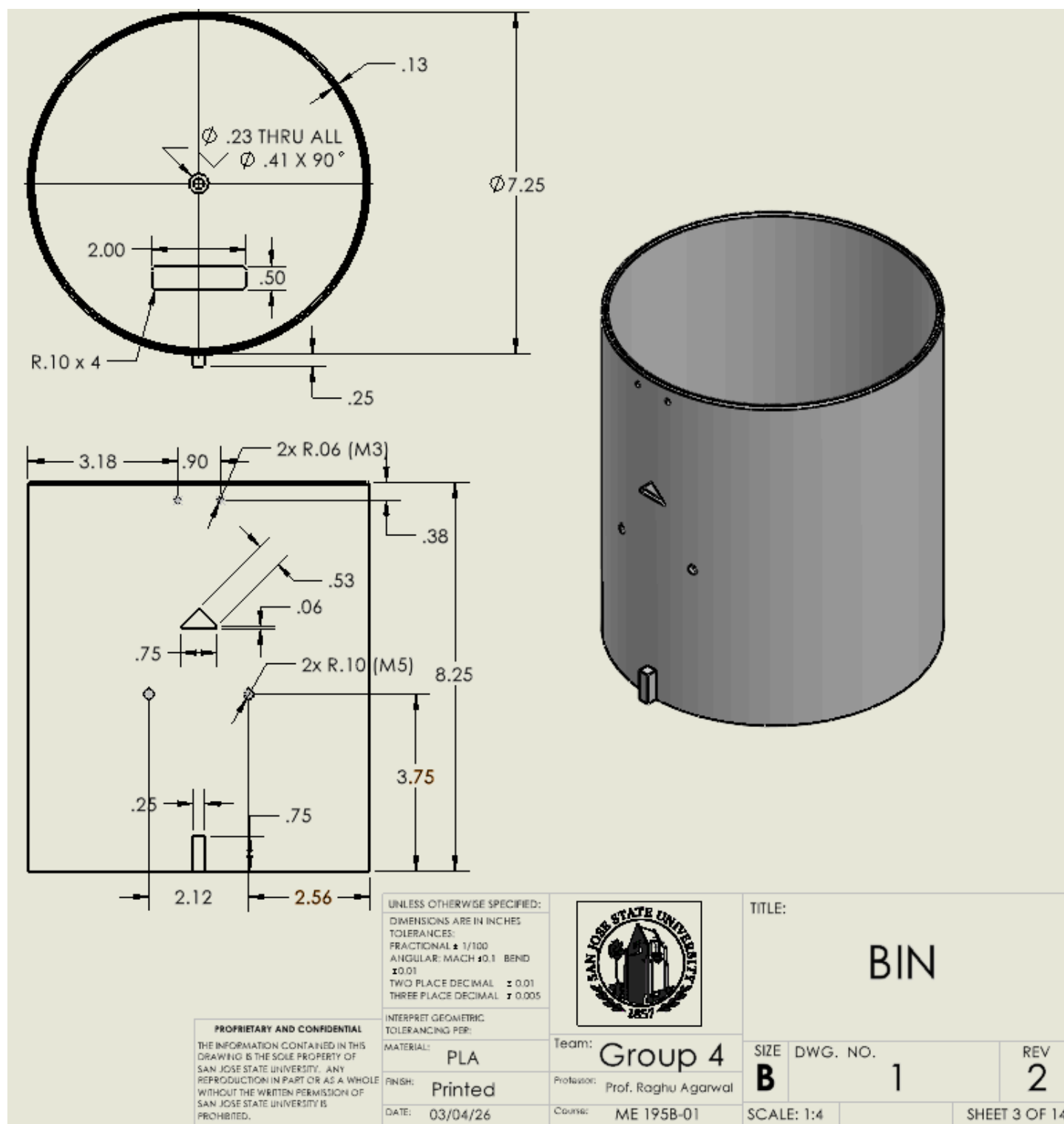


Figure C-3: CAD Drawing of Bin

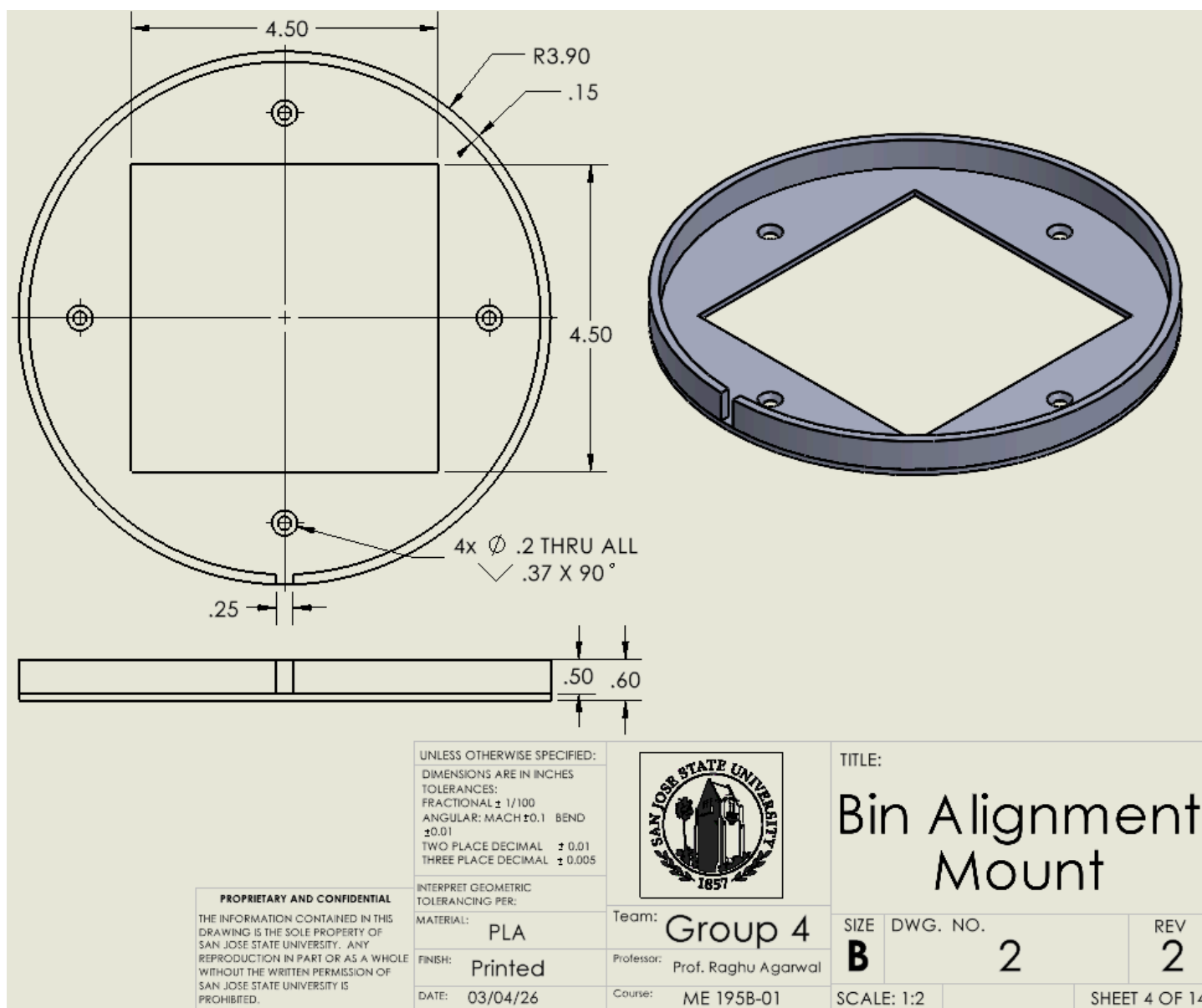


Figure C-4: CAD Drawing of Bin Alignment Mount

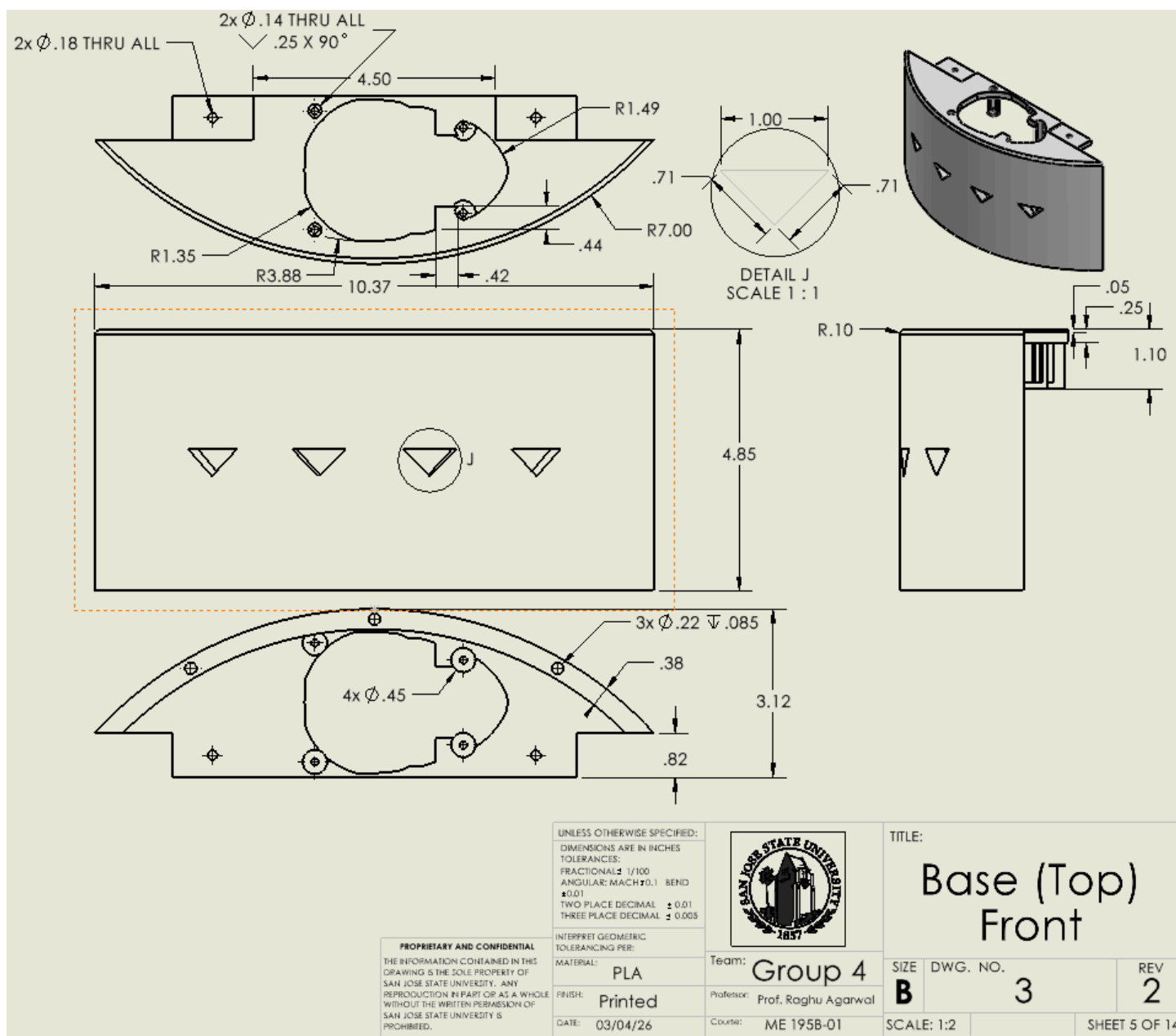


Figure C-5: CAD Drawing of Robot Base (Top) Front

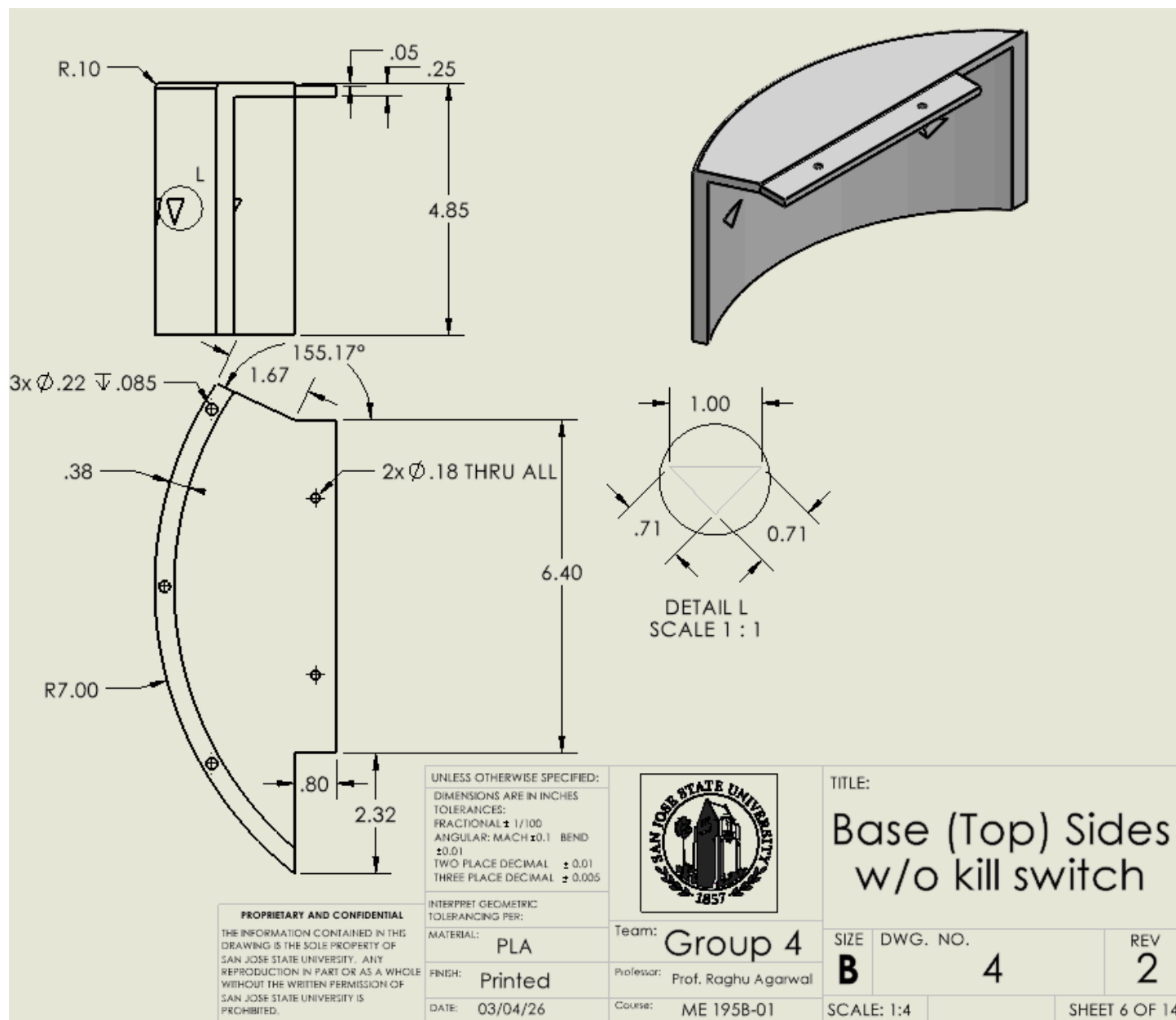


Figure C-6: CAD Drawing of Robot Base (Top) Sides w/o Kill Switch

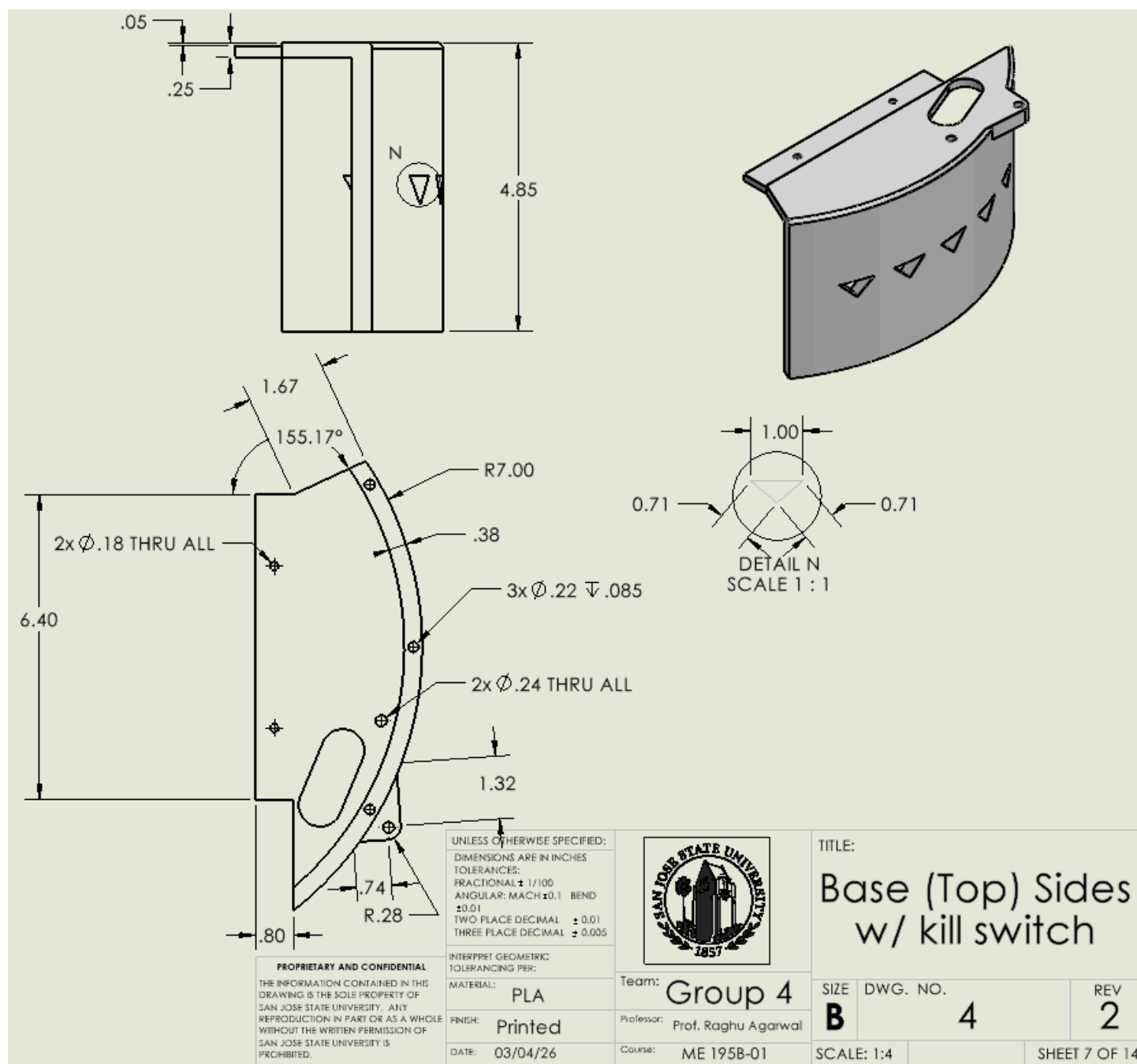


Figure C-7: CAD Drawing of Robot's Base (Top) Sides w/ Kill Switch

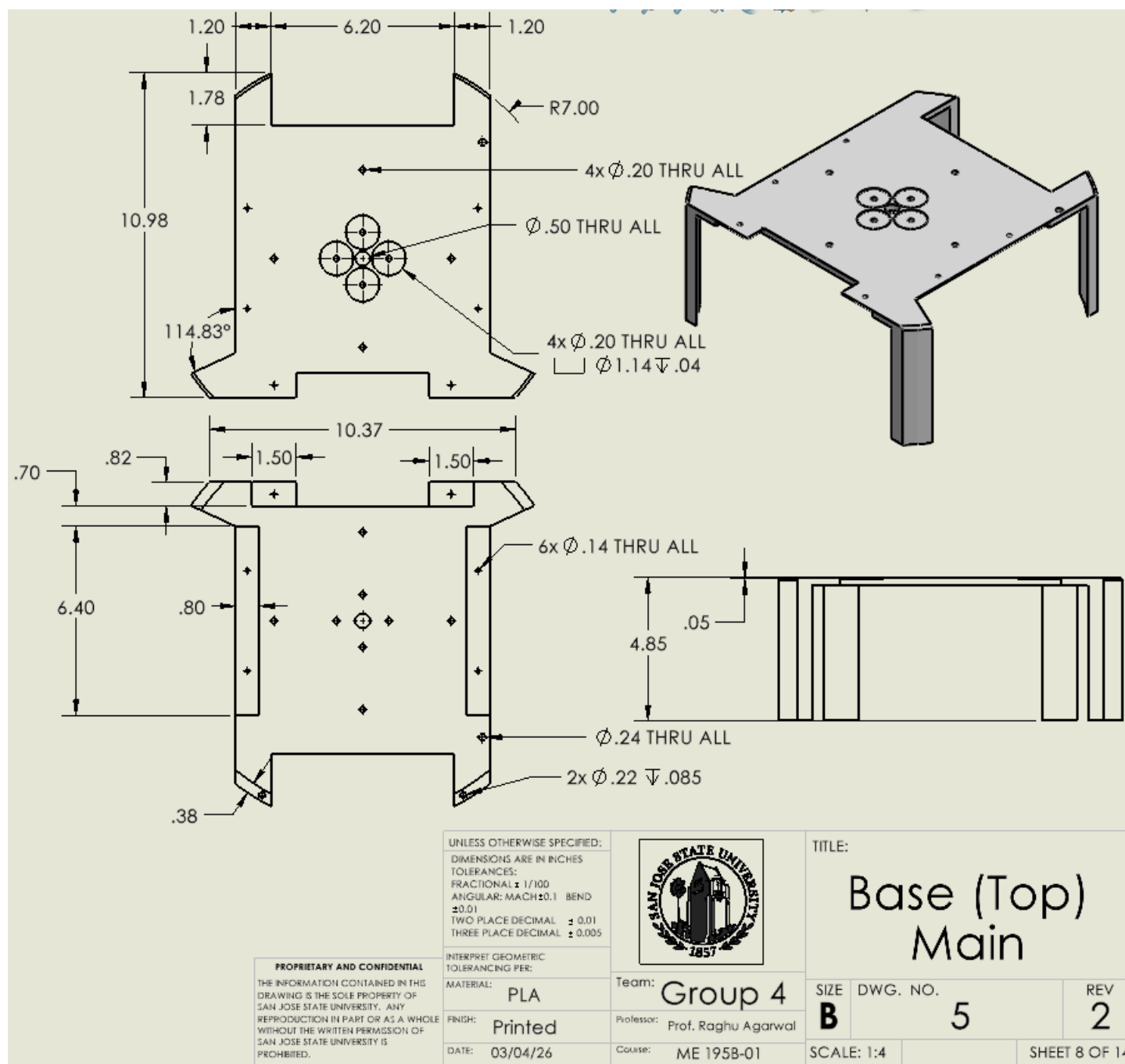


Figure C-8: CAD Drawing of Robot's Base (Top) Main

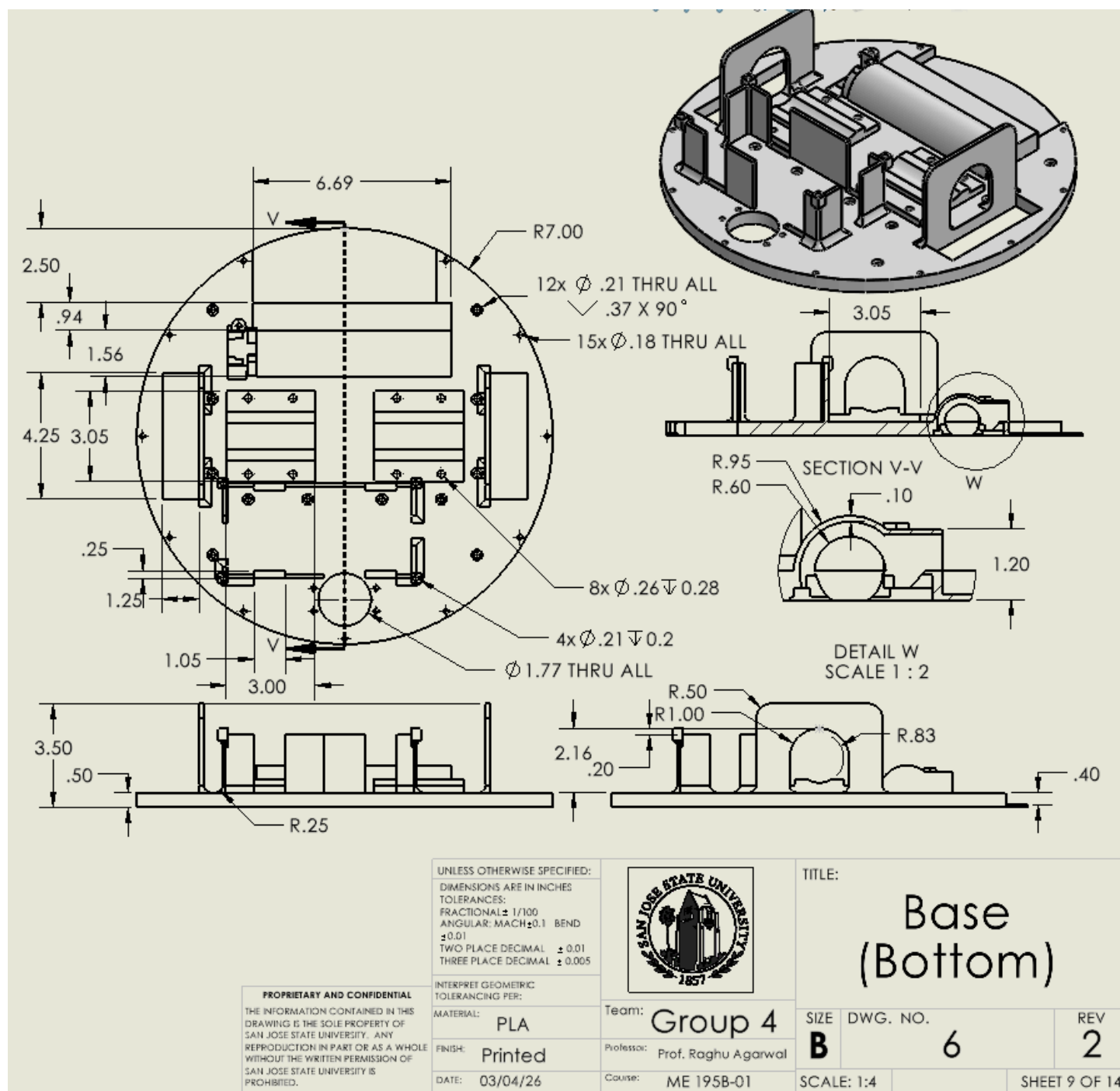


Figure C-9: CAD Drawing of Robot's Base (Bottom)

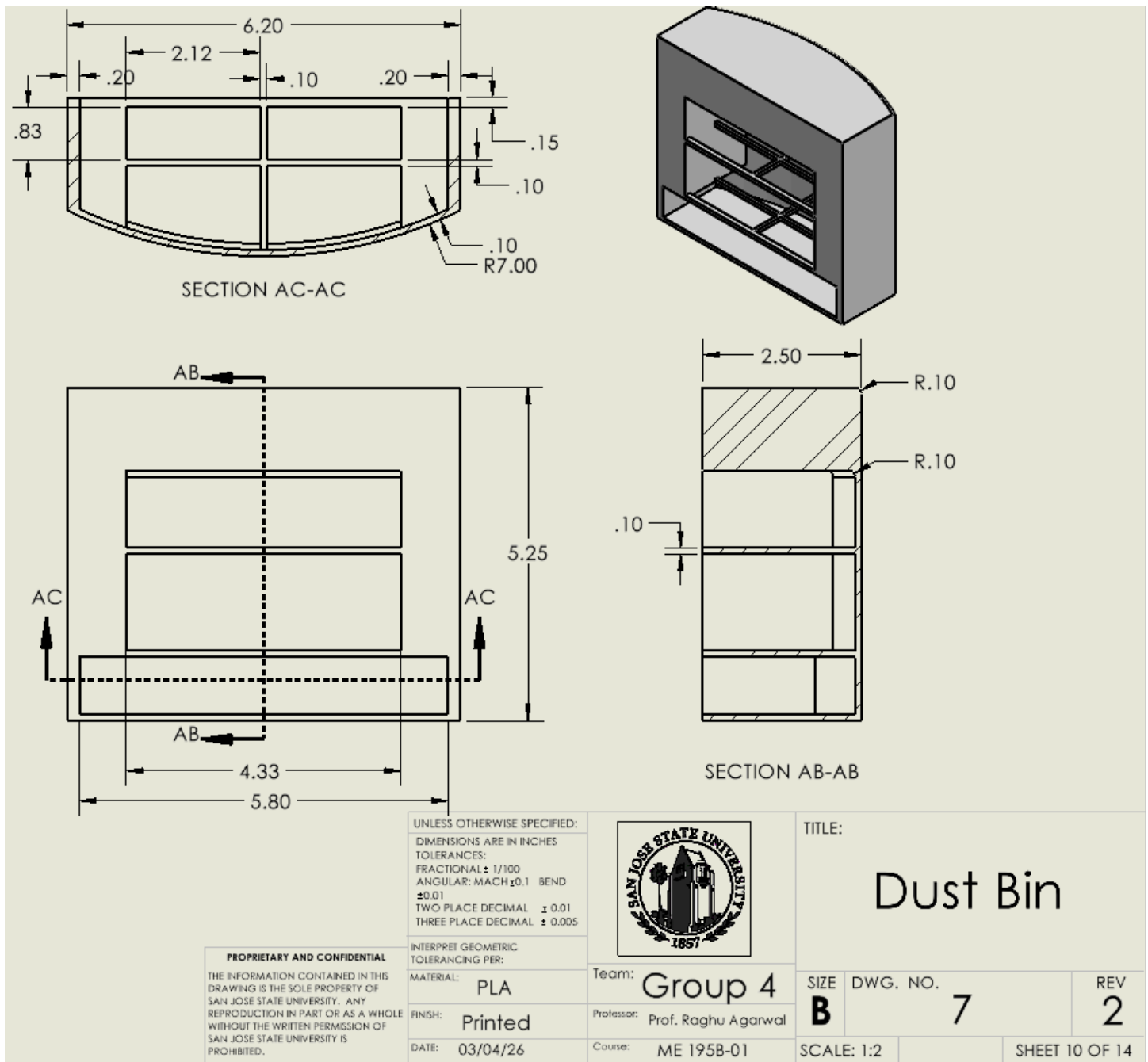


Figure C-10: CAD Drawing of Dust Bin

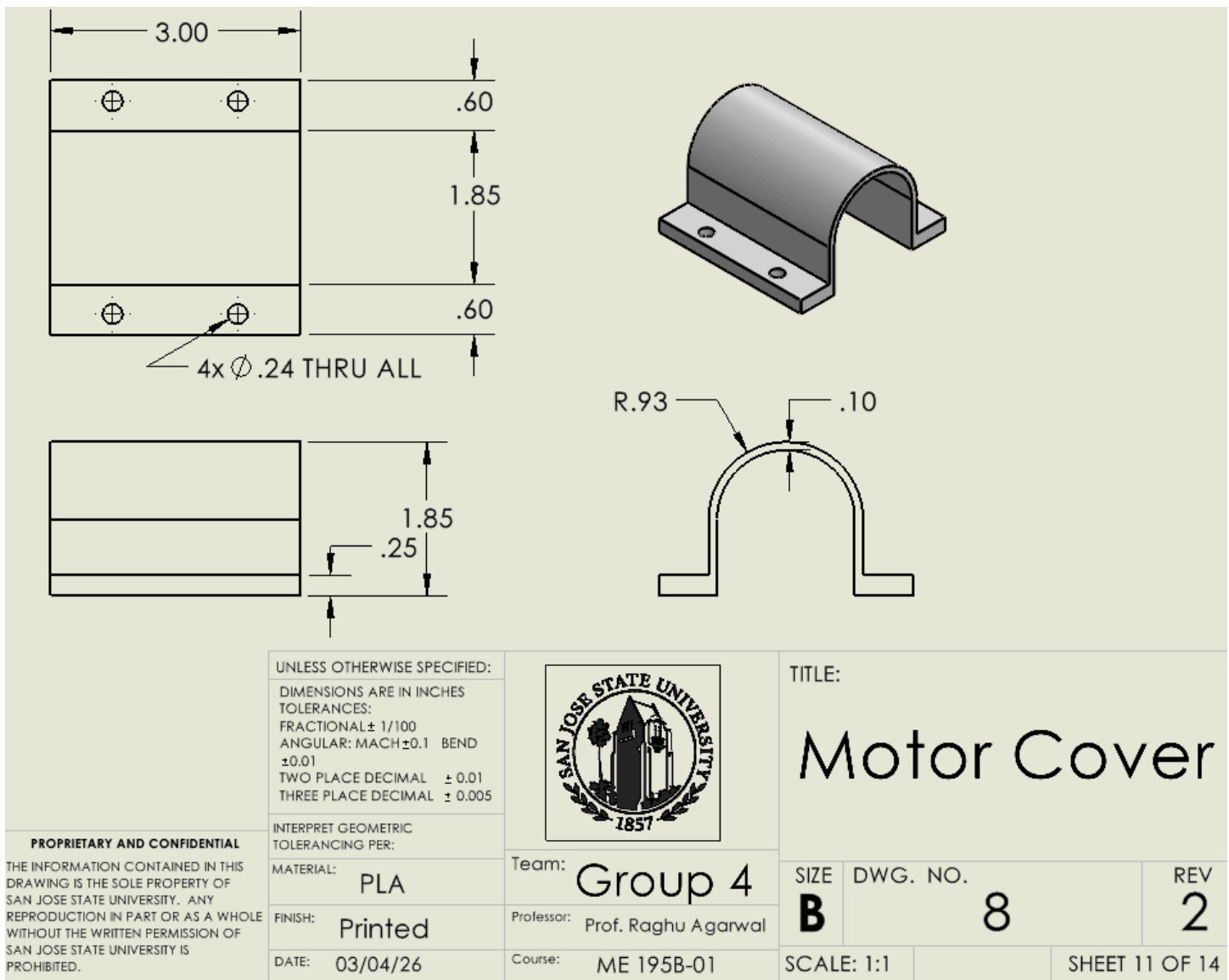


Figure C-11: CAD Drawing of Motor Cover

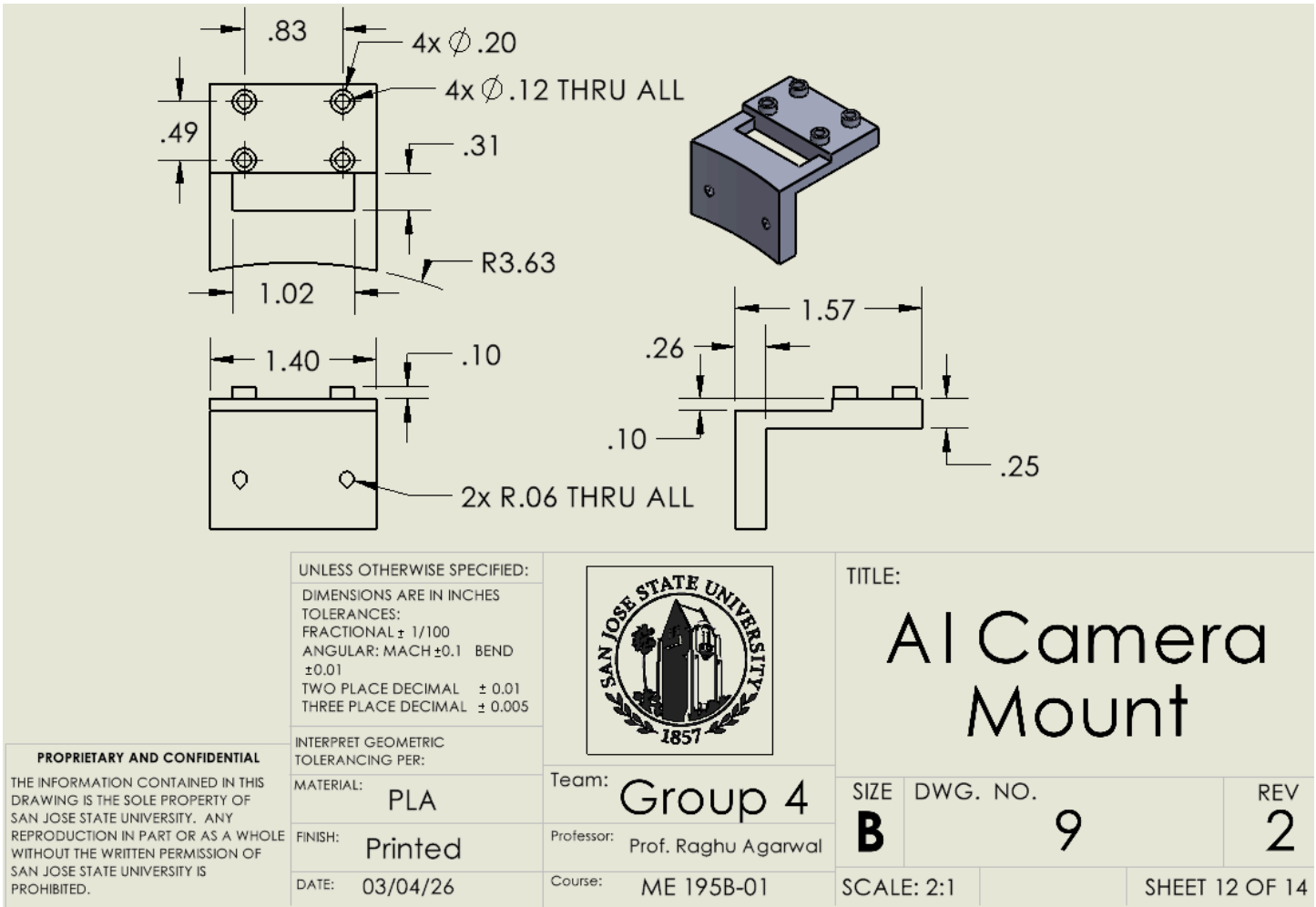


Figure C-12: CAD Drawing of AI Camera Mount

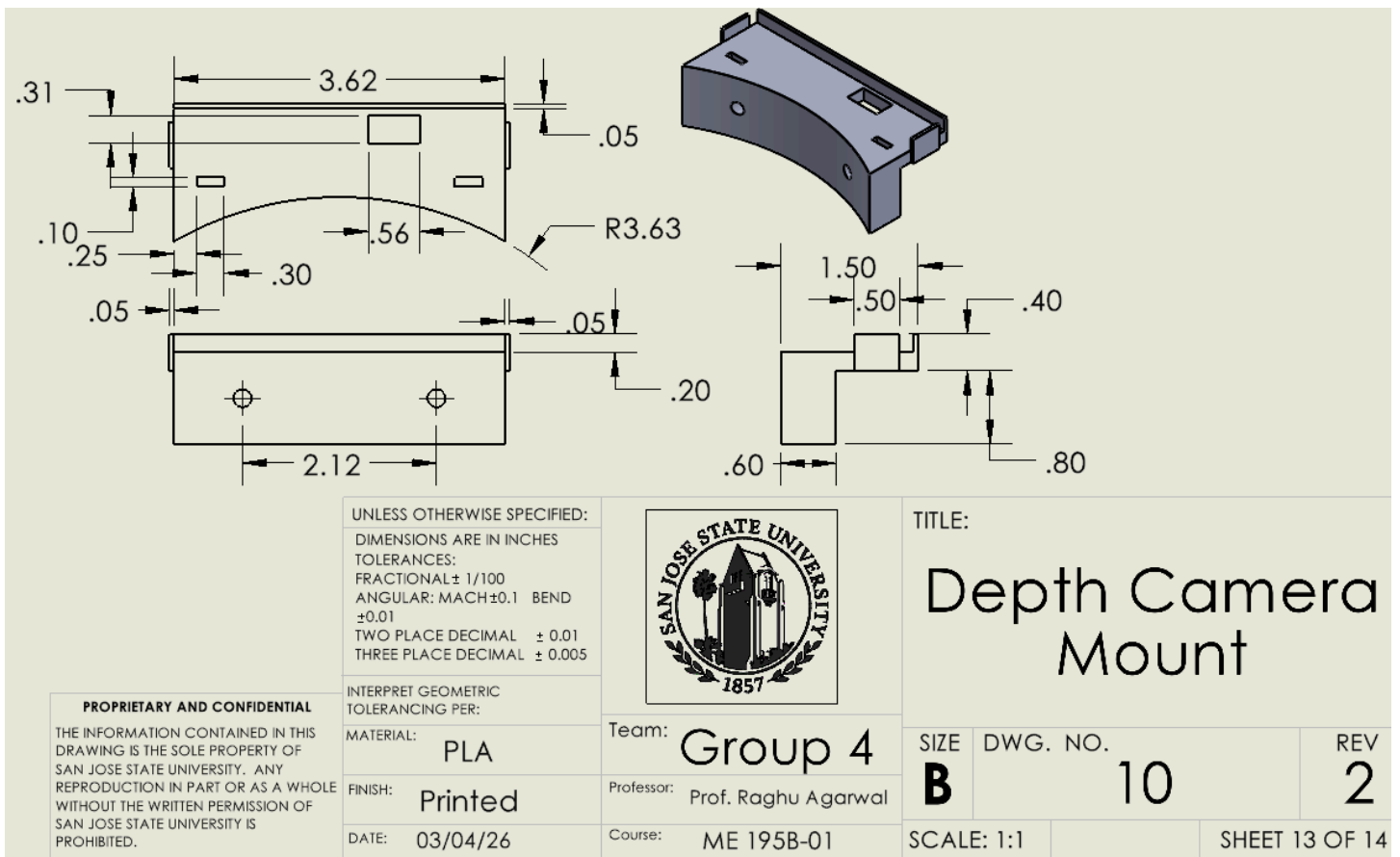


Figure C-13: CAD Drawing of Depth Camera Mount

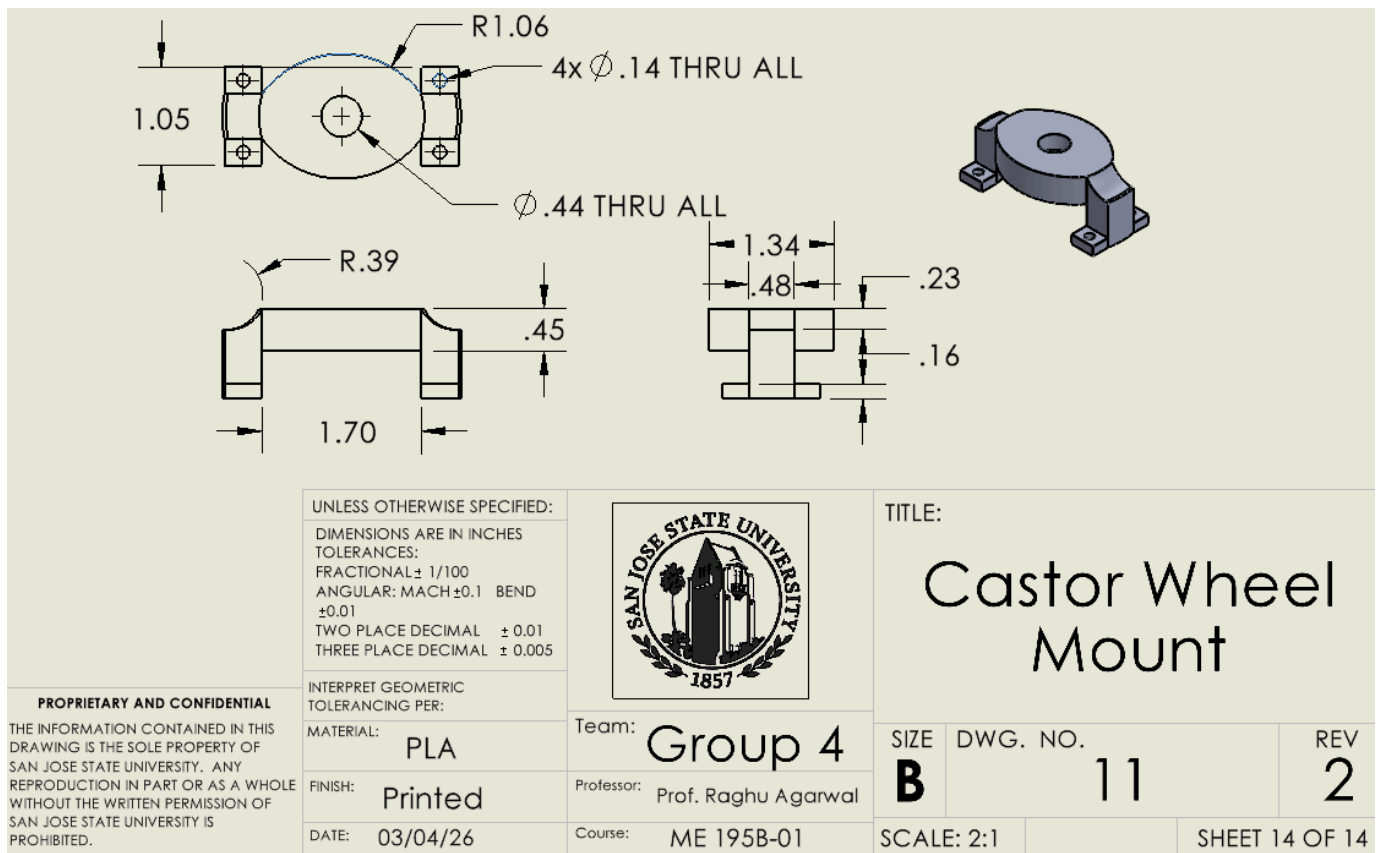


Figure C-14: CAD Drawing of Castor Wheel Mount

Appendix D: Excerpts from Datasheets of Components

Voltage: DC24.0V max power 150W											
Reduction ratio	1:3.7	1:5	1:13	1:25	1:49	1:67	1:92	1:125	1:246	1:336	1:625
number of stages	1	1	2	2	3	3	3	3	4	4	4
No-load current (mA)	≤700	≤700	≤700	≤700	≤700	≤700	≤700	≤700	≤700	≤700	≤700
No-load speed (rpm)	2200	1640	630	330	165	120	90	64	33	24	13
Rated torque(Kg.cm)	11	15	30	30	40	50	60	60	90	90	90
Rated torque(N.m)	1.2	1.3	3	3	4	5	6	6	9	9	9
Rated-load speed (rpm)	1100	820	300	250	130	105	80	55	30	22	12
Rated-load current (A)	≤13	≤13	≤13	≤9	≤7	≤6	≤5	≤4	≤4	≤4	≤9
Stall current (A)	24.5	24.5	24.5	24.5	24.5	24.5	24.5	24.5	24.5	24.5	24.5
Gearbox Length (L)	33.5	33.5	43.5	43.5	53.5	53.5	53.5	53.5	63.5	63.5	63.5

Figure D-1: Spec Sheet for Motor Selection

Product details

Current Rating	75 Amps
Item Pitch	17.45 Millimeters
Connector Type	Screw
Number Of Contacts	10
Mounting Type	Panel Mount
Material	Copper
Color	75A/600V, M5
Brand	GLOSO
Voltage	600 Volts
Number Of Poles	10

Figure D-2: Datasheet from Motor Selection

Technical data sheet PLA



Description

PLA (Polylactic Acid) is a biodegradable, sustainable and food safe polymer made from organic sources.

It is the most common used filament in FFF 3D printers for its ease of use and a wide range of applications, especially those not mechanically or thermally demanding. Definitely a good starting point to learn about the 3D Printing manufacturing process.

Properties

- Detailed and glossy surface quality
- Good tensile strength
- Rigid, fragile behaviour
- Good UV resistance
- Withstand operating temperatures up to 50 °C.
- Odor-free, ideal for educational and office environments
- Compatible with PVA supports
- Low humidity resistance

Recommendations

Plastics absorb moisture from the air. For long periods of time without printing, it is recommended to keep the PLA spools in a box or airtight container with desiccant to keep them dry.

PLA emits low levels of gasses and particles when printed. We recommend printing it in a well-ventilated area to ensure a healthy environment.

Filament specifications

Diameter	Ø 2.85 mm
Max roundness deviation	≥ 95%
Net filament weight	750 g
Specific gravity (ISO 1183)	1.24 g/cm³

Mechanical properties

	Typical value	Test method
MFR 210°C/2.16 kg	9.56 gr/10 min	ISO 1133
Tensile strength at yield	70 Mpa	ISO 527
Strain at yield	5 %	ISO 527
Strain at break	20 %	ISO 527
Tensile Modulus	3120 MPa	ISO 527
Impact strength-Charpy method 23°C	3.4 kJ/m²	ISO 179
Moisture absorption	1968 ppm	ISO 62

Thermal properties

	Typical value	Test method
Melting temp.	115±35°C	ISO 11357
Vicat softening temp.	60 °C	ISO 306
Glass transition temp.	57 °C	ISO 11357

Printing settings

Extruder temperature	190 °C - 220 °C
Bed temperature	65 °C
Speed	10-70 mm/s
Retraction speed	40 mm/s
Retraction distance	4 mm
Cooling fan	Yes
Minimum layer height	0.05 mm

More information about PLA: <https://www.bcn3dtechnologies.com/en/3d-printer/filaments/#pla>

Figure D-3: Technical Data Sheet on PLA

Research on Intellectual Property

Link to article:

<https://www.ambercycle.com/news/inditex-signs-a-three-year-agreement-to-buy-textile-to-textile-recycled-polyester-cycora-r-for-over-eu70-million>

According to an article, “Inditex and Ambercycle sign a three-year agreement to buy regenerated polyester for over €70 million”. Inditex, the company behind Zara, has partnered with Ambercycle in a €70 million, three-year deal to buy recycled polyester made from textile waste, called Cycora. This collaboration not only helps Ambercycle scale up its recycling technology but also supports Inditex’s commitment to producing fully sustainable textiles by 2030. The key takeaway from this is that the partnership is protecting intellectual property. Ambercycle’s patented recycling process and its brand are essential for innovation and growth. This allows both parties to profit while the startup is able to keep its own individuality. This is important for mechanical engineers for this project because it is a clear reminder to startups of the importance of safeguarding ideas, documenting the work, and seeking collaborations that drive the funding to the product.

Ownership and Confidentiality

Considering this is an independent project from the team members of this project, with no company affiliations or university backing, the members plan to hold on to ownership of all designs and assemblies.

Research

A full project report will fully outline the methodology, results, research, and future search depending on the complications that may arise. Please review the Gantt Chart and the project research and development timeline.

IP Protection

Our team is the original creator of this project and its design. All concepts, schematics, software, and prototypes developed are considered proprietary intellectual property. To safeguard our work, we will document all designs and maintain dated records, keep project materials confidential, and use non-disclosure agreements when sharing details outside the team.

ME 195A Trash Catching Vacuum Robot Non-Disclosure Agreement

It is understood and agreed to that the below identified disclosure of confidential information may provide certain information that is and must be kept confidential. To ensure the protection of such information, and to preserve any confidentiality necessary under patent and/or trade secret laws, it is agreed that

1. The Confidential Information to be disclosed can be described as and includes:

Invention description(s), technical and business information relating to proprietary ideas and inventions, ideas, patentable ideas, trade secrets, drawings and/or illustrations, patent searches, existing and/or contemplated products and services, research and development, production, costs, profit and margin information, finances and financial projections, customers, clients, marketing, and current or future business plans and models, regardless of whether such information is designated as "Confidential Information" at the time of its disclosure.

2. The Recipient shall limit disclosure of confidential Information within its own organization to its directors, officers, partners, members, employees and/or independent contractors (collectively referred to as "affiliates") having a need to know. The Recipient and affiliates will not disclose the confidential information obtained from the disclosure unless required to do so by law.

3. This Agreement imposes no obligation upon Recipient with respect to any Confidential Information (a) that was in Recipient's possession before receipt from Discloser; (b) is or becomes a matter of public knowledge through no fault of Recipient; (c) is rightfully received by Recipient from a third party not owing a duty of confidentiality to the Discloser; (d) is disclosed without a duty of confidentiality to a third party by, or with the authorization of, Discloser; or (e) is independently derived by Recipient.

4. This Agreement states the entire agreement between the parties concerning the disclosure of Confidential Information. Any addition or modification to this Agreement must be made in writing and signed by the parties.

5. If any of the provisions of this Agreement are found to be unenforceable, the remainder shall be enforced as fully as possible and the unenforceable provision(s) shall be deemed modified to the limited extent required to permit enforcement of the Agreement as a whole.

WHEREFORE, the parties acknowledge that they have read and understand this Agreement and voluntarily accept the duties and obligations set forth herein.

Disclosure of Confidential Information:

Name (Print or Type): Patrick Wong

Signature: _____



Date: 12/12/2025

Disclosure of Confidential Information:

Name (Print or Type): Gustavo Ornelas Ledezma

Signature: *Gustavo Ornelas*

Date: 12/12/2025

Disclosure of Confidential Information:

Name (Print or Type): Branden Tran

Signature: *Branden*

Date: 12/12/2025

Disclosure of Confidential Information:

Name (Print or Type): Alejandro Cedenio

Signature: *Alejandro Cedenio*

Date: 12/12/2025

Disclosure of Confidential Information:

Name (Print or Type): Kristopher Do

Signature: *Kristopher Do*

Date: 12/12/2025

Signatures

Faculty Advisor: Dr. Raghu Agarwal Signature: _____ Date: _____

Team Members:

1. Patrick Wong Signature: _____ Date: _____

2. Branden Tran Signature: _____ Date: _____

3. Gustavo Ornelas Signature: _____ Date: _____

4. Alejandro Cedeno Signature: _____ Date: _____

5. Kristopher Do Signature: _____ Date: _____